



สถาบันพัฒนาฝีมือ
แรงงาน 3 ชลบุรี

อบรมเชิงปฏิบัติการหลักสูตร ระบบขับเคลื่อนและส่งกำลังยานยนต์ไฟฟ้า



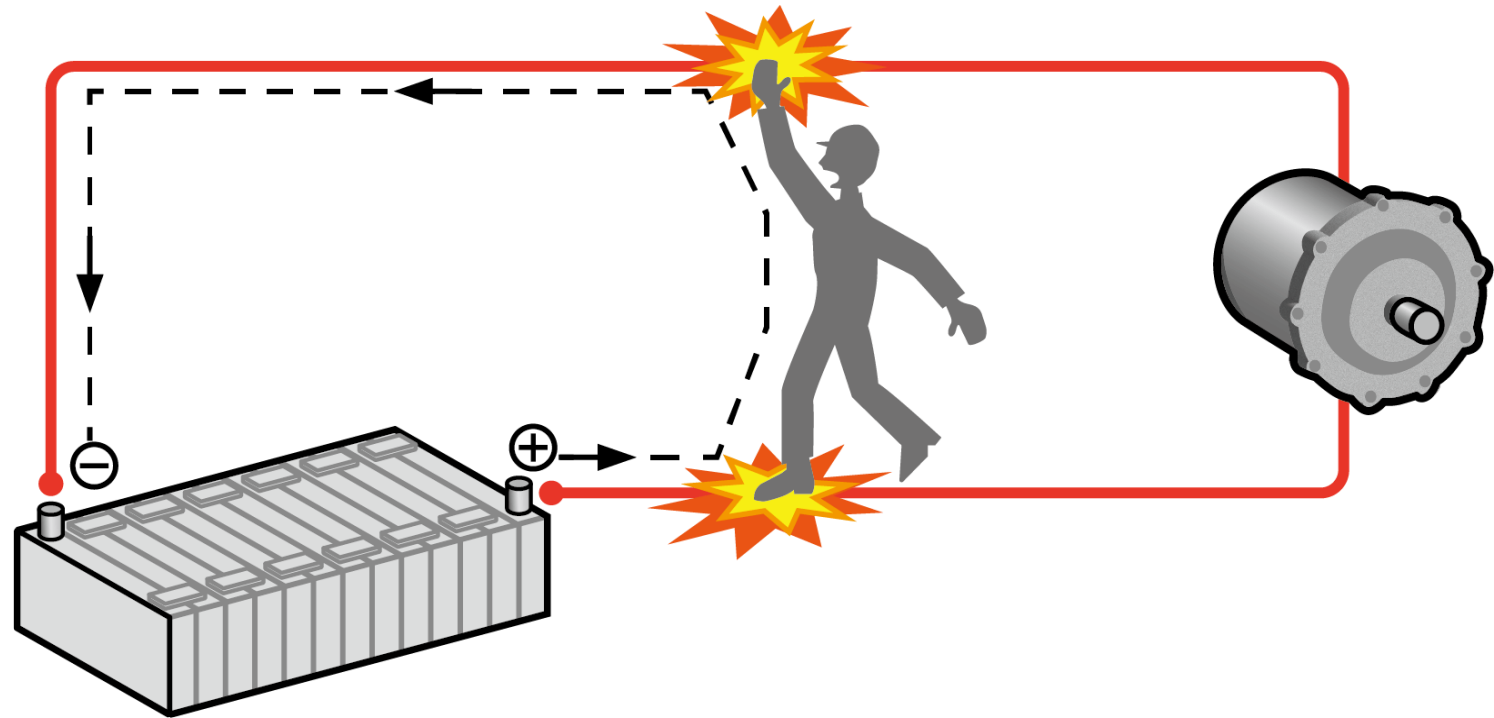
วันที่ 3-6 AUG 69

ณ.สถาบันพัฒนาฝีมือแรงงาน 3 ชลบุรี

โดย ผู้ช่วยศาสตราจารย์สุวิทย์ วงษ์เย็น

ผู้ช่วยศาสตราจารย์สุวิทย์ วงษ์เย็น

1.ความปลอดภัยในการปฏิบัติงานการซ่อมบำรุงรถยนต์ไฟฟ้า



1.1 นิยามความปลอดภัยของรถยนต์ไฟฟ้า (EV)

นิยามความปลอดภัยของรถยนต์ไฟฟ้า (EV)

ความปลอดภัยของรถยนต์ไฟฟ้า (EV) ครอบคลุมถึงแนวปฏิบัติ กฎระเบียบ และเทคโนโลยีที่ออกแบบมาเพื่อให้มั่นใจในความปลอดภัยและสวัสดิภาพของบุคคลและสิ่งแวดล้อมที่เกี่ยวข้องกับรถยนต์ไฟฟ้า แนวคิดนี้มีความเกี่ยวพันอย่างลึกซึ้งกับ HSE (สุขภาพ ความปลอดภัย และสิ่งแวดล้อม) โดยมุ่งเน้นที่การลดความเสี่ยงที่เกี่ยวข้องกับการใช้งาน การบำรุงรักษา และการชาร์จรถยนต์ไฟฟ้า



1.2 การระบุอันตรายและความเสี่ยงที่เกี่ยวข้องกับรถยนต์ไฟฟ้า

การระบุอันตรายและความเสี่ยงที่เกี่ยวข้องกับรถยนต์ไฟฟ้า

(Identifying Hazards and Risks Associated with EVs)

แม้ว่ารถยนต์ไฟฟ้าจะมีข้อดีมากมาย แต่ก็มีอันตรายเฉพาะตัวที่ต้องพิจารณาอย่างรอบคอบ

1. อันตรายจากระบบไฟฟ้า(Electrical Hazards)

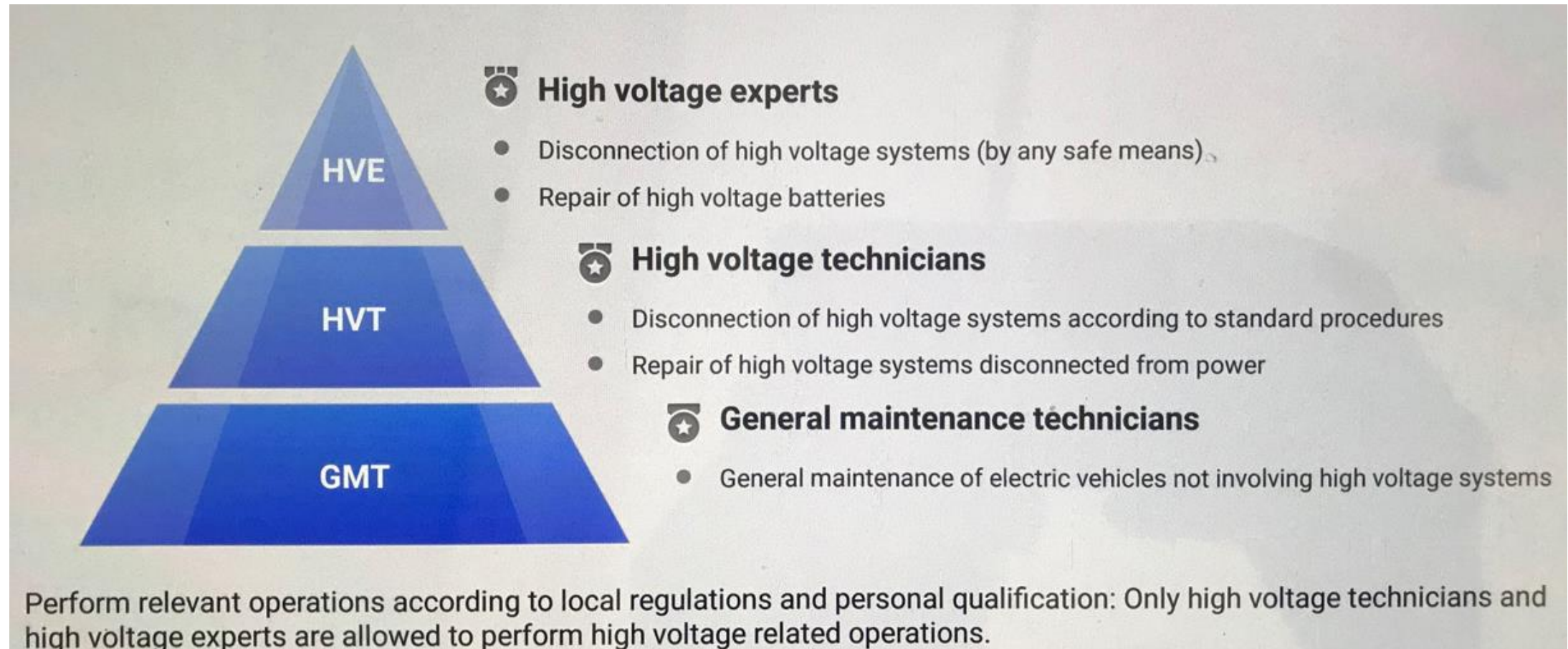
ความเสี่ยงที่เห็นได้ชัดที่สุดในรถยนต์ไฟฟ้ามาจากระบบไฟฟ้า ส่วนประกอบแรงดันสูง ซึ่งมักทำงานระหว่าง 200-800 โวลต์ อาจก่อให้เกิดอันตรายจากไฟฟ้าดูดหรือลัดวงจรอย่างรุนแรง การลัดวงจรหรือสายเคเบิลที่ชำรุดอาจนำไปสู่ไฟฟ้าดูดและไฟไหม้ ตัวอย่างเช่น กรณีที่รายงานโดยสมาคมป้องกันอัคคีภัยแห่งชาติ (NFPA) เน้นย้ำถึงเหตุการณ์ที่เกิดไฟไหม้เนื่องจากสายแบตเตอรี่ชำรุด

2. ความเสี่ยงของแบตเตอรี่(Battery Risks)

รถยนต์ไฟฟ้าใช้แบตเตอรี่ลิเทียมไอออน ซึ่งแม้จะมีประสิทธิภาพ แต่ก็อาจเกิดอันตรายได้หากไม่ได้รับการจัดการอย่างเหมาะสม การชาร์จไฟเกิน ความเสียหายทางกายภาพ หรือข้อบกพร่องในการผลิต อาจนำไปสู่ภาวะความร้อนสูงเกินไป ซึ่งเป็นสถานะที่แบตเตอรี่ร้อนจัดและอาจเกิดไฟไหม้หรือระเบิดได้

1.3 Professional technicians

The repair and maintenance of battery system and high-voltage system of new energy vehicles must be carried out by professional technicians with battery professional knowledge, electrician certificate and technically



1.4 มาตรฐานฝีมือแรงงานแห่งชาติ สาขาอาชีพช่างเครื่องกล สาขาช่างซ่อมบำรุงรถยนต์ไฟฟ้า

ข้อ 2 มาตรฐานฝีมือแรงงานแห่งชาติ สาขาอาชีพช่างเครื่องกล สาขาช่างซ่อมบำรุงรถยนต์ไฟฟ้า แบ่งออกเป็น 3 ระดับ ดังต่อไปนี้

2.1 ระดับ 2 หมายถึง ผู้ที่มีความรู้ ความสามารถในการทำงานซ่อมบำรุง รถยนต์ไฟฟ้า งานระบบไฟฟ้าแรงเคลื่อนสูงในรถยนต์ไฟฟ้า งานระบบไฟฟ้าแรงเคลื่อนต่ำในรถยนต์ไฟฟ้า งานระบบเครื่องปรับอากาศในรถยนต์ไฟฟ้า งานระบบระบายความร้อนในรถยนต์ไฟฟ้า งานระบบส่งกำลัง และรองรับน้ำหนักในรถยนต์ไฟฟ้า และงานทั่วไป

2.2 ระดับ 3 หมายถึง ผู้ที่มีความรู้ ความสามารถในการทำงานซ่อมบำรุงรถยนต์ไฟฟ้า งานระบบไฟฟ้าแรงเคลื่อนสูงในรถยนต์ไฟฟ้า งานระบบไฟฟ้าแรงเคลื่อนต่ำในรถยนต์ไฟฟ้า งานระบบเครื่องปรับอากาศในรถยนต์ไฟฟ้า งานระบบระบายความร้อนในรถยนต์ไฟฟ้า งานระบบส่งกำลังและรองรับน้ำหนักในรถยนต์ไฟฟ้า และการให้คำปรึกษาในการปฏิบัติงาน

2.3 ระดับ 4 หมายถึง ผู้ที่มีความรู้ ความสามารถในการทำงานซ่อมบำรุงรถยนต์ไฟฟ้า งานวินิจฉัยขั้นสูงของระบบไฟฟ้าแรงสูง งานวินิจฉัยขั้นสูงของระบบไฟฟ้าแรงต่ำ งานวินิจฉัยขั้นสูงของระบบปรับอากาศในรถยนต์ไฟฟ้า งานวินิจฉัยขั้นสูงของระบบระบายความร้อน ในรถยนต์ไฟฟ้า งานวินิจฉัยขั้นสูงของระบบส่งถ่ายกำลังและรองรับน้ำหนักในรถยนต์ไฟฟ้า งานวินิจฉัยขั้นสูงที่เกี่ยวเนื่องกันหลายระบบ และการให้คำปรึกษาแก้ไขปัญหา

1.5 สิ่งที่ต้องปฏิบัติเพื่อป้องกันเกี่ยวกับ อันตรายและคำเตือน



DANGER

- During repair and maintenance, please use insulating and special tools, insulating gloves and insulating shoes.
- During operation and repair, do not wear metal ornaments such as gold and silver jewelry and watches.
- It is strictly forbidden for operators to operate by hand without wearing protective equipment.

CAUTION



1. The repair and maintenance of battery system and high-voltage system of new energy vehicles must be carried out by professional technicians with battery professional knowledge, electrician certificate and technically trained and authorized by our company, so as to avoid casualties, battery burning, fire and other accidents.
2. Before dismantling the high-voltage system, it is necessary to have the insulating tools needed in hand.

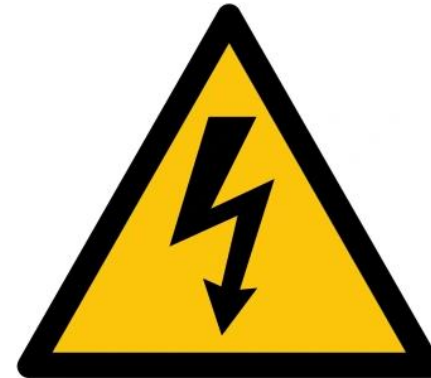
3. During removal and installation, the bolts must be tightened in accordance with the torque specified in "Removal and installation" to ensure assembly reliability.
4. The electronic components inside the drive motor are sensitive to static electricity, so prevent foreign matters from falling into the interior or coming into contact with the circuit board.
5. The motor housing shall be reliably grounded, otherwise there is a risk of electric shock or fire.
6. Make sure that the motor and MCU are compatible, otherwise the motor or MCU may be damaged.



1.6.ความปลอดภัยในการปฏิบัติงานกับรถยนต์ไฟฟ้า

ยานพาหนะ ทุกชนิดสามารถทำให้เกิดอันตรายกับผู้ขับขี่และผู้ปฏิบัติงานซ่อมบำรุงได้
ถ้าผู้เกี่ยวข้องขาดความระมัดระวัง ขณะปฏิบัติงาน

ในรถยนต์ไฟฟ้ามีแหล่งกักเก็บพลังงานที่เป็นไฟฟ้าแรงสูง คือแบตเตอรี่ไฟฟ้าแรงสูง ที่มีแรงดันไฟฟ้าตั้งแต่ 200 V – 800V ดังนั้นจึงต้องปฏิบัติงานด้วยความระมัดระวัง ใช้อุปกรณ์ป้องกันตนเอง(PPE)และเครื่องมือซ่อมบำรุงที่ป้องกันไฟฟ้าแรงสูง (VDE Tools) ต้องปฏิบัติตามขั้นตอนที่ปลอดภัย และ ป้ายเตือนอันตราย



1.7 แผ่นป้ายเตือนสถานะ(Status cards)



Attention!
High-Voltage
System Deactivated



Attention!
High-Voltage
Active



Attention!
High-Voltage
Active with
Electrical Fault

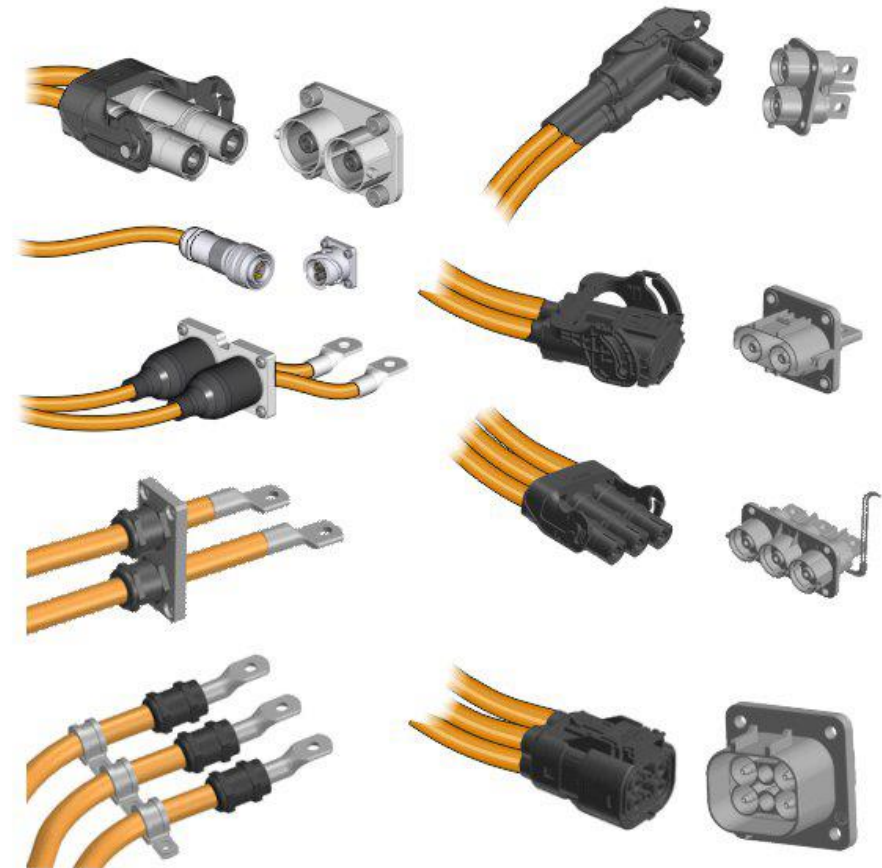
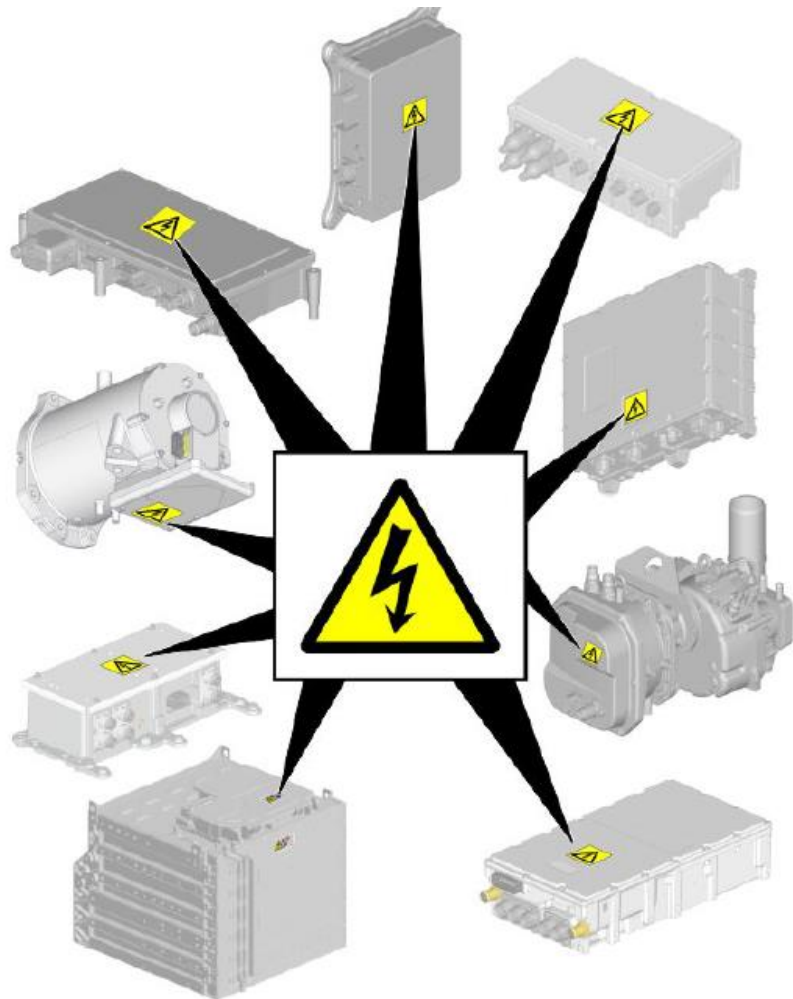


Status Cards:

- Green: HV is deactivated
- Yellow: HV is still active
- Red: HV is active with electrical fault

1.8 Warning label and identification of high voltage cables

Most of the high voltage components are marked with the warning label for hazardous voltage.



Traction voltage cables are orange in colour
for easy identification

สิ่งที่ต้องปฏิบัติเพื่อป้องกันเกี่ยวกับ อันตรายและค่าเตือน



DANGER

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- During operation and repair, do not wear metal ornaments such as gold and silver jewelry and watches.
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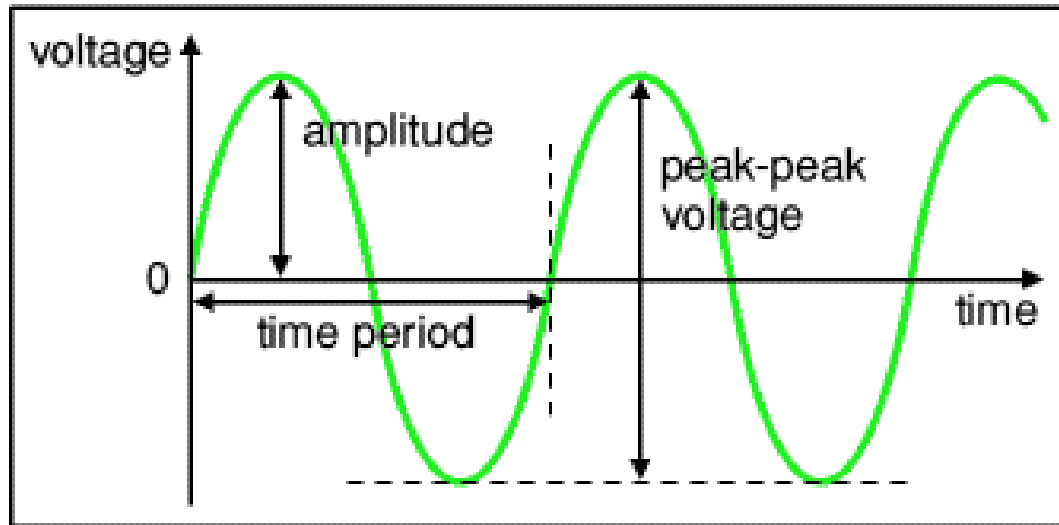
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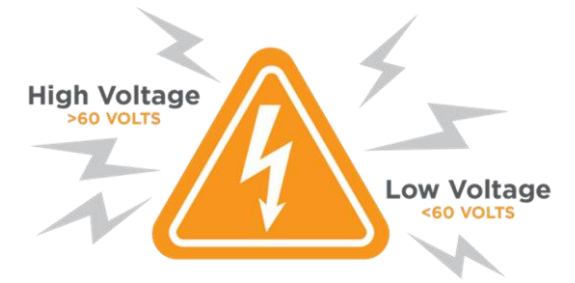
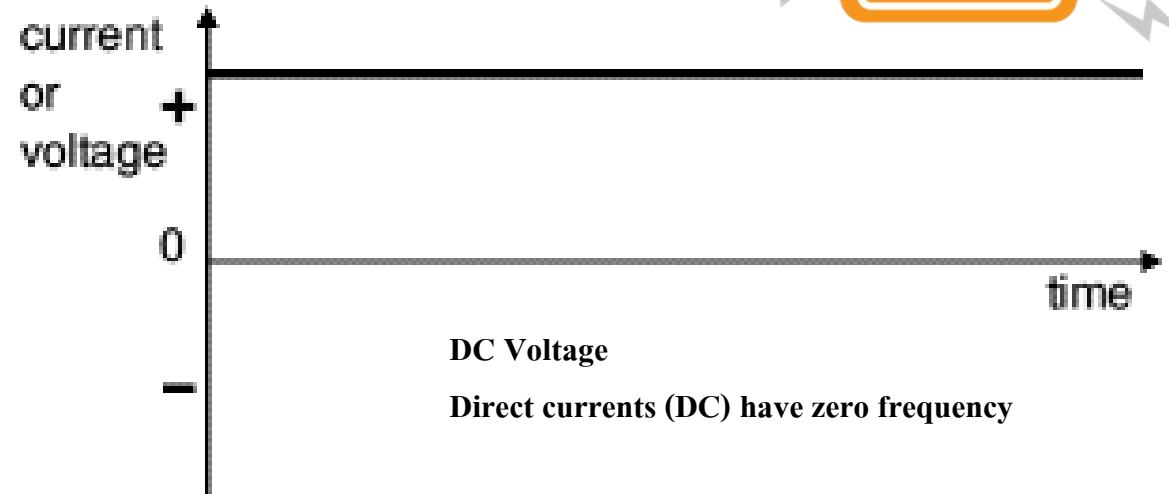
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6. Make sure that the motor and MCU are compatible, otherwise the motor or MCU may be damaged.



ชนิดของกระแสไฟฟ้า(Electric type)



AC Voltage /AC current 50–60 Hz



อันตรายจากไฟฟ้าดูด(Electric shock)

Dangerous from electric up to

- size of the current
- duration
- frequency.

Main causes of electrical injury or electrocution

Electric contact can occur in many ways and situations, we listed the most common below:

- defective or damaged tools, badly maintained equipment
- inadequate wiring, improper grounding
- unsafe work practice or environment
- unexpected exposition to electric parts
- contact with overhead power lines, high voltage circuits
- overloaded circuits, improper insulation

ข้อกำหนดที่เป็นมาตรฐานเกี่ยวกับการอนุมัติยานพาหนะตาม ข้อกำหนดเฉพาะของระบบส่งกำลังไฟฟ้า

(Uniform provisions concerning the approval of vehicles with regard to specific requirements for the electric power train)

UN ECE R100



UNITED NATIONS

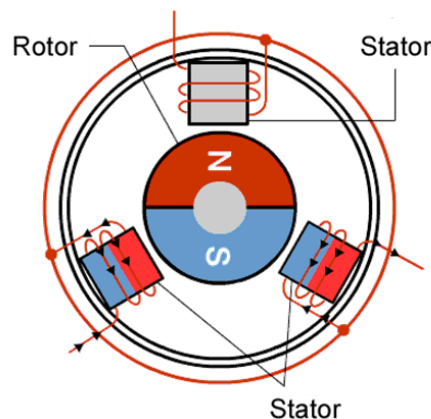
Regulation No. 100

Uniform provisions concerning the approval of vehicles with regard to specific requirements for the electric power train

1. Scope

The following prescriptions apply to safety requirements with respect to the electric power train of road vehicles of categories M and N, with a maximum design speed exceeding 25 km/h, equipped with one or more traction motor(s) operated by electric power and not permanently connected to the grid, as well as their high voltage components and systems which are galvanically connected to the high voltage bus of the electric power train.

This regulation does not cover post crash safety requirements of road vehicles.



M Passenger Vehicles



M1: ≤ 9 Seats

M2: > 9 Seats $\leq$
5000 kg



M3: > 9 Seats $>$
5000 kg

N Commercial Vehicles



N1: ≤ 3500 kg

N2: > 3500 kg $\leq$
12000 kg



N3: > 12000 kg

- 2.13. "*External electric power supply*" means an alternating current (AC) or direct current (DC) electric power supply outside of the vehicle.
- 2.14. "*High Voltage*" means the classification of an electric component or circuit, if its working voltage is $> 60 \text{ V}$ and $\leq 1500 \text{ V DC}$ or $> 30 \text{ V}$ and $\leq 1000 \text{ V AC}$ root mean square (rms).
- 2.15. "*High voltage bus*" means the electrical circuit, including the coupling system for charging the RESS that operates on high voltage.
- 2.16. "*Indirect contact*" means the contact of persons with exposed conductive parts.
- 2.17. "*Live parts*" means the conductive part(s) intended to be electrically energized in normal use.

5.1.3. Isolation resistance

5.1.3.1. Electric power train consisting of ***separate Direct Current- or Alternating Current-buses***

If AC high voltage buses and DC high voltage buses are galvanically isolated from each other,
isolation resistance between the high voltage bus and the electrical chassis shall have a minimum value of 100 Ω /volt of the working voltage for DC buses, and a minimum value of 500 Ω /volt of the working voltage for AC buses.

The measurement shall be conducted according to Annex 4A "Isolation resistance measurement method for vehicle based tests".

5.1.3.2. Electric power train consisting of ***combined DC- and AC-buses***

If AC high voltage buses and DC high voltage buses are galvanically connected isolation resistance between the high voltage bus and the electrical chassis shall have a minimum value of 500 Ω /volt of the working voltage.

However, if all AC high voltage buses are protected by one of the 2 following measures, isolation resistance between the high voltage bus and the electrical chassis shall have a minimum value of 100 Ω /V of the working voltage:

4.เครื่องมือวัดทางไฟฟ้าสำหรับการซ่อมบำรุงรถยนต์ไฟฟ้า



เครื่องมือวัดทางไฟฟ้าที่จำเป็นในการซ่อมบำรุงรถยนต์ไฟฟ้า



Digital multi-meter



Insulation Resistance Tester



Internal Resistance Battery Cell



Mili-Ohm meter



EVSE Adapter test



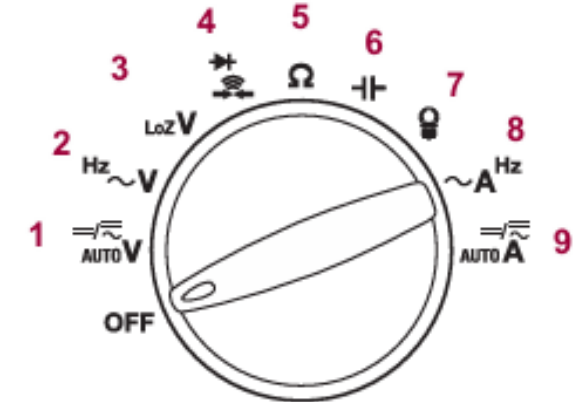
Clamp Meter



Thermal Image Camera

4.1 มาตรฐานดิจิทัล(Digital multimeter)

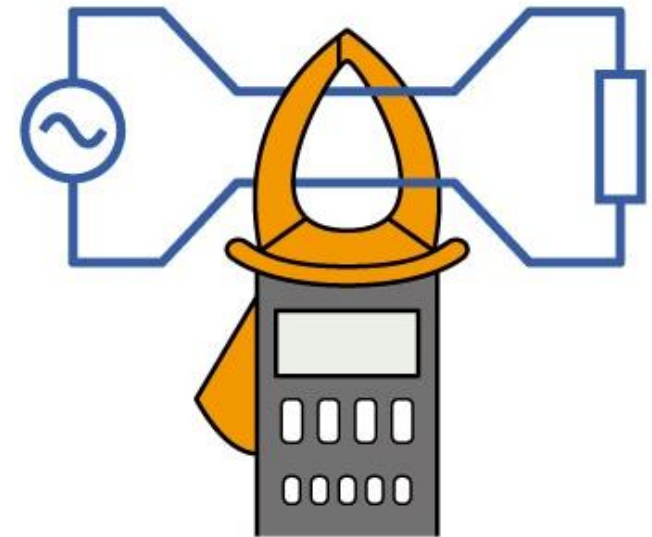
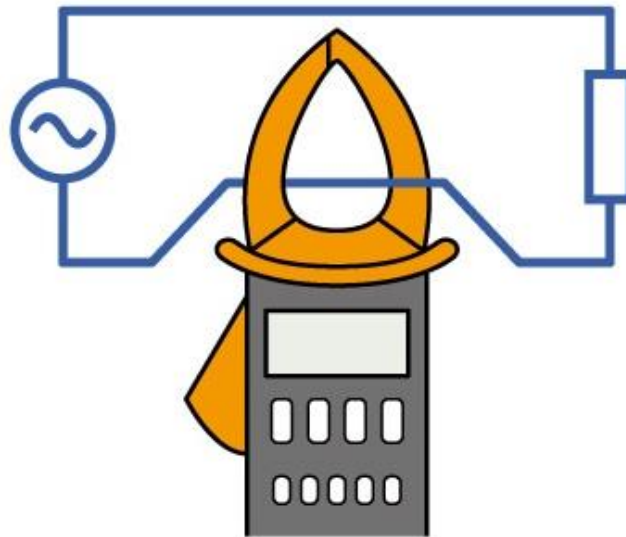
Measurement ,Voltage ,Resistance, Current ,Frequency



	OFF	Turns OFF the power to the instrument.
1	AUTO V	AC/DC voltage measurement (automatic judgment), DC voltage measurement, AC+DC voltage measurement (Input impedance 10 MΩ or more)
2	Hz ~ V	AC voltage measurement, frequency measurement
3	LoZ V	AC/DC voltage measurement (automatic judgment) (Input impedance 1.0 MΩ ±20%)
4	Diode	Continuity check, diode test
5	Ω	Resistance measurement
6	Capacitor	Electrostatic capacity measurement
7	Clamp sensor	AC current measurement (with clamp sensor)
8	Hz ~ A	AC current (A) measurement, frequency measurement
9	AUTO A	AC/DC current measurement (automatic judgment), DC current measurement, AC+DC current measurement

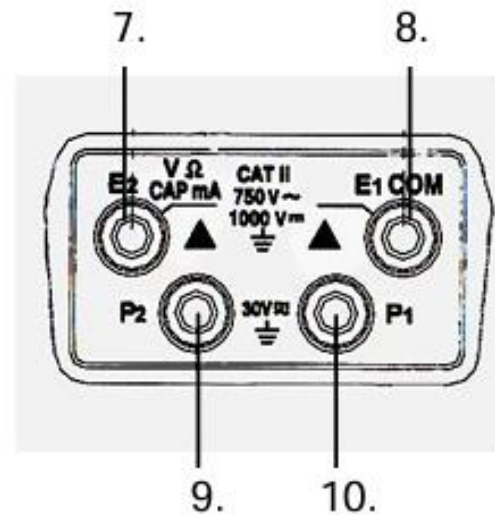
4.2 Clamp meters

Measurement ,AC And DC Currents and voltage

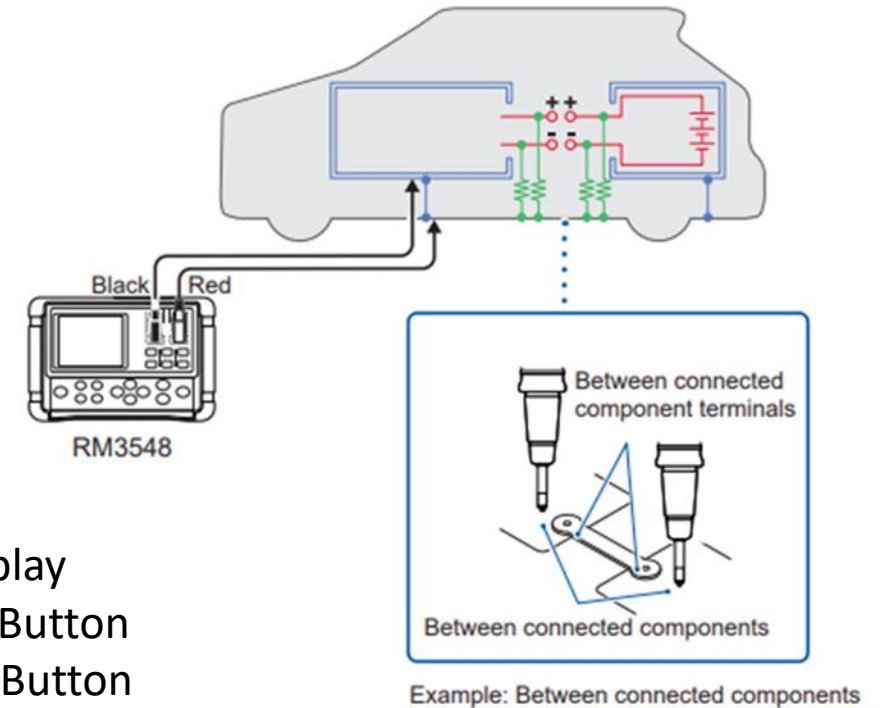


4.3 มิเตอร์วัดความต้านทานต่ำ (Digital Milli-Ohmmeter)

Measurement ,Voltage , Low resistance ,surface resistance

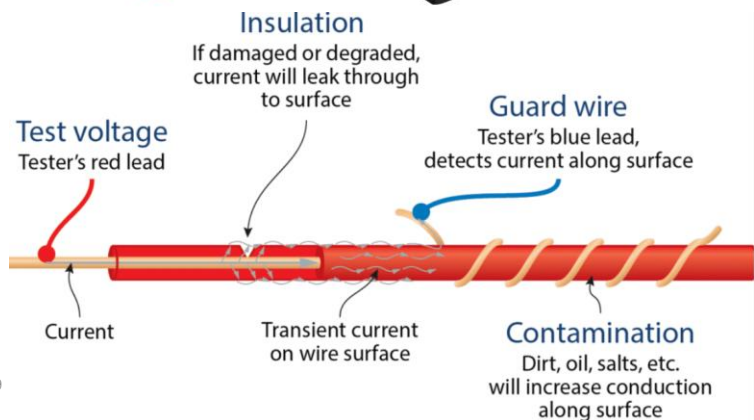


- 1.Digital Display
- 2.Data Hold Button
- 3.MAX/MIN Button
- 4.Backlight Button
- 5.Mode/REL Button
- 6.Rotary Function switch
- 7.V Ω Cap mA E2 Jack
- 8.COM E1 jack
- 9.P2 Jack
10. P1 jack



4.4 มิเตอร์วัดความต้านทานฉนวนไฟฟ้า(Insulation Resistance Tester)

Measurement Insulation resistance and isolation resistance



5 ranges

Rated output voltage (DC)
Effective maximum indicated value

50 V / 100 MΩ

125 V / 250 MΩ

250 V / 500 MΩ

500 V / 2000 MΩ

1000 V / 4000 MΩ

4.5 มิเตอร์วัดค่าความต้านทานภายในเซลล์แบตเตอรี่ (Internal Resistance Tester)

- 1, Battery internal resistance measurement;
- 2, Battery voltage measurement



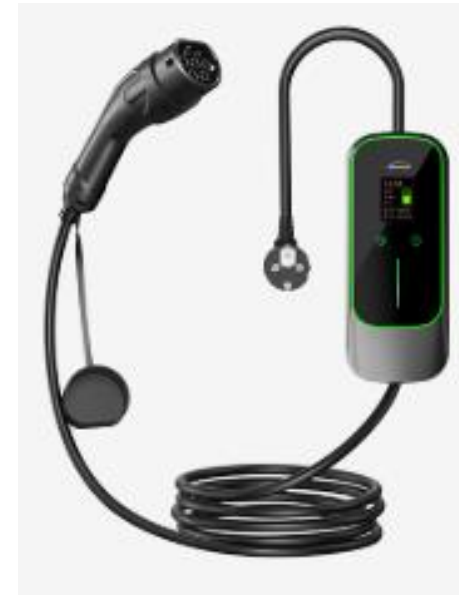
4.6 กล้องถ่ายภาพความร้อน(Thermal Imager Camera)

Measurement , High voltage battery temperature ,motor temperature



4.7 Electric vehicle supply equipment tester (EVSE)

EVSE Safety Inspection, PP signal, CP Signal, PE



5. เครื่องมือหุ้มฉนวนป้องกันไฟฟ้า(Insulated tools.)

	Electrical and Electronic Technology	All industrial sectors except electrical and electronic technology
International standards (world standard)	IEC standards	ISO Standards
European standards (regional level)	EN standard (CENELEC standard)	EN standards (CEN standard)

Conforms to IEC 60900:2004 standard, the only international standard for insulated tools.

**国際的な認証機関
VDE 認証絶縁ドライバー**

Insulated Screwdrivers Series certified by VDE,
International certification authority.

1000V AC, 1500V DC

VDE GS 1000V IEC60900

VDE は ヨーロッパで権威ある第三者認証機関です
VDE is an authorized third-party certification organization.

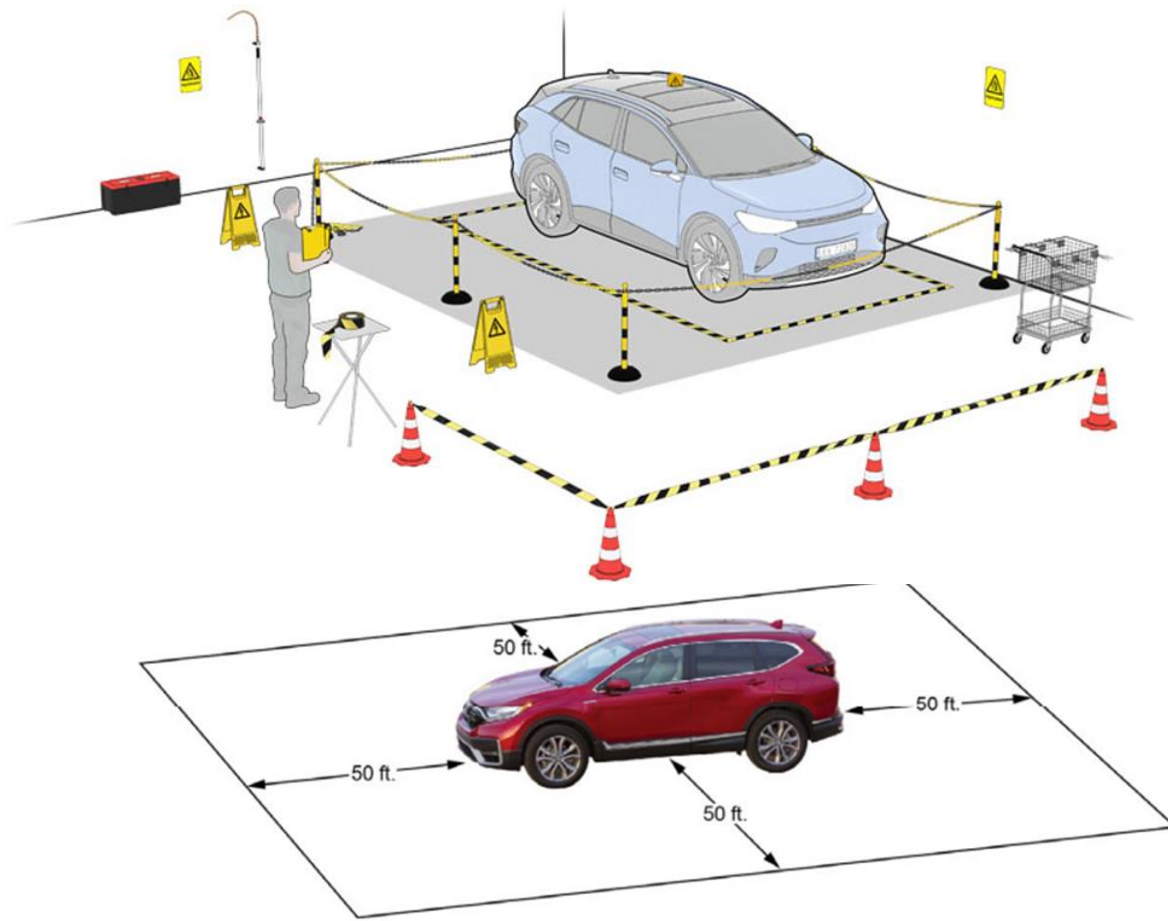


5.1 ประแจวัดแรงบิดหุ้มฉนวน(INSULATED TORQUE WRENCH)



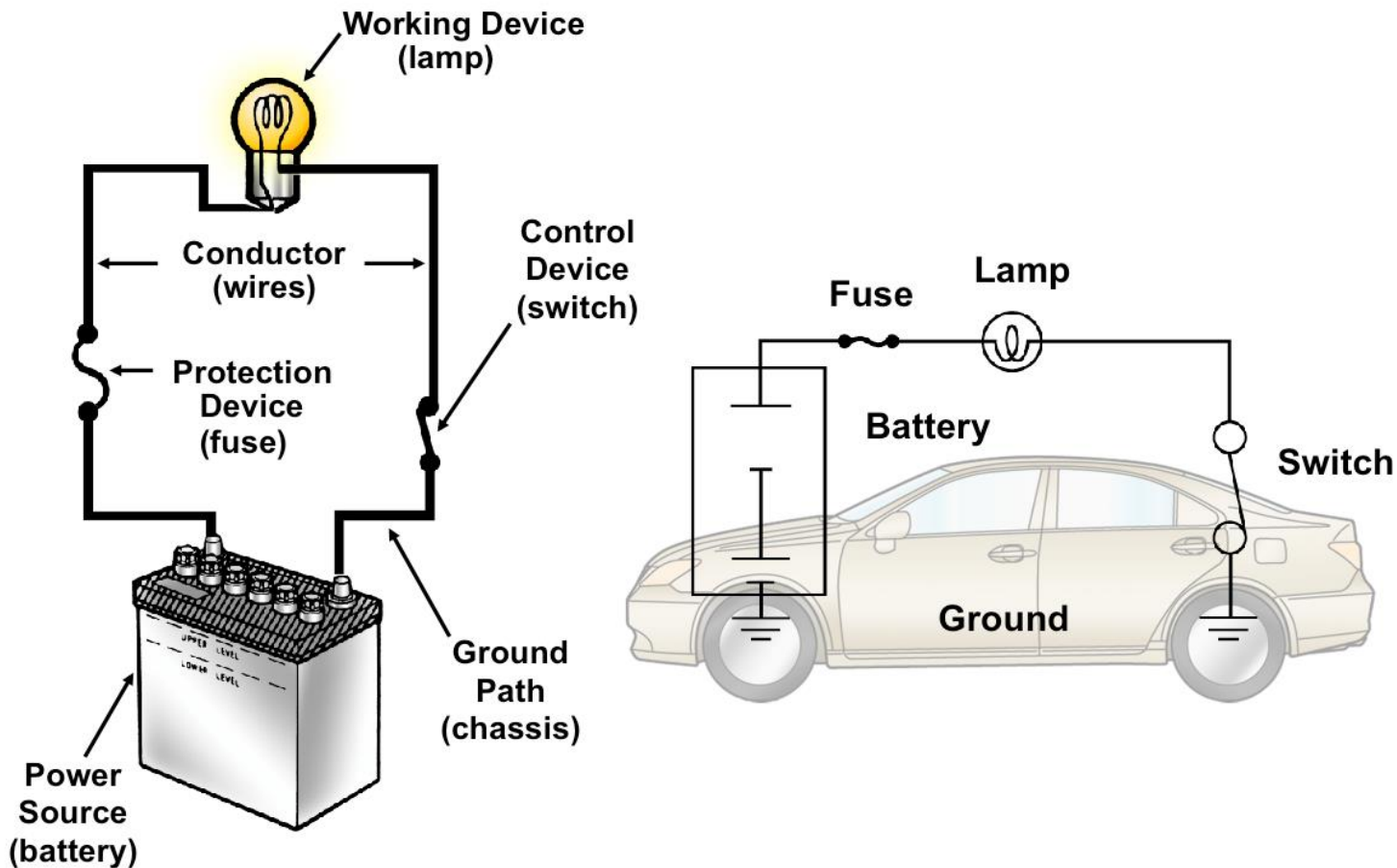
6.การเตรียมพื้นที่เพื่อปฏิบัติงานซ่อมบำรุงรถยนต์ไฟฟ้า

การซ่อมบำรุงรถยนต์ไฟฟ้าของสถานประกอบการ งานซ่อมทั่วไป งานซ่อมตัวถังและสี สถานที่จัดเก็บรถยนต์ที่ได้รับอุบัติเหตุสถานประกอบการต้องจัดเตรียมสถานที่ปฏิบัติงานให้ปลอดภัย เช่น กั้นพื้นที่เฉพาะ ป้ายเตือนอันตราย และผู้ปฏิบัติงานต้องมีความรู้ ความเข้าใจเกี่ยวกับรถยนต์ไฟฟ้าผ่านการฝึกอบรมและมีทักษะในส่วนเกี่ยวข้อง ได้รับการรับรองมาตรฐานช่างซ่อมรถยนต์ไฟฟ้า ระดับ 2 จากกรมพัฒนาฝีมือแรงงาน



Electrical Circuits

Electrical Circuits



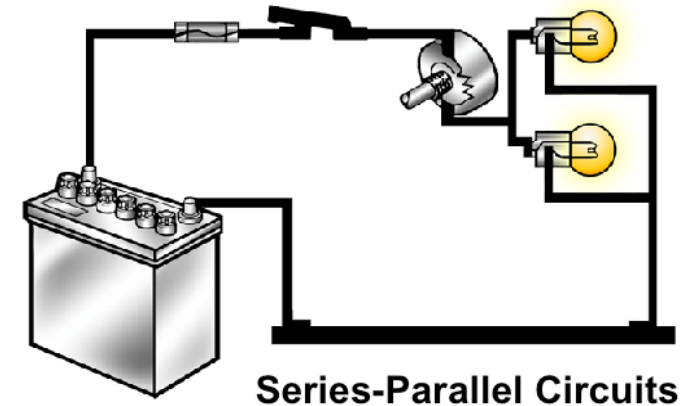
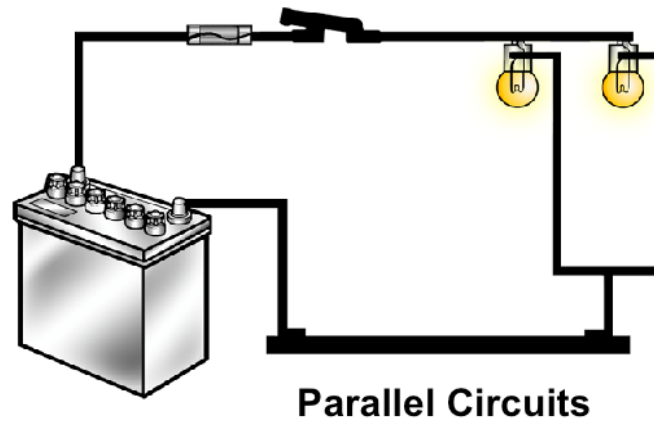
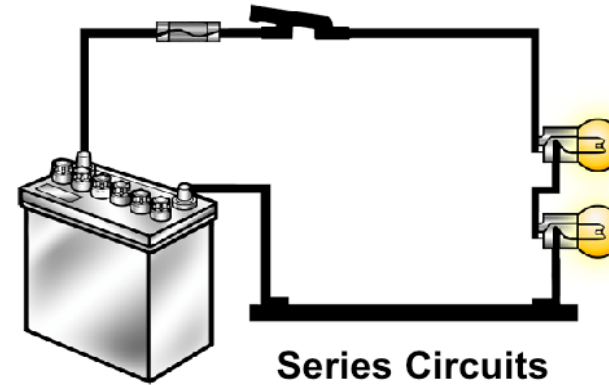
From Lexus Electrical Circuit Diagnosis

complete path, or circuit, is required for voltage to create current flow through resistances and perform work. Every electrical circuit on a vehicle contains:

- **Power source** to produce voltage or electrical potential
 - A battery or an alternator
- **Conductors** to provide a path for current flow
 - Wires or printed circuit boards (on the power side of the circuit)
 - Vehicle frame or chassis (on the ground side of the circuit)
- **Loads or working devices** to convert electrical energy into another form of energy to perform work
 - Lamps, motors, horn, defogger, cigarette lighter, etc.
- **Protection devices** to interrupt the current path and prevent damage to conductors and working devices from too much current
 - Fuses, circuit breakers, fusible links
- **Control devices (switches)** to open or close the current path, or to vary the amount of current in the circuit and regulate how the load works (dimmer switches)
 - On/off switches, potentiometers, relays, transistors, ECUs

Types of Electric Circuits

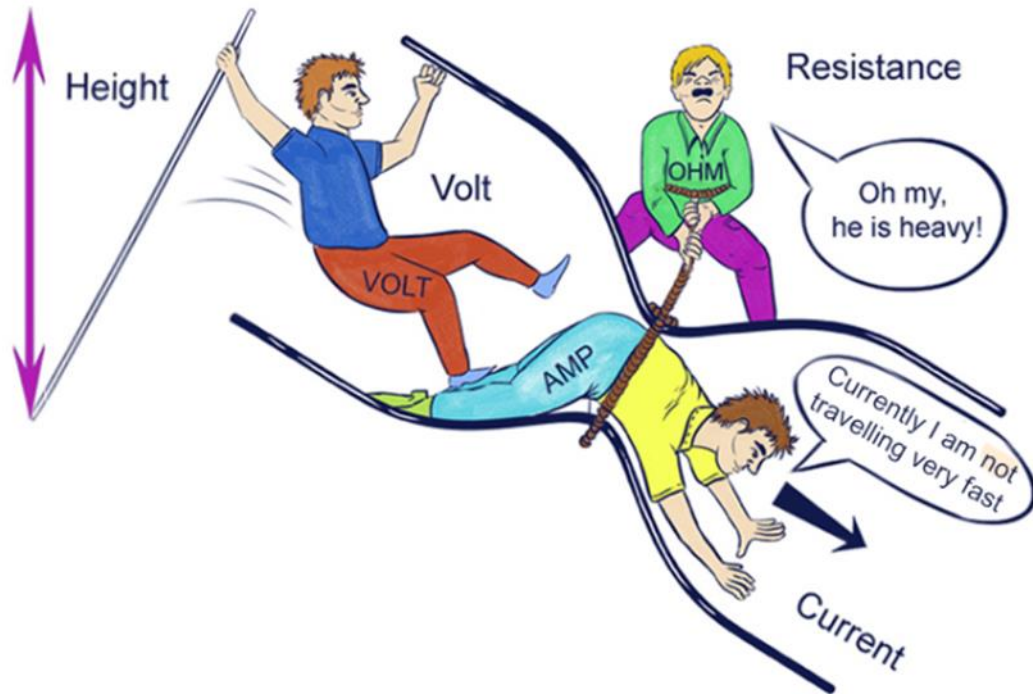
Types of Circuits



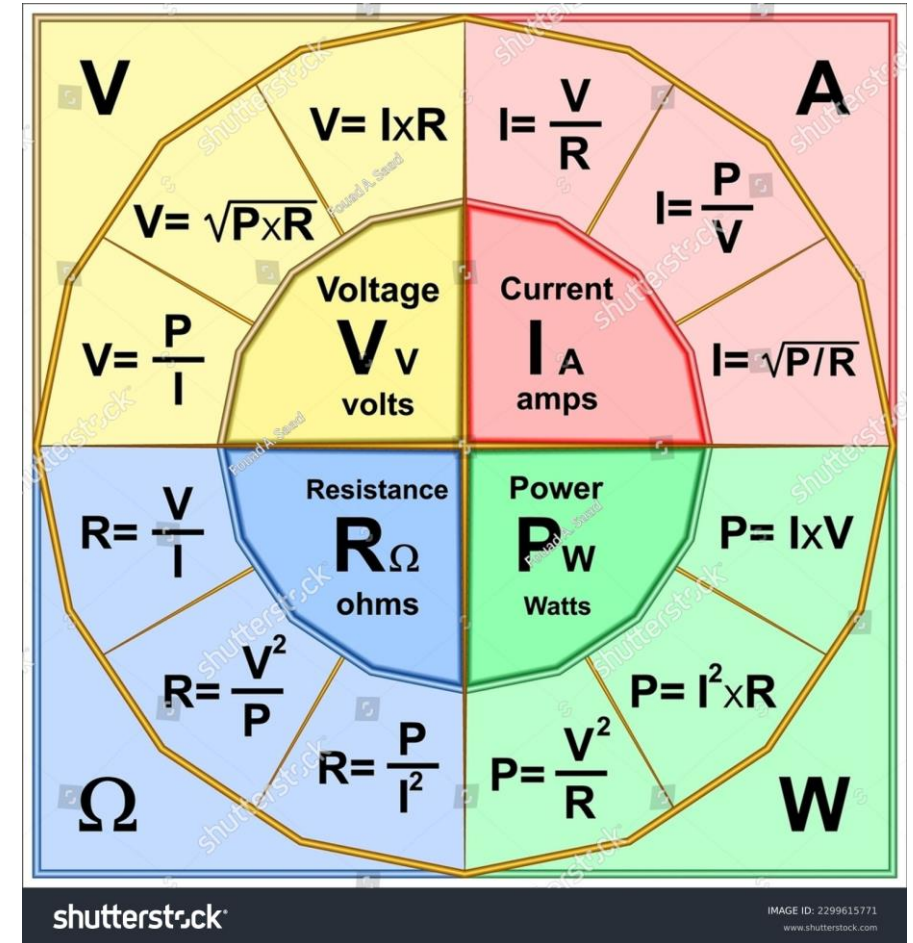
There are three types of circuits:

- Series – loads connected together with a common single path to ground
- Parallel – each load has its own alternate path to ground
- Series-Parallel – some loads on a single path, and others on alternate paths

Electric formula



กฎของโอห์ม(Ohm's Law)



<https://www.inchcalculator.com/ohms-law-calculator/>

เครื่องมือวัดทางไฟฟ้าที่จำเป็นในการซ่อมบำรุงรถยนต์ไฟฟ้า



Digital multi-meter



Insulation Resistance Tester



Internal Resistance Battery Cell



Mili-Ohm meter



EVSE Adapter test



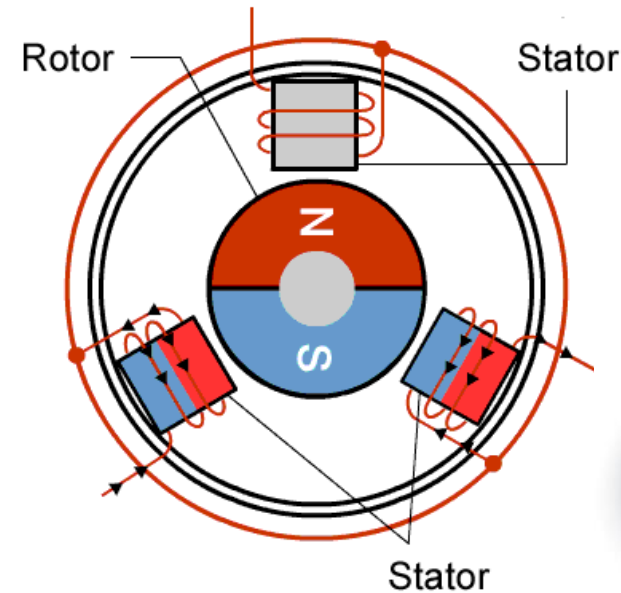
Clamp Meter



Thermal Image Camera

รถยนต์ไฟฟ้าคืออะไร(What is electric vehicle?)

	Front axle axially parallel (APA) i = 9.205	Rear axle coaxial (AKA / ATA) i = 9.08
Audi e-tron 50 quattro Audi e-tron 55 quattro	APA250  $P_{EM} = 125 \text{ kW}$ $P_{EM, boost} = 135 \text{ kW}$ $T_{EM} = 247 \text{ Nm} / T_{axle} = 2200 \text{ Nm}$ $T_{EM, boost} = 309 \text{ Nm} / T_{axle} = 2750 \text{ Nm}$	AKA320  $P_{EM} = 140 \text{ kW}$ $P_{EM, boost} = 165 \text{ kW}$ $T_{EM} = 314 \text{ Nm} / T_{axle} = 2800 \text{ Nm}$ $T_{EM, boost} = 355 \text{ Nm} / T_{axle} = 3120 \text{ Nm}$
Audi e-tron S 	APA320  $P_{EM} = 129 \text{ kW}$ $P_{EM, boost} = 157 \text{ kW}$ $T_{EM} = 314 \text{ Nm} / T_{axle} = 2800 \text{ Nm}$ $T_{EM, boost} = 355 \text{ Nm} / T_{axle} = 3120 \text{ Nm}$	ATA250  $P_{EM} = 2 \times 102 \text{ kW}$ $P_{EM, boost} = 2 \times 138 \text{ kW}$ $T_{EM} = 2 \times 247 \text{ Nm} / T_{axle} = 2 \times 2200 \text{ Nm}$ $T_{EM, boost} = 2 \times 309 \text{ Nm} / T_{axle} = 2 \times 2750 \text{ Nm}$



“An electric vehicle (EV) is a vehicle that uses one or more electric motors for propulsion”

<https://precisionevr.com/Electric-Vehicle-Motor-Repair>

ชนิดของรถยนต์ไฟฟ้า(Type of electric vehicle)



BEV
BATTERY ELECTRIC
VEHICLE



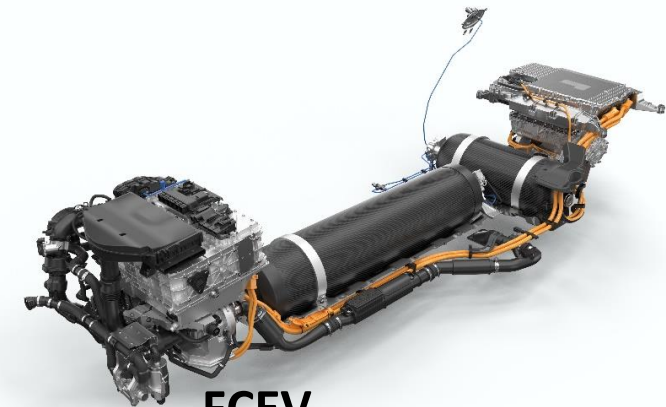
EREV
EXTENDED RANGE
ELECTRIC VEHICLE



PHEV
PLUG-IN HYBRID
ELECTRIC VEHICLE



HEV
HYBRID ELECTRIC
VEHICLE



FCEV
Fuel cell electric
vehicle

HEV = PHEV = BEV = EREV Or REEV = FCEV = Electric Vehicle

ไดอะแกรมของรถยนต์ไฟฟ้าชนิดต่างๆ

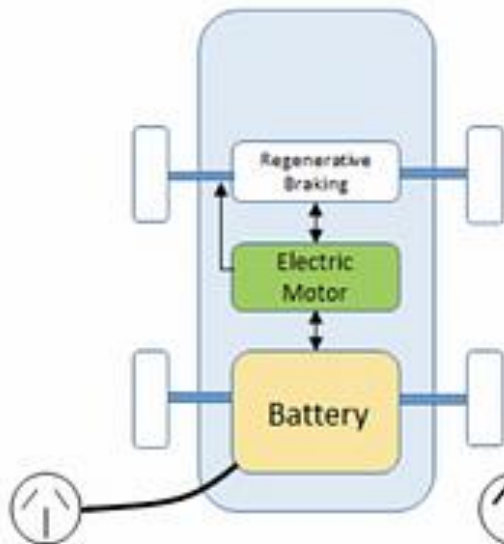
PEV

Plug-in Electric Vehicle



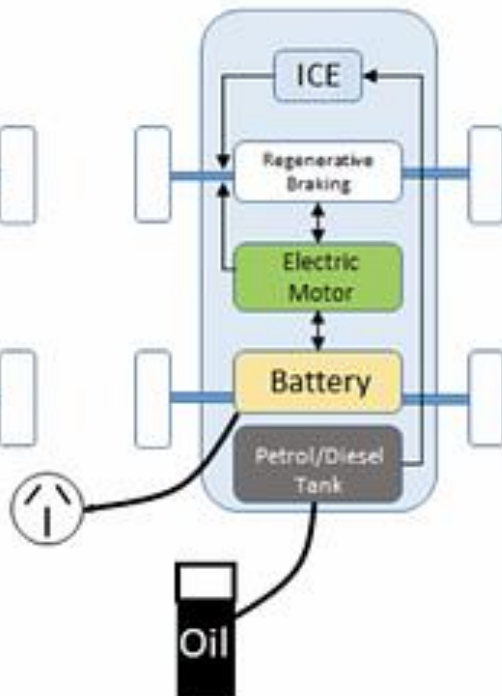
BEV

Battery Electric Vehicle



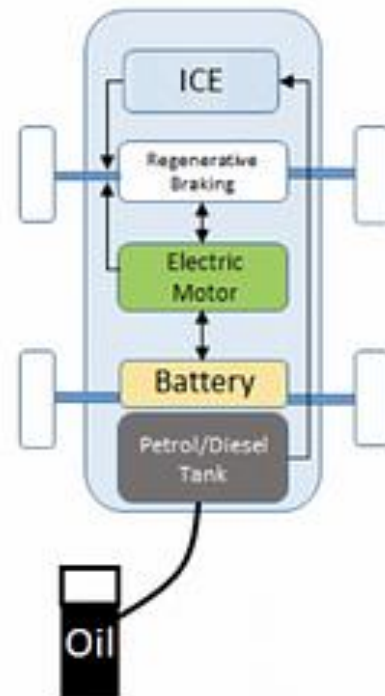
PHEV

Plug-in Hybrid Electric Vehicle



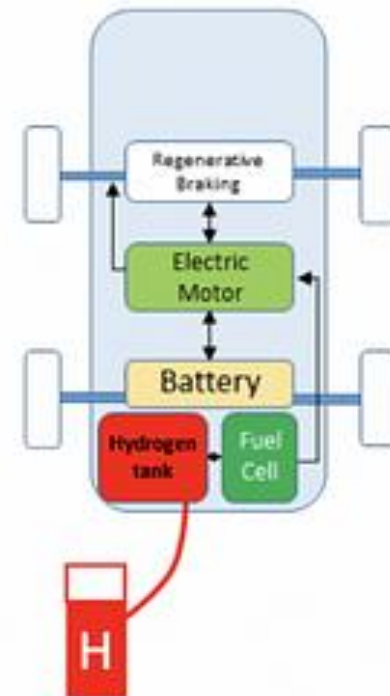
HEV

Hybrid Electric Vehicle

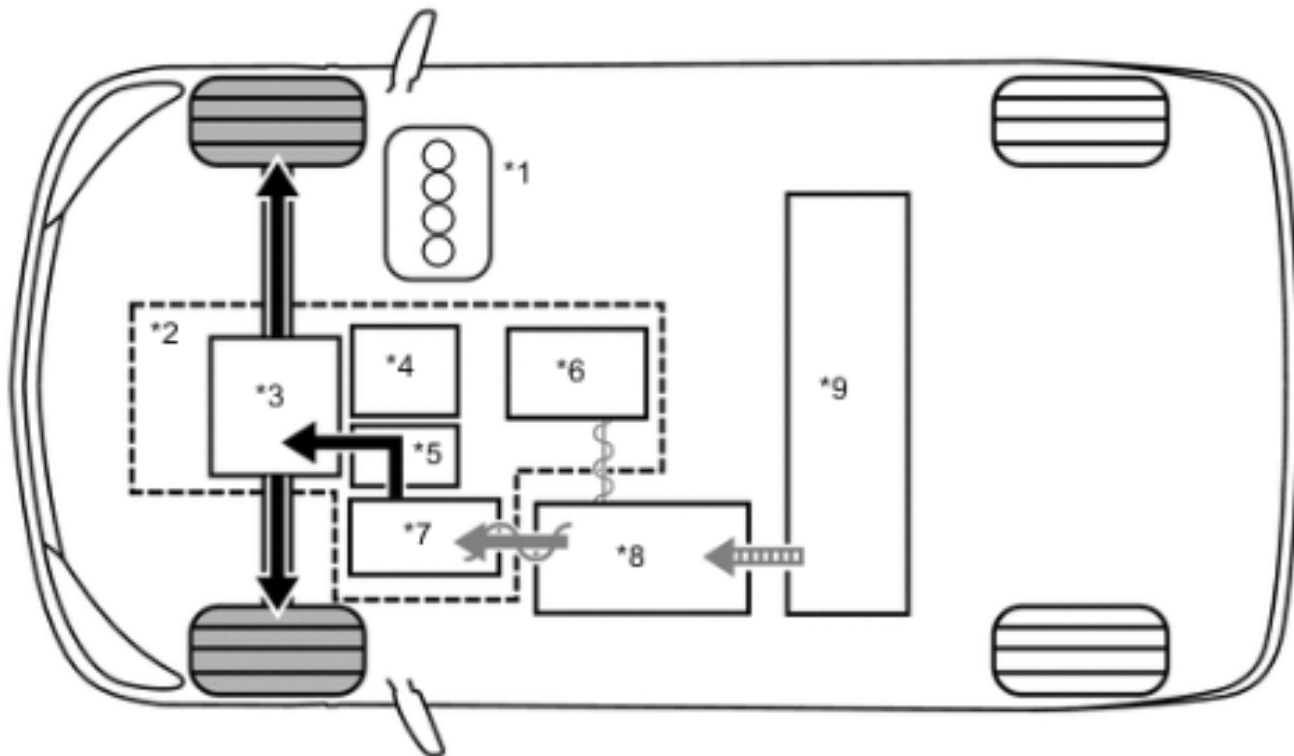


FCEV

Fuel Cell Electric Vehicle



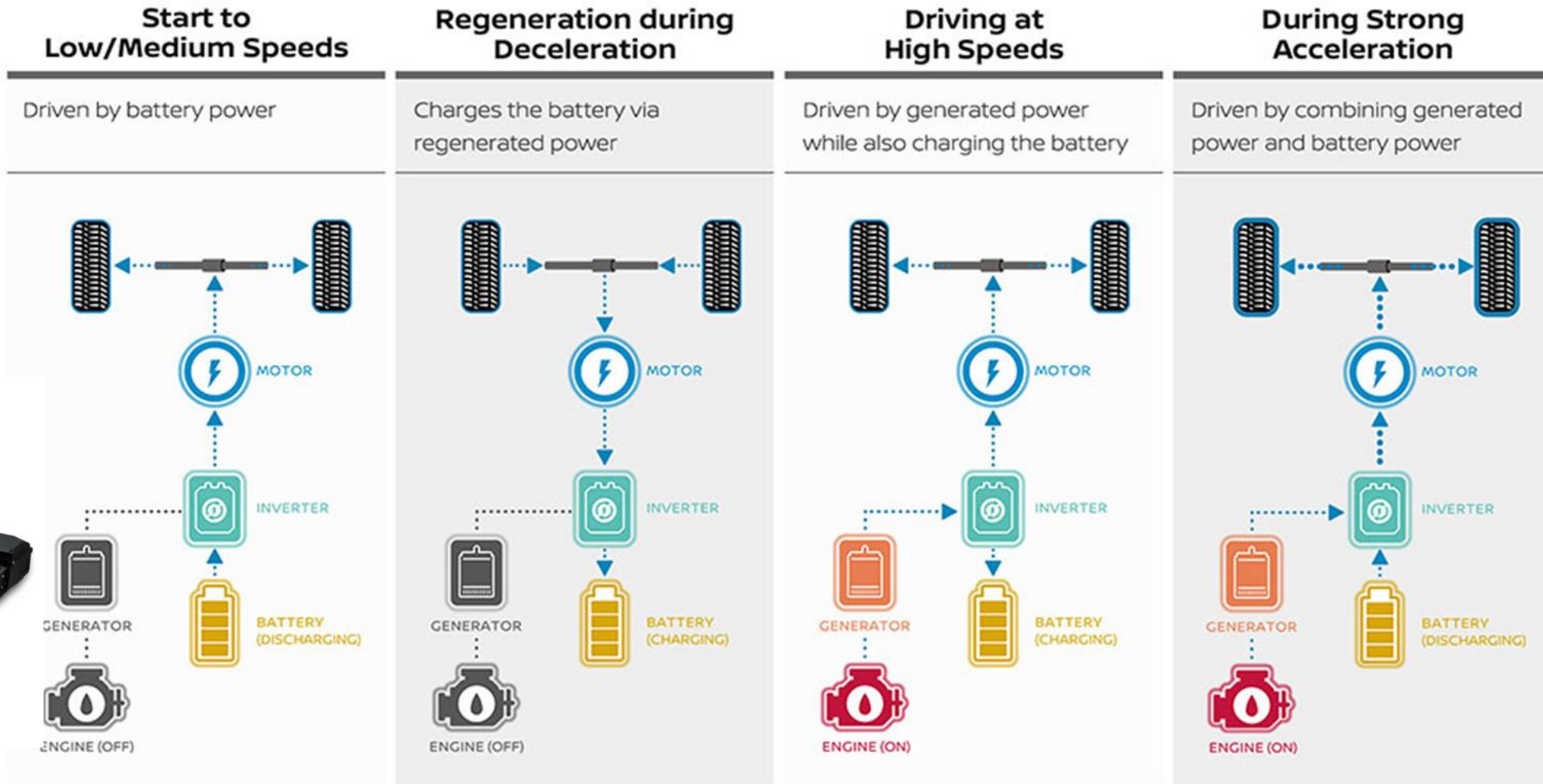
รถยนต์ไฟฟ้าแบบไฮบริด(HYBRID ELECTRIC VEHICLE) : HEV



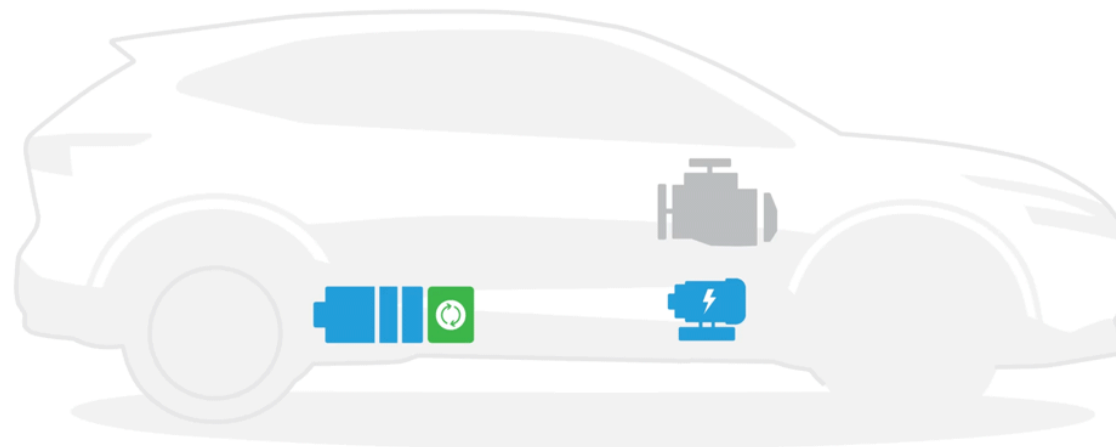
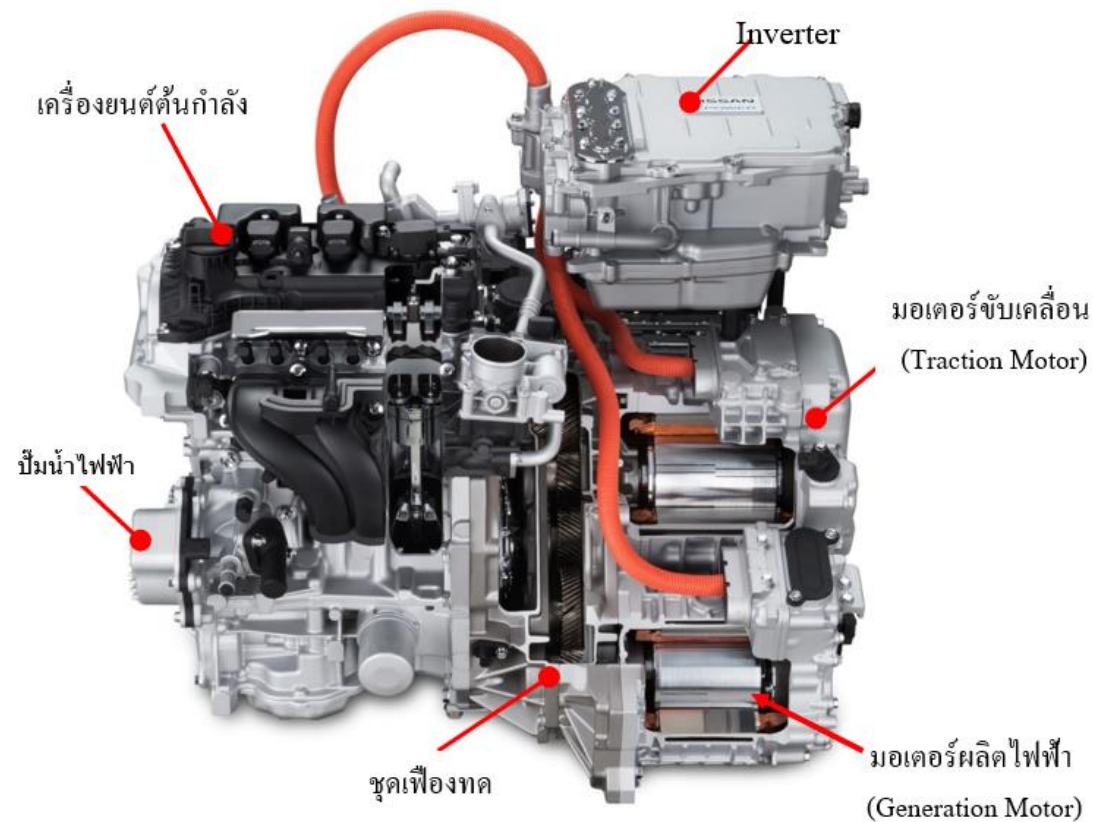
Starting Off

- *1 Engine
- *2 Hybrid Transaxle (Hybrid Vehicle Transaxle Assembly)
- *3 Differential
- *4 Power Split Planetary Gear
- *5 Motor Speed Reduction Gear
- *6 MG1
- *7 MG2
- *8 Inverter with Converter Assembly
- *9 Hybrid Battery Assembly
-
- *a Alternating Current
- *b Electrical Power Path (AC)
- *c Direct Current (DC 201.6 V)
- *d Electrical Power Path (DC)
- *e Power Transmission Mechanical Power Path
- *f Engine Stopped
- *g Engine Running

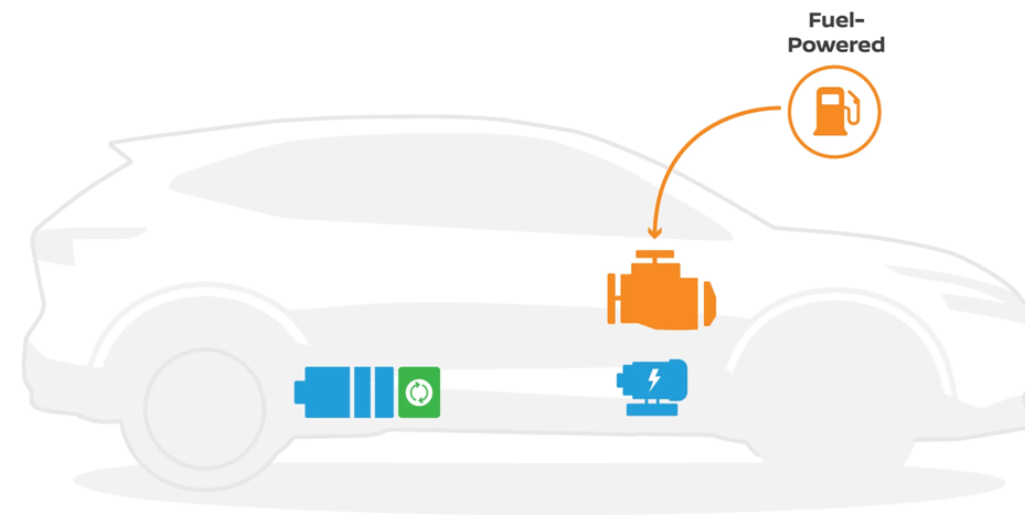
HEV Example : How Nissan e-Power system work?



HEV : e – Power components



Fuel-Powered Engine/Generator Battery Inverter Electric Motor

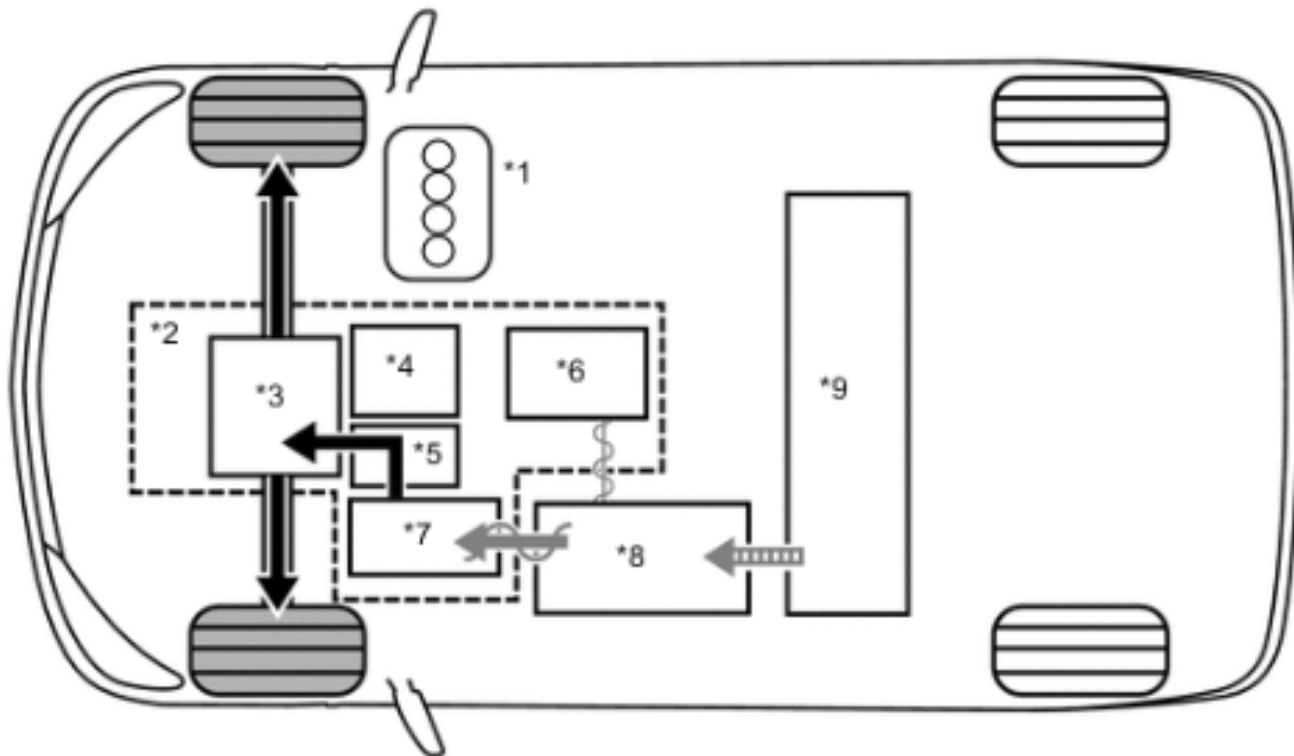


Fuel-Powered Engine/Generator Battery Inverter Electric Motor

<https://www.nissan.co.nz/about-nissan/e-power.html>

รถยนต์ไฮบริด(HYBRID ELECTRIC VEHICLE CONTROL SYSTEM)

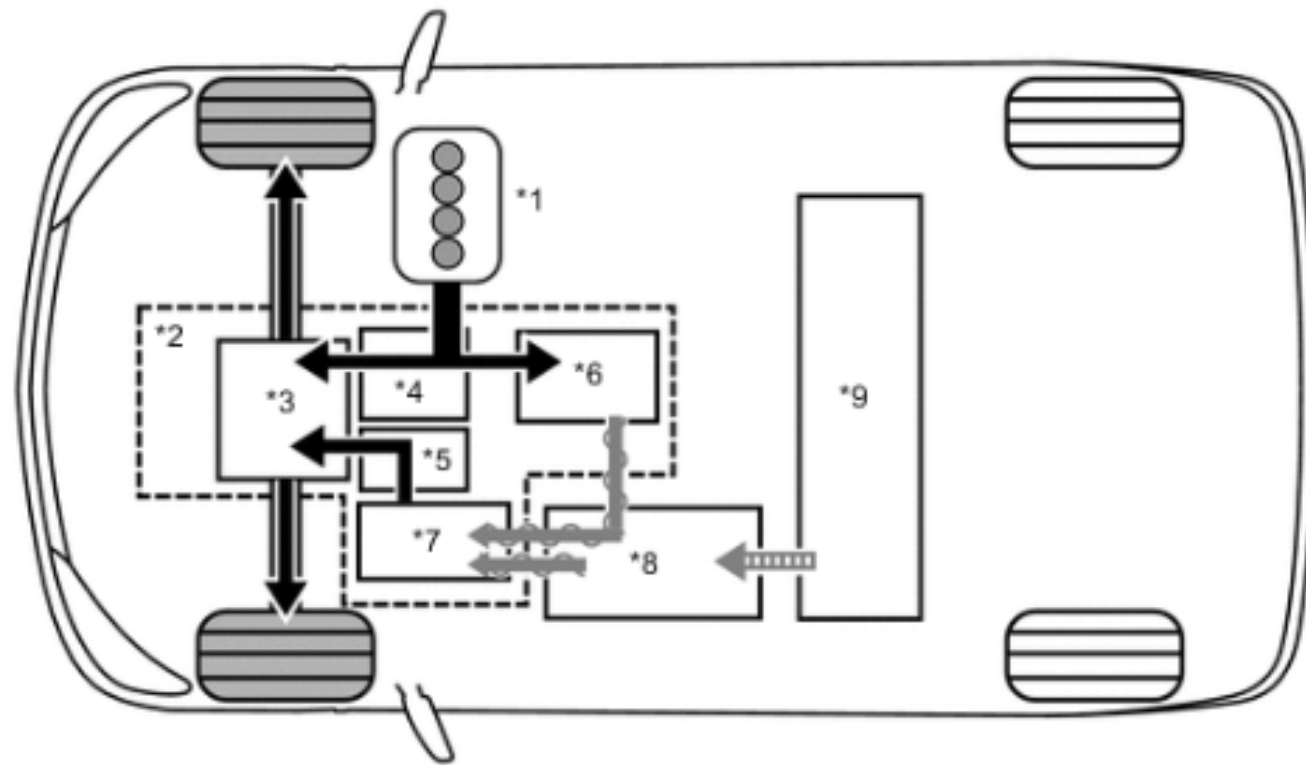
Fully hybrid electric vehicle



Starting Off

- *1 Engine
- *2 Hybrid Transaxle (Hybrid Vehicle Transaxle Assembly)
- *3 Differential
- *4 Power Split Planetary Gear
- *5 Motor Speed Reduction Gear
- *6 MG1
- *7 MG2
- *8 Inverter with Converter Assembly
- *9 Hybrid Battery Assembly
-
- *a Alternating Current
- *b Electrical Power Path (AC)
- *c Direct Current (DC 201.6 V)
- *d Electrical Power Path (DC)
- *e Power Transmission Mechanical Power Path
- *f Engine Stopped
- *g Engine Running

Driving Condition D: during Full Throttle Acceleration



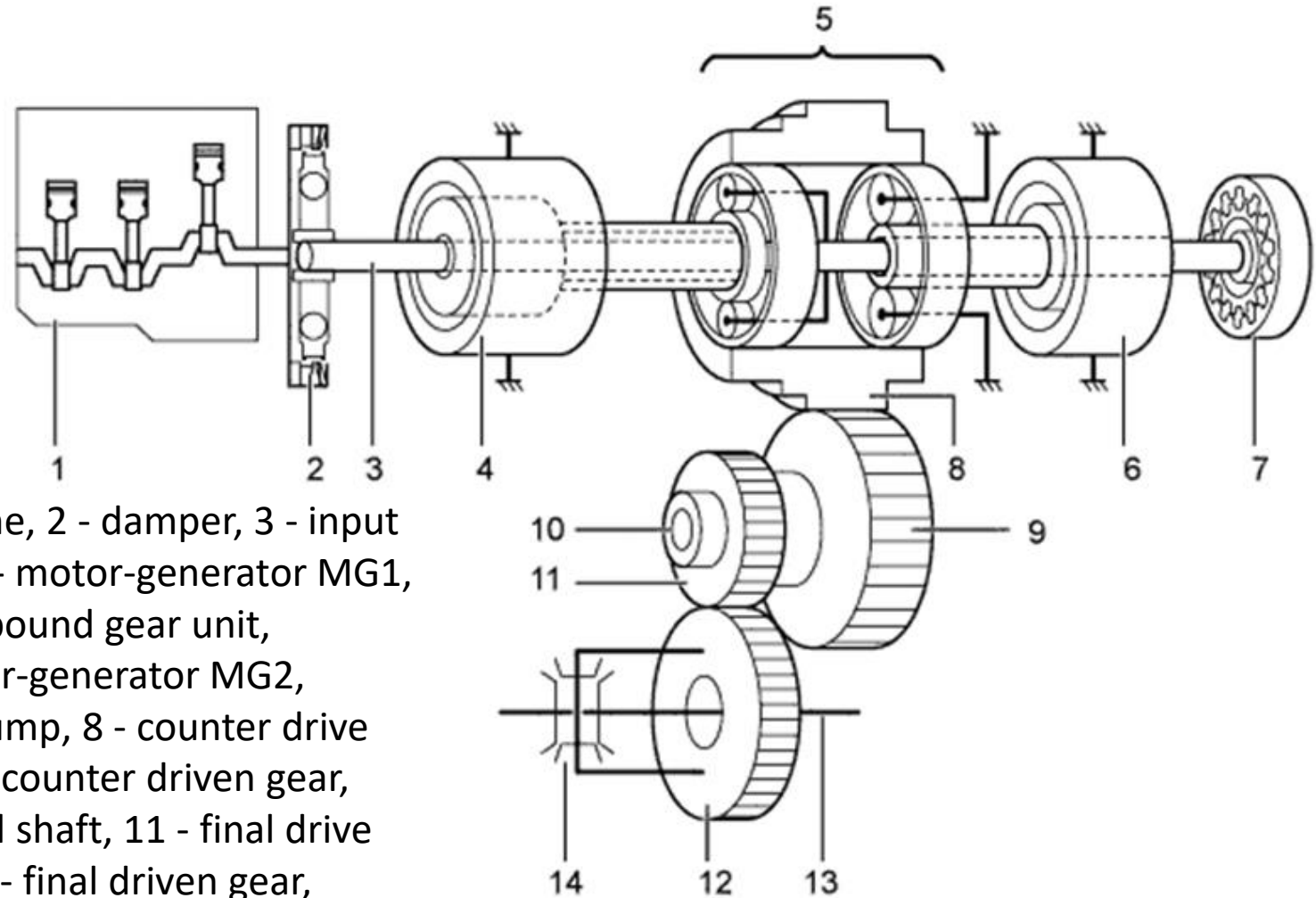
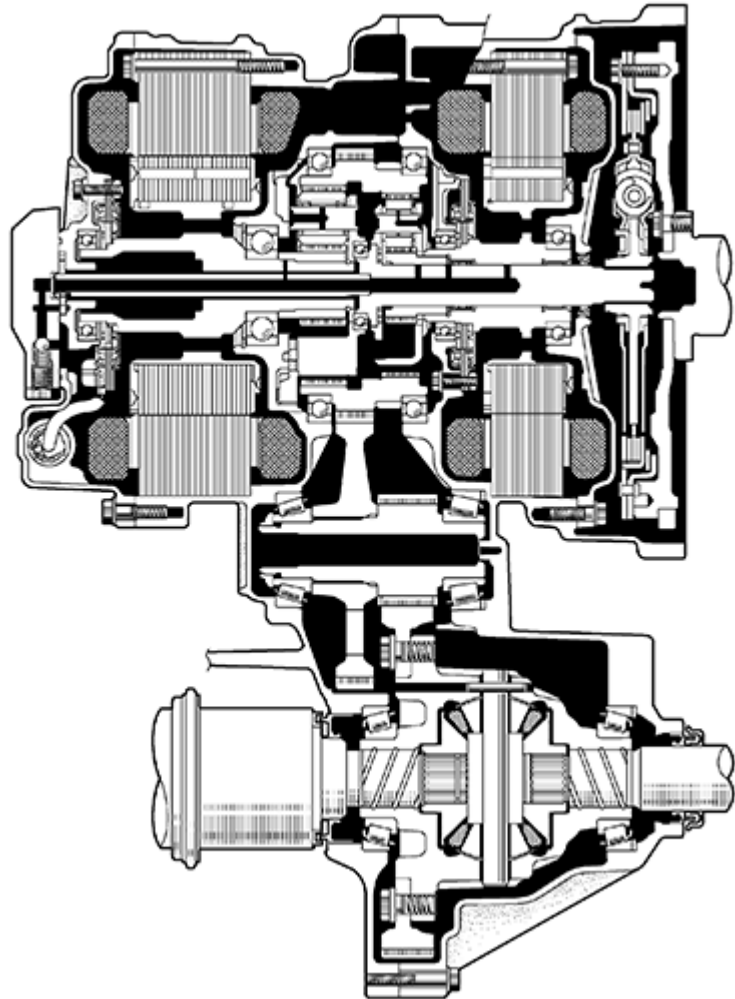
During Full Throttle Acceleration



- *1.Engine
- *2.Hybrid Transaxle (Hybrid Vehicle Transaxle Assembly)
- *3.Differential
- *4.Power Split Planetary Gear
- *5.Motor Speed Reduction Gear
- *6.MG1
- *7.MG2
- *8.Inverter with Converter Assembly
- *9.Hybrid Battery Assembly
- *a.Alternating Current
- *b.Electrical Power Path (AC)
- *c.Direct Current (DC 201.6 V)
- *d.Electrical Power Path (DC)
- *e.Power Transmission Mechanical Power Path
- *f.Engine Stopped
- *g.Engine Running

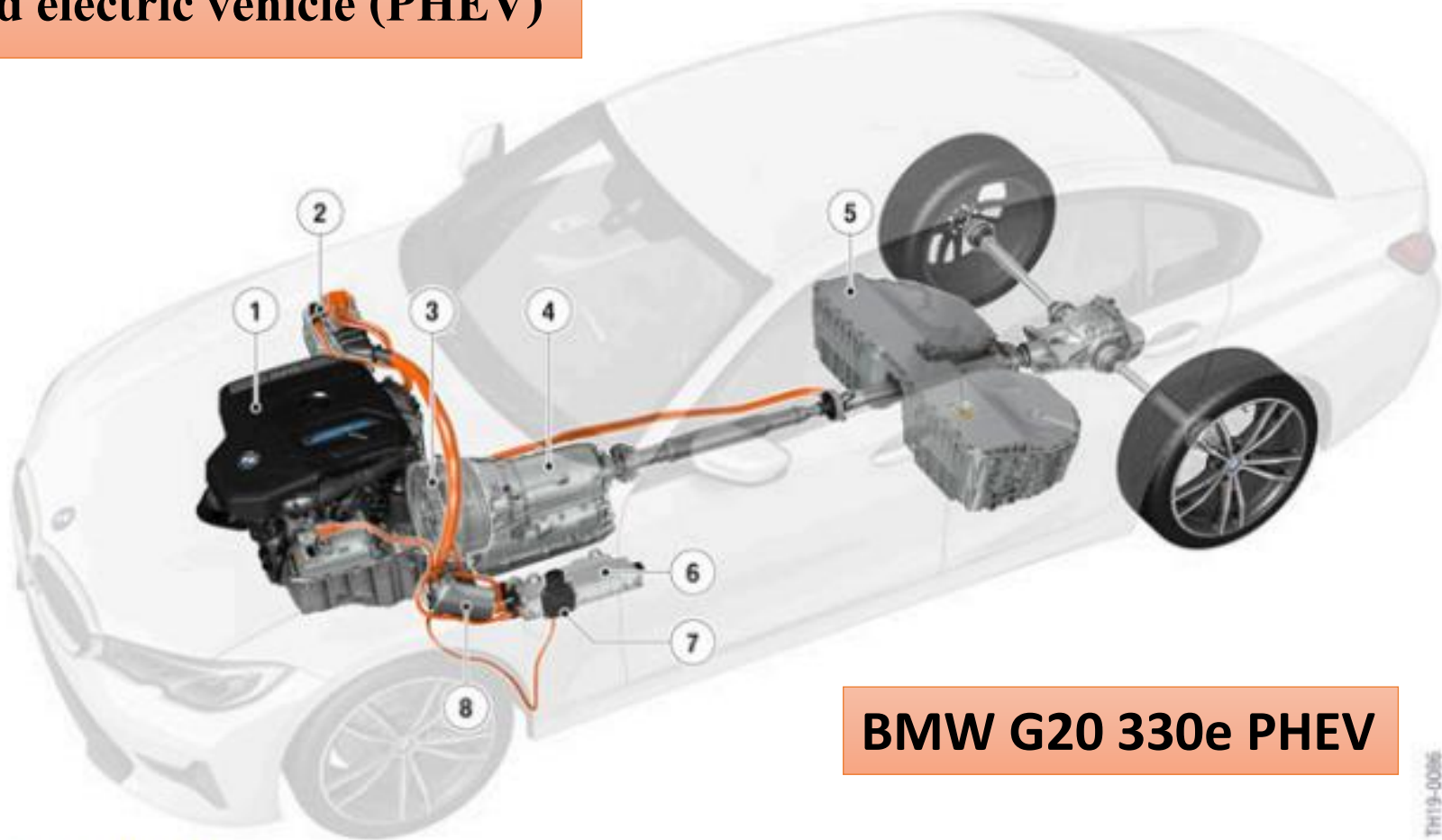
TOYOTA HEV P310-P410-P510

The operating principles of P310-P410-P510 transmissions are similar.



1 - engine, 2 - damper, 3 - input shaft, 4 - motor-generator MG1, 5 - compound gear unit, 6 - motor-generator MG2, 7 - oil pump, 8 - counter drive gear, 9 - counter driven gear, 10 - 2nd shaft, 11 - final drive gear, 12 - final driven gear, 13 - 3rd shaft, 14 - differential gear unit

Plug – in hybrid electric vehicle (PHEV)



BMW G20 330e PHEV

G20 PHEV hybrid drive

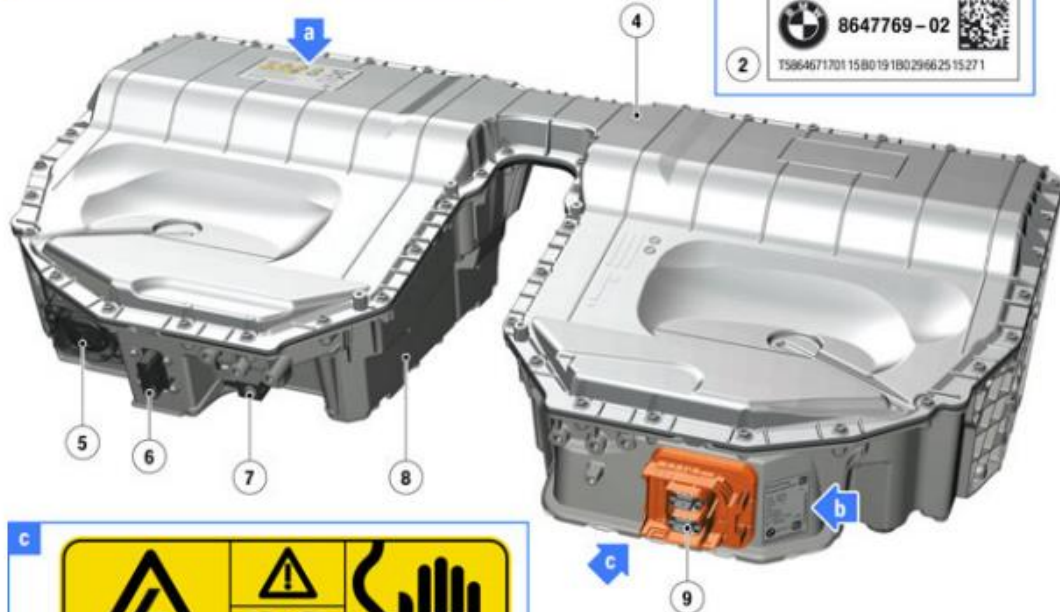
Index

- 1. Combustion engine B46B20O1
- 3. Electric motor
- 5. High-voltage battery unit
- 7. Charging socket

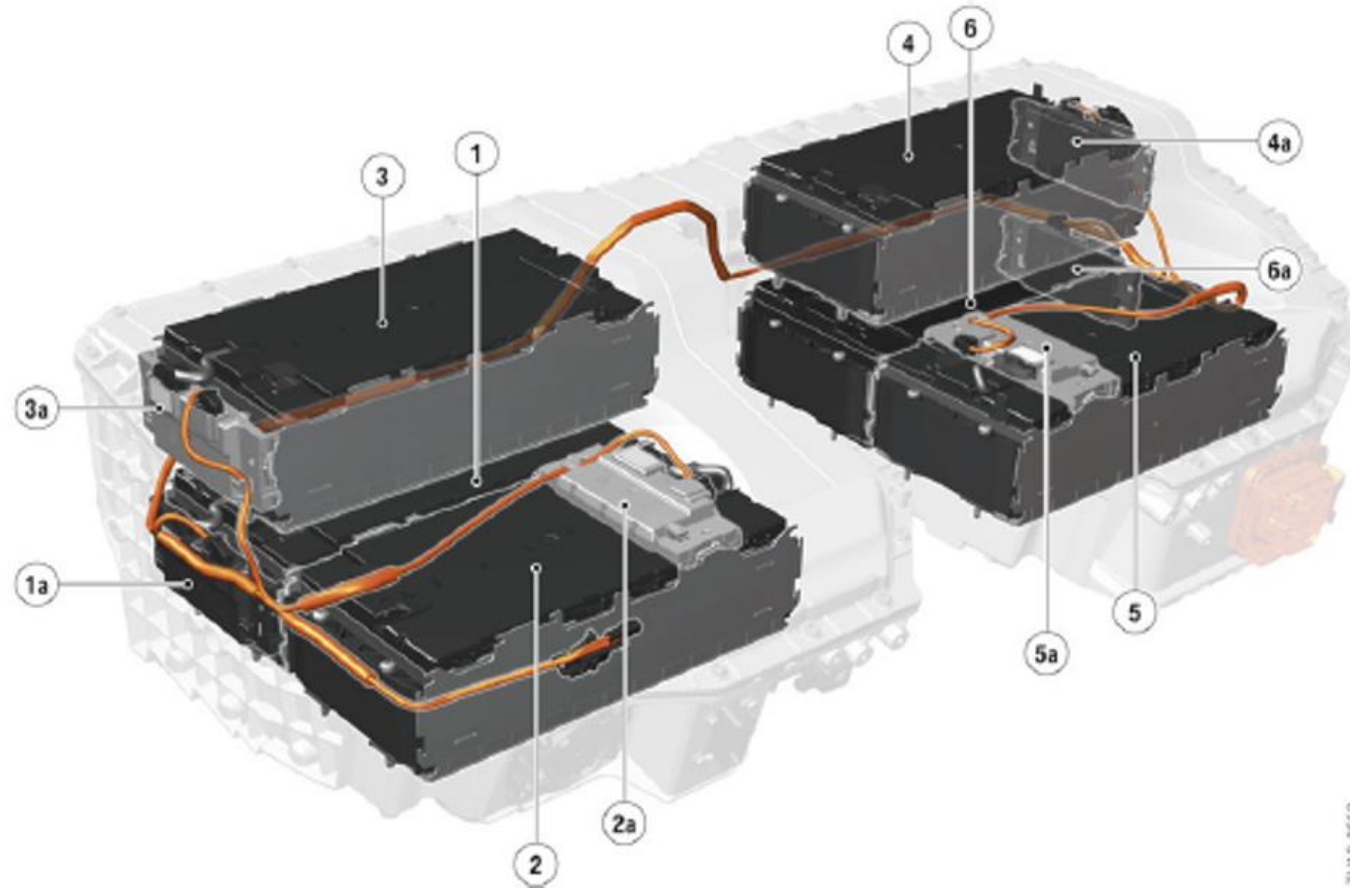
Explanation

- 2. Electric Motor Electronics (EME)
- 4. Automatic transmission GA8P75HZ
- 6. Convenience charging electronics (KLE)
- 8. Electrical Heating (EH)

PHEV Battery energy capacity

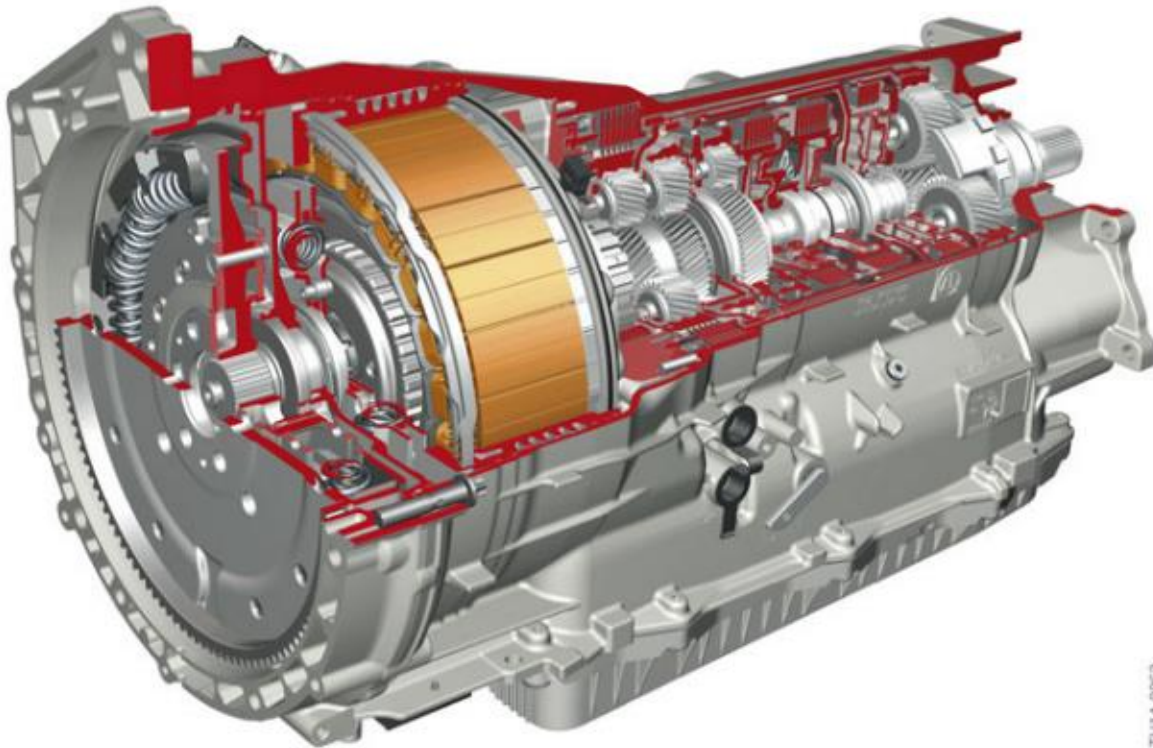


Signs on the housing of the SP41 high-voltage battery



BMW G20 /330e :12 kWh

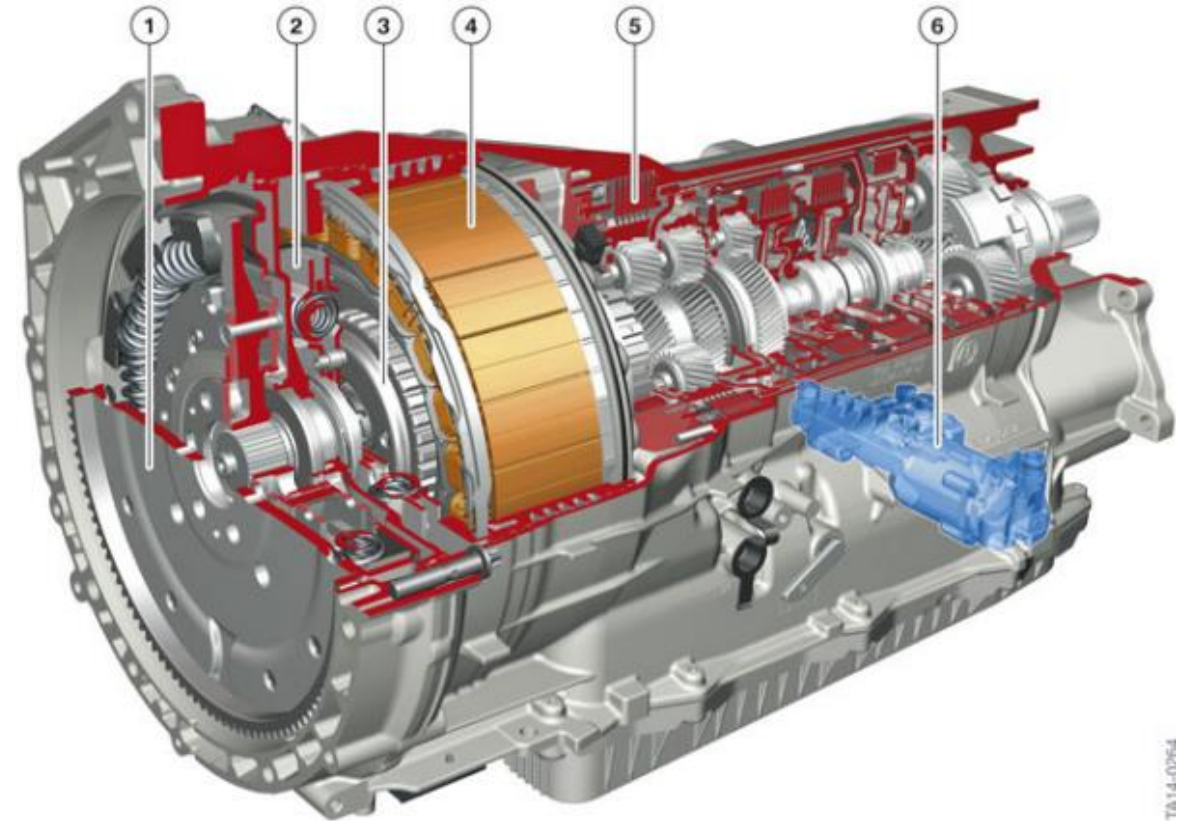
Automatic transmission :GA8P75HZ of the G20 PHEV



TH14-0263

Index

- 1 Dual-mass flywheel (including torsional vibration damper and centrifugal pendulum)
- 3. Separation clutch
- 5. Multidisc brake B

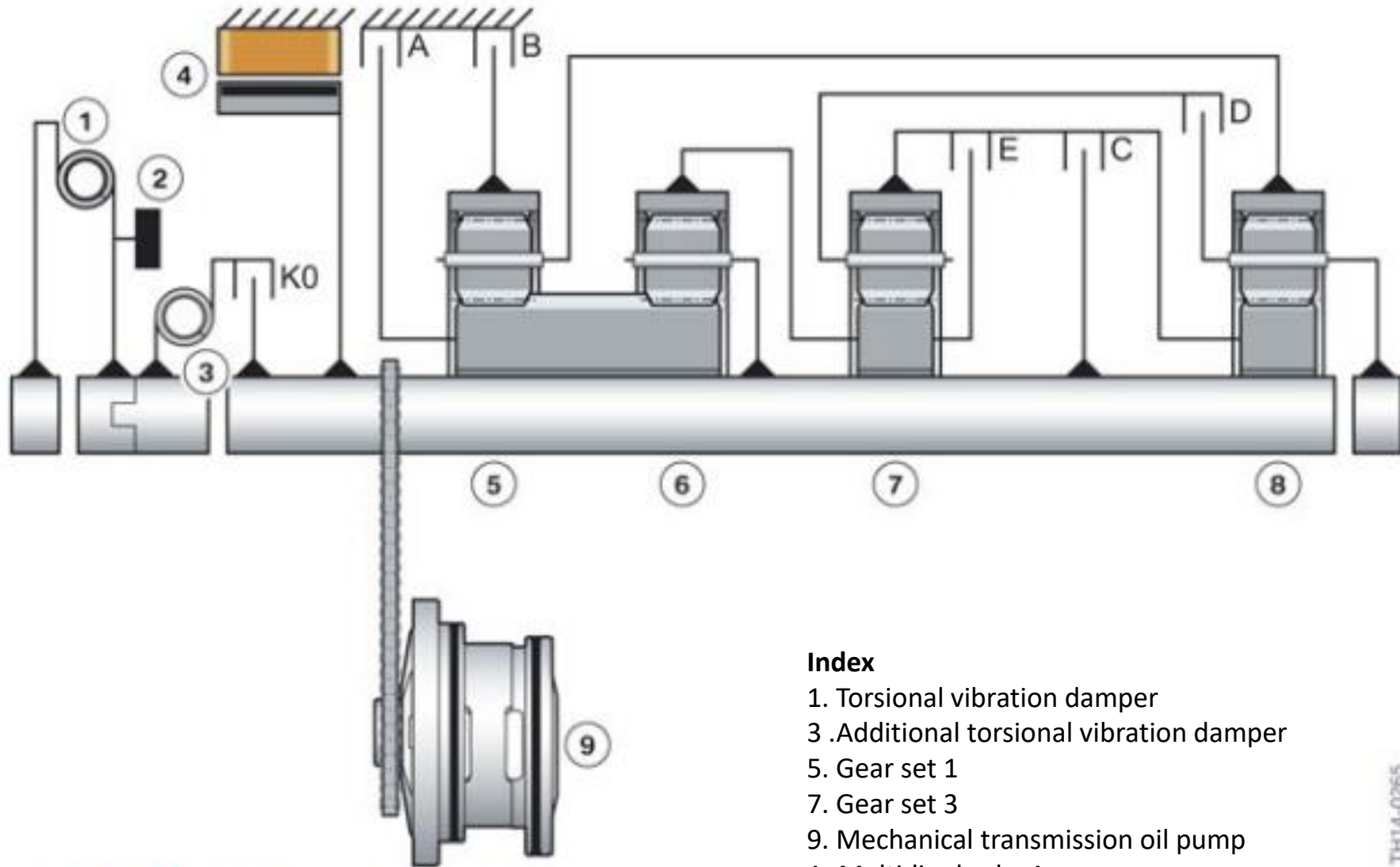


TH14-0263

Explanation

- 2. Additional torsional vibration damper
- 4. Electric motor
- 6. Electrical transmission oil pump

Drive Components



Schematic structure of the GA8P75HZ transmission

Index

- 1. Torsional vibration damper
- 3 .Additional torsional vibration damper
- 5. Gear set 1
- 7. Gear set 3
- 9. Mechanical transmission oil pump
- A. Multidisc brake A
- C. Multidisc clutch C
- E. Multidisc clutch E

Explanation

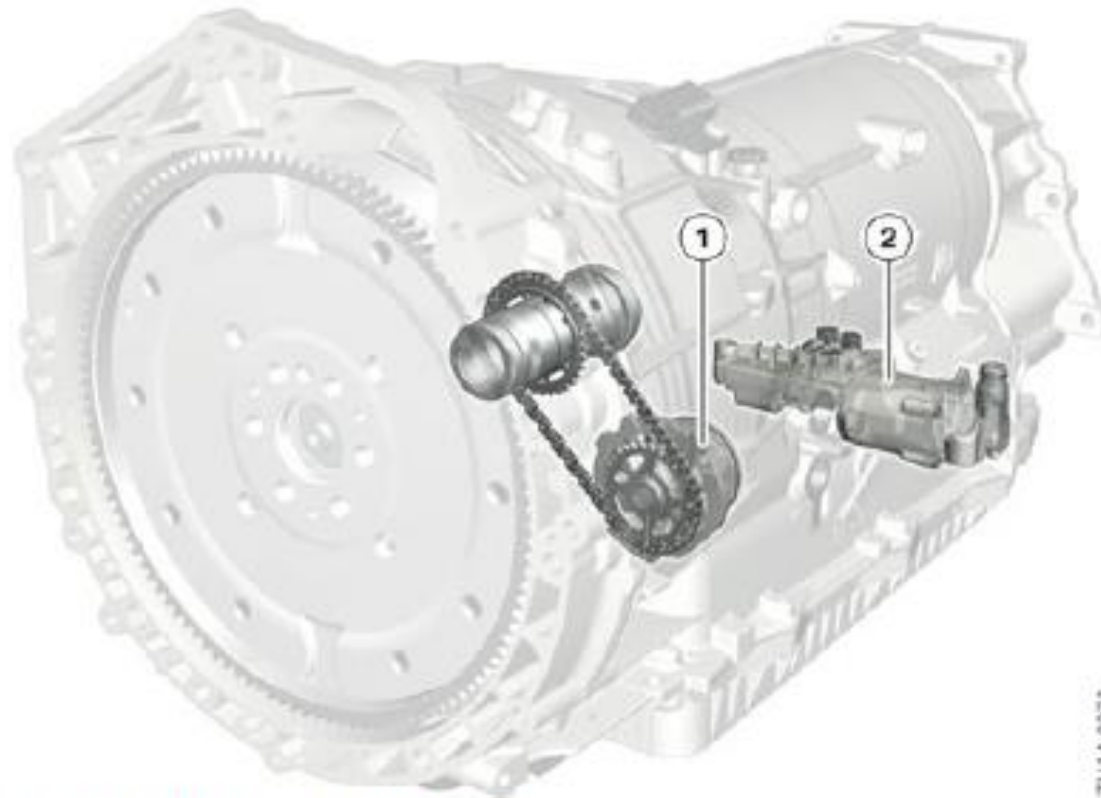
- 2. Centrifugal pendulum
- 4. Electric motor
- 6. Gear set 2
- 8 .Gear set 4
- B. Multidisc brake B
- D.Multidisc clutch D
- K0. Separation clutch

Oil supply

Oil supply

The basic functions of the oil circuit of the GA8P75HZ transmission correspond to those of the GA8HP70Z transmission. The oil has the following tasks:

- Lubrication
- Control of the shift elements
- Cooling



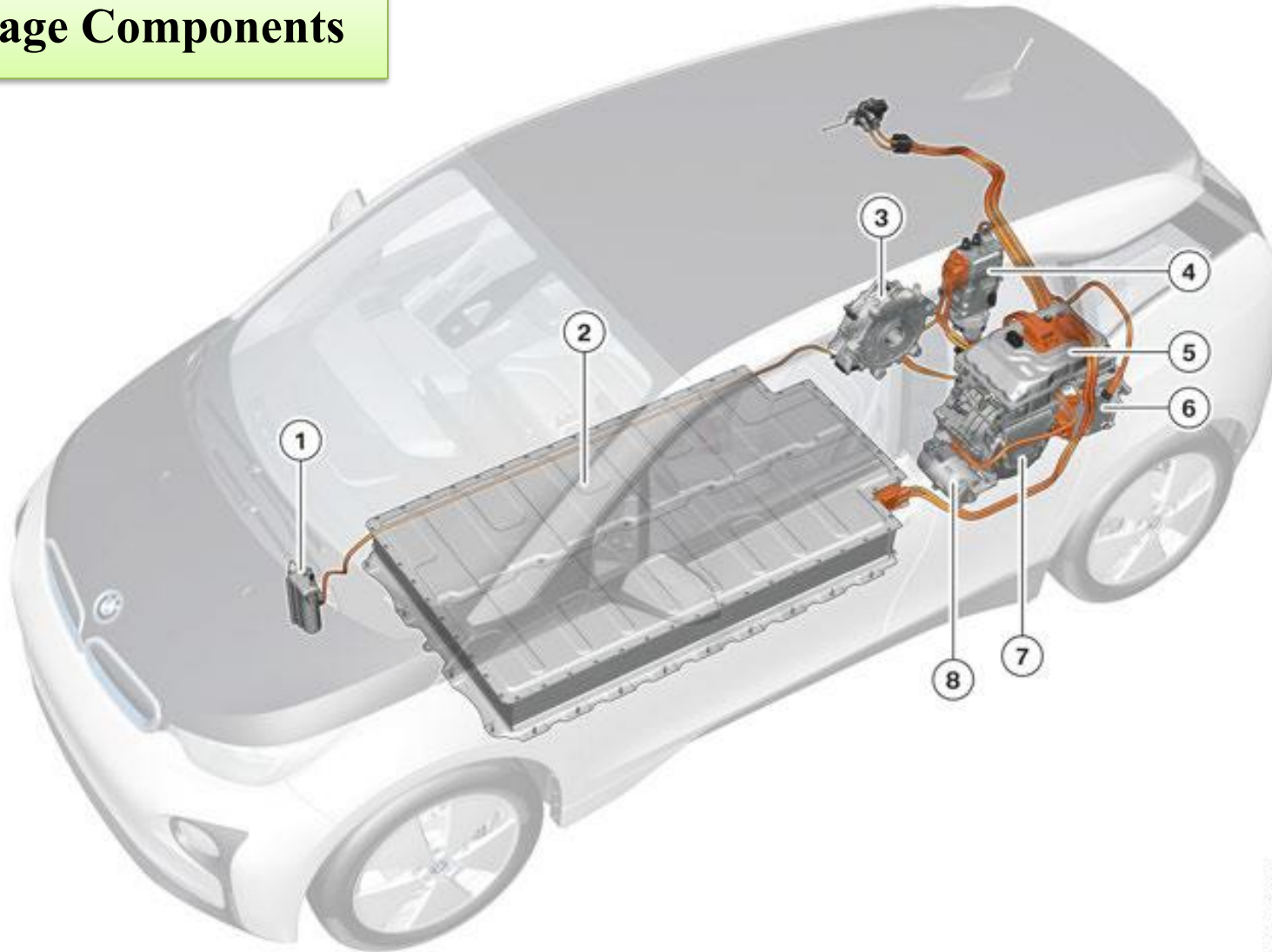
Transmission oil pump GA8P75HZ transmission

Index Explanation

- 1 Mechanical transmission oil pump
- 2 Electrical transmission oil pump

REEV(RANGE-EXTENDED ELECTRIC VEHICLE)

High-voltage Components

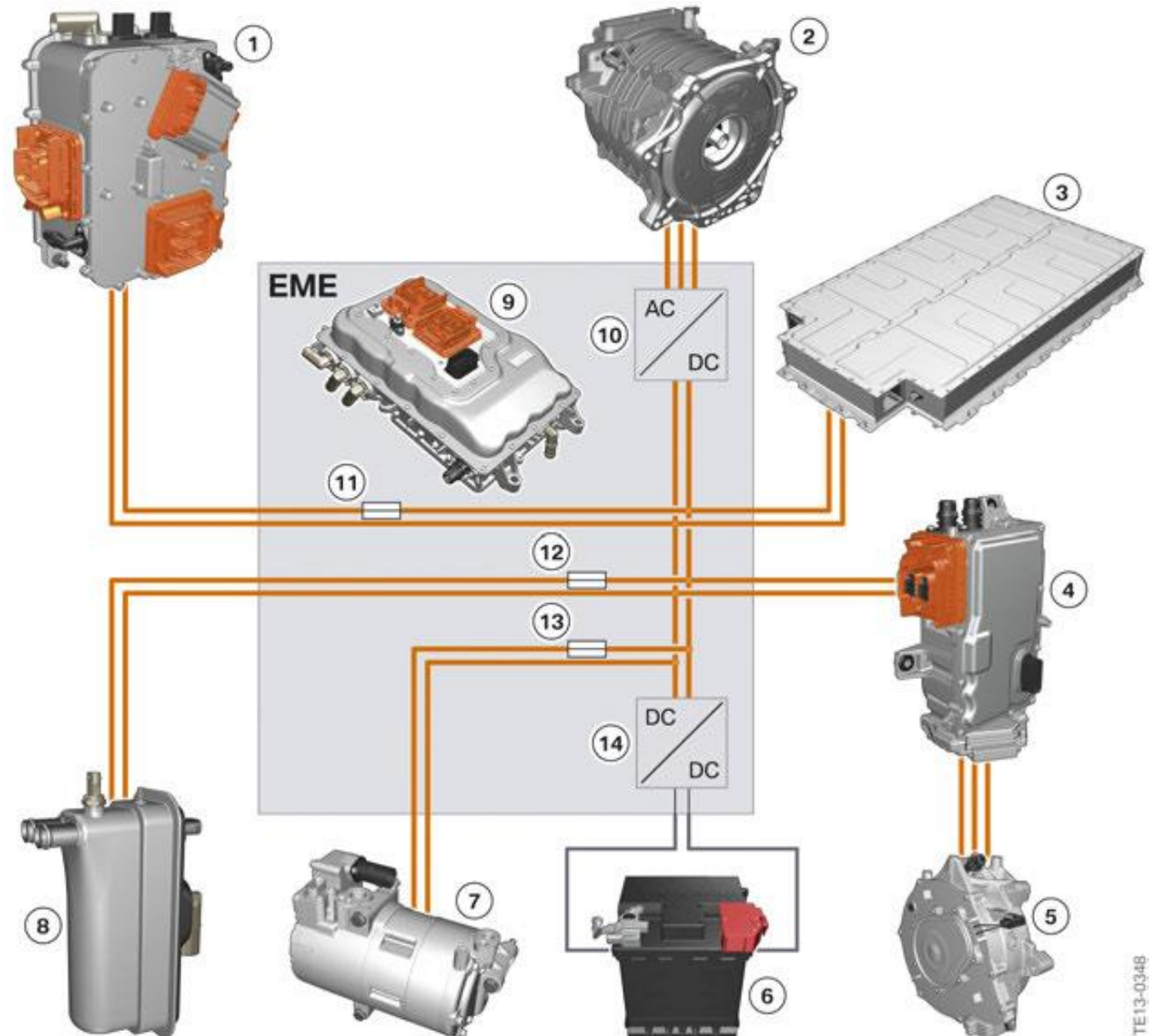


- 1 Electric heating
- 2 High-voltage battery
- 3 Range extender electrical machine
- 4 Range Extender Electrical Machine Electronics
- 5 Electrical machine electronics
- 6 Convenience charging electronics
- 7 Electrical machine
- 8 EKK

REEV(RANGE-EXTENDED ELECTRIC VEHICLE)

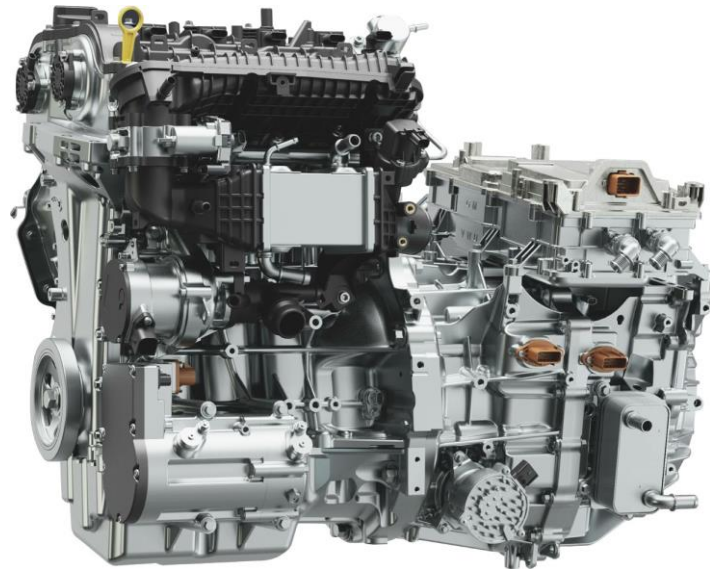
High voltage overview

- 1 Convenience charging electronics
- 2 Electrical machine
- 3 High-voltage battery
- 4 Range Extender Electrical Machine Electronics(REME)
- 5 Range extender electrical machine
- 6 12 V battery
- 7 EKK
- 8 Electric heating
- 9 Electrical machine electronics(entirety)



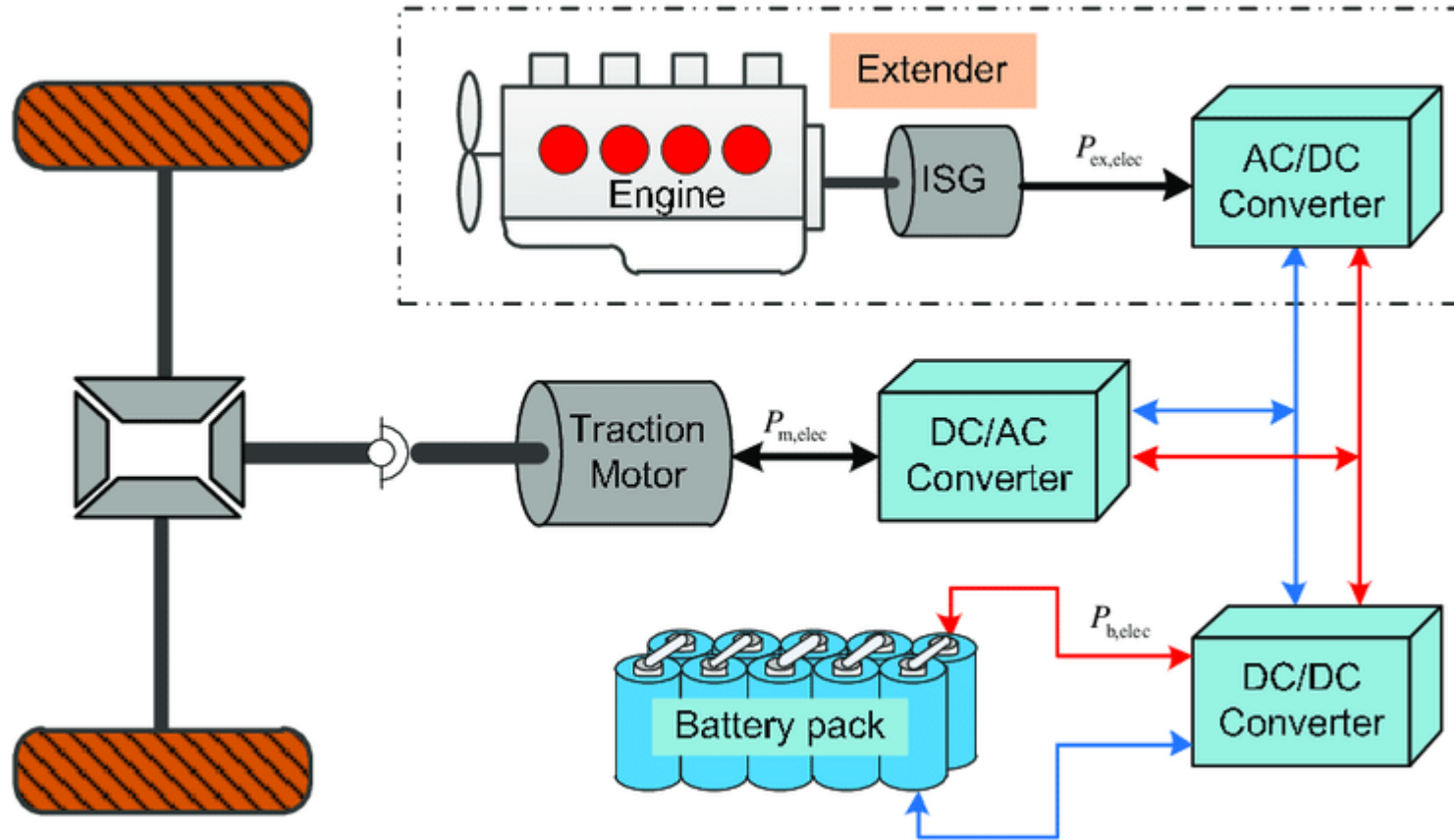
REEV(RANGE-EXTENDED ELECTRIC VEHICLE)

เทคโนโลยีรถยนต์ไฟฟ้าขยายระยะทาง (Range-Extended Electric Vehicle หรือ REEV) เป็นระบบขับเคลื่อนแบบไฮบริดชนิดหนึ่งที่ผสมผสานระบบขับเคลื่อนไฟฟ้าเข้ากับเครื่องยนต์สันดาปภายใน (ICE) และเครื่องกำเนิดไฟฟ้าในตัวรถ เพื่อเพิ่มระยะทางการขับขี่ของรถให้ไกลกว่าที่แบตเตอรี่เพียงอย่างเดียว



Range extended electric vehicle (REEV)

REEV Diagram

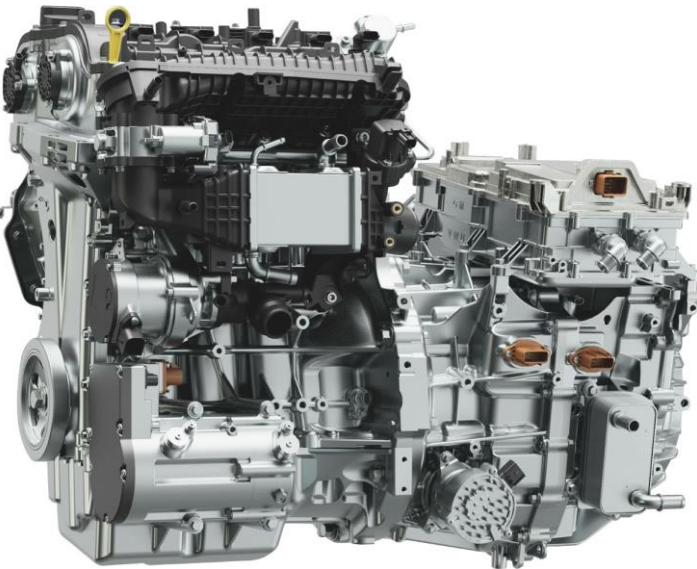


<https://diyguru.org/electric-vehicle/erev/>

ผู้ช่วยศาสตราจารย์สุวิทย์ วงษ์เย็น

REEV(RANGE-EXTENDED ELECTRIC VEHICLE)

เทคโนโลยีรถยนต์ไฟฟ้าขยายระยะทาง (Range-Extended Electric Vehicle หรือ REEV) เป็นระบบขับเคลื่อนแบบไฮบริดชนิดหนึ่งที่ผสมผสานระบบขับเคลื่อนไฟฟ้าเข้ากับเครื่องยนต์สันดาปภายใน (ICE) และเครื่องกำเนิดไฟฟ้าในตัวรถ เพื่อเพิ่มระยะทางการขับขี่ของรถให้ไกลกว่าที่แบตเตอรี่เพียงอย่างเดียว



ภาพรวมระบบของรถยนต์ไฟฟ้า(System overview electric vehicle)

LV Net Work

CANBUS

Current Sensor

Voltage Sensor

IR Sensor

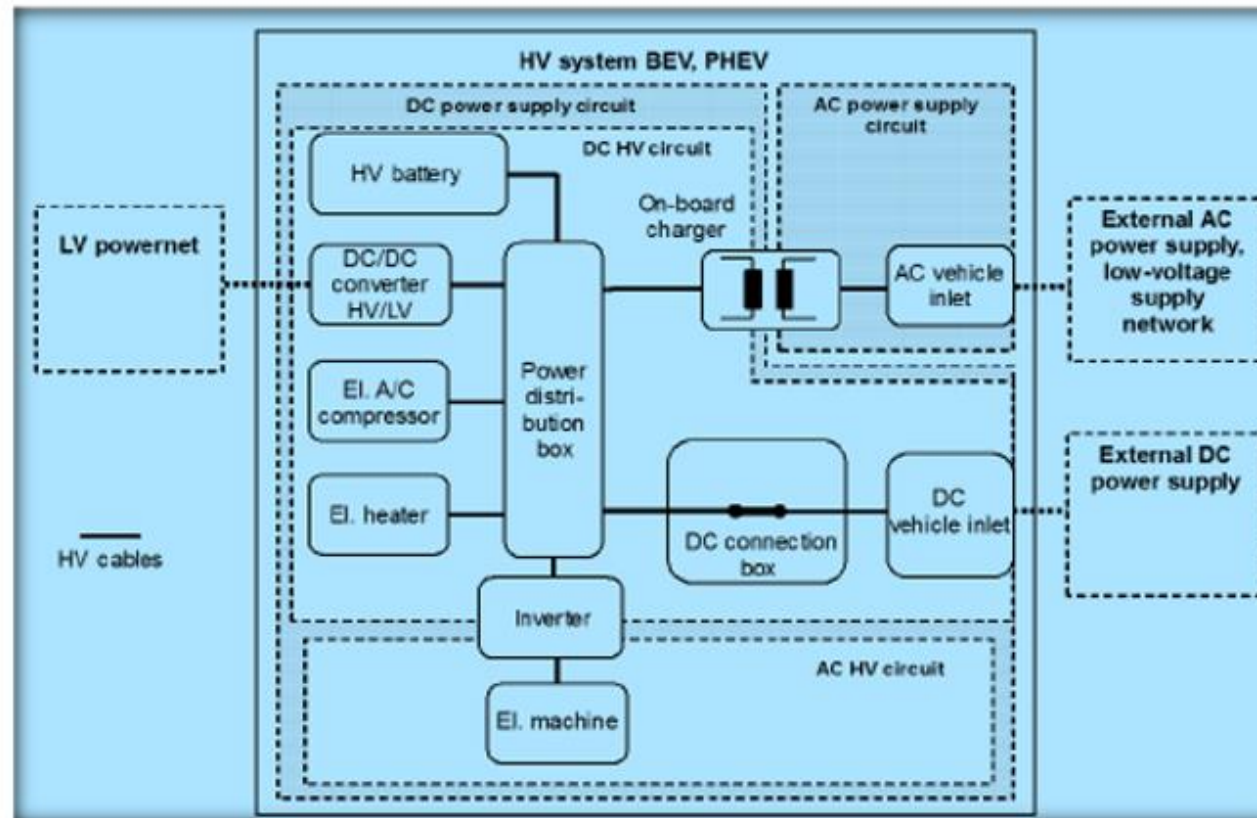
Temp. Sensor

Crash Sensor

Lamps...

Pumps...

HV system



HV Components

Charging port

RESS

Inverter

Motor

PDU

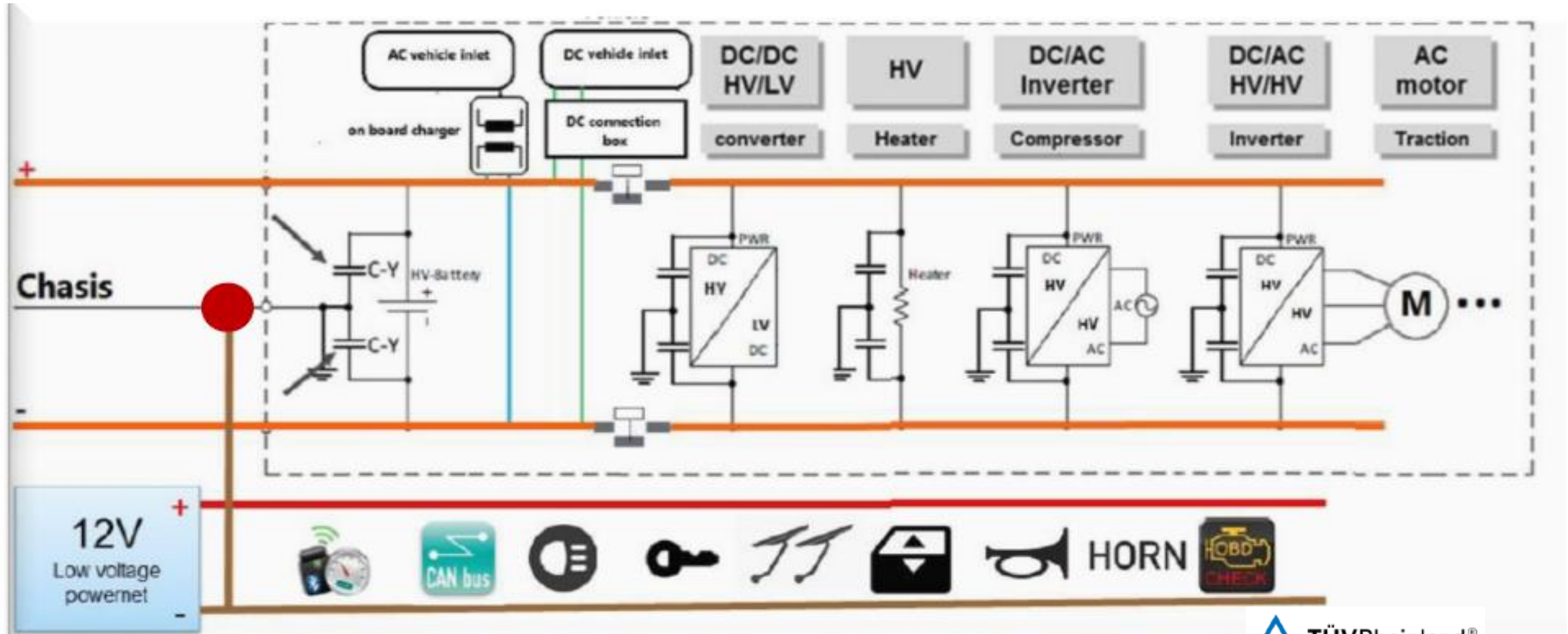
PTC

EACP

DC-DC Converter

On-board charger

ไดอะแกรมไฟฟ้าแรงสูงรถยนต์ไฟฟ้า(EV Block diagram)



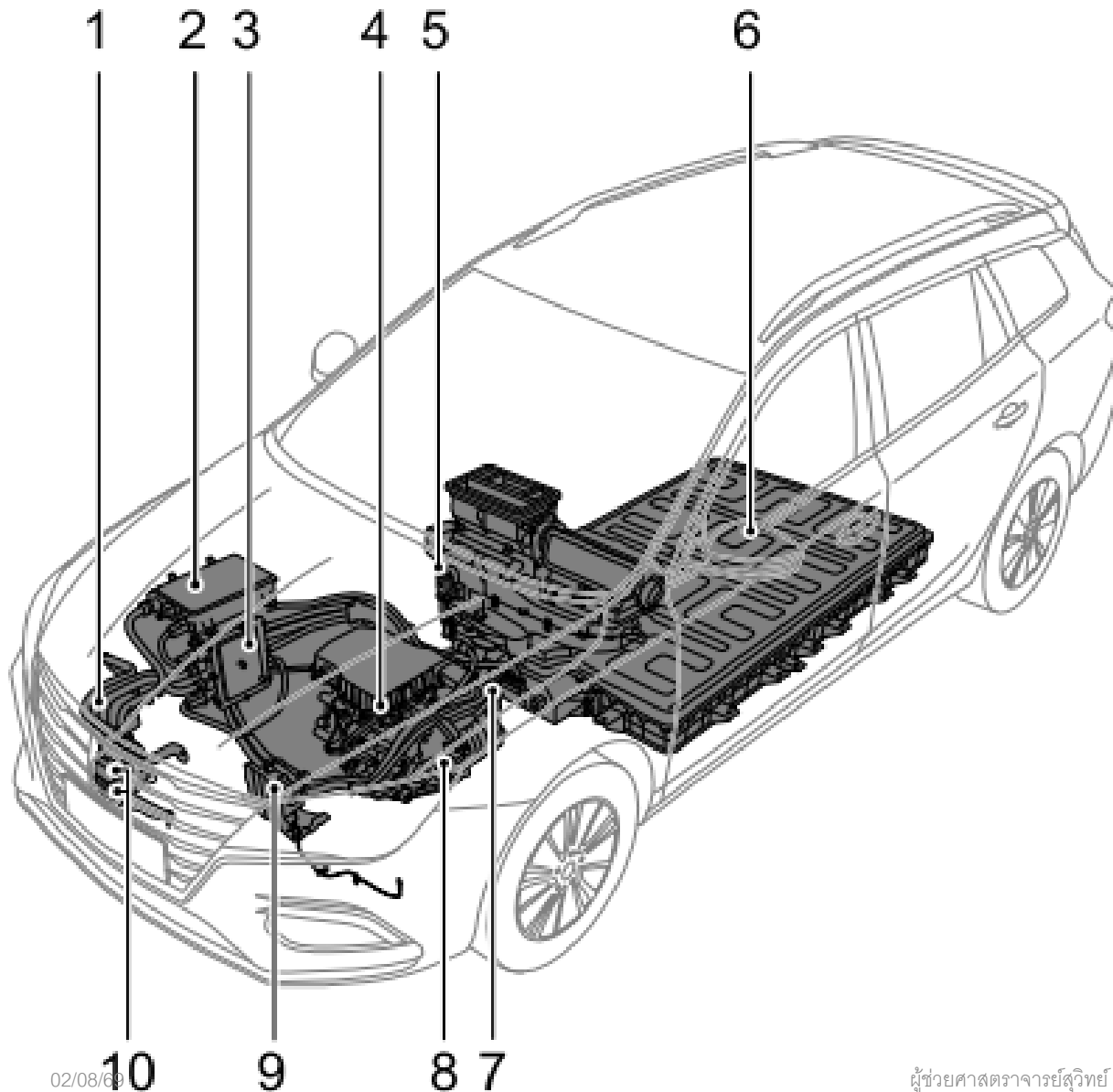
Electric Vehicle Components Location:MG EP



A: Cooling Tank For OBC/DC-DC/Traction Motor/Inverter
 C: Inverter With Converter
 E: High Voltage Cable
 G: Electric Vehicle Communication Controller(EVCC)
 I: 12VDC Battery

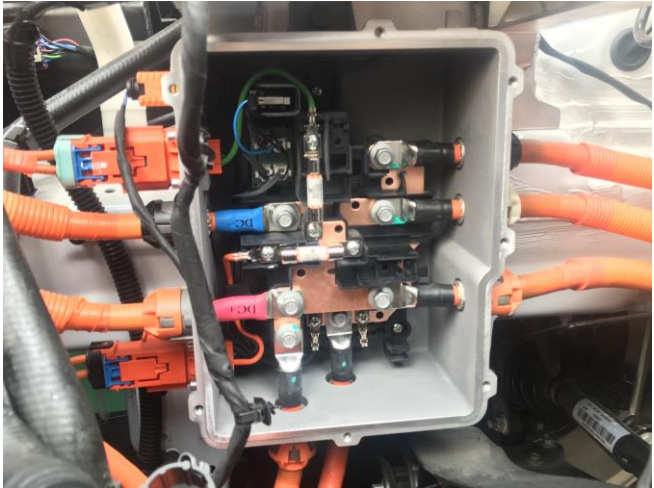
B: OBC/DC-DC Converter
 D: HVB Chiller
 F: A/C Compressor
 H: Cooling Tank For HVB
 J: 220VAC 50Hz Power Supply For OBC

High Voltage System Component : MG EP



- 1 High Voltage Harness
- 2 Combined Charging Unit (CCU)
- 3 Power Distribution Unit (PDU)
- 4 Electric Drive Transmission
- 5 Electric Heater
- 6 High Voltage Battery (ESS)
- 7 Manual Service Disconnect (MSD)
- 8 Electric A/C Compressor
- 9 HV Battery Heater
- 10 Charging Port

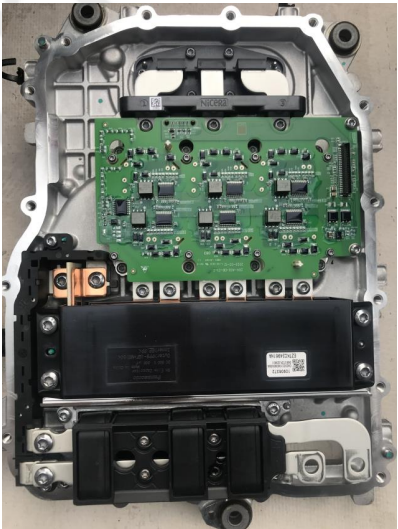
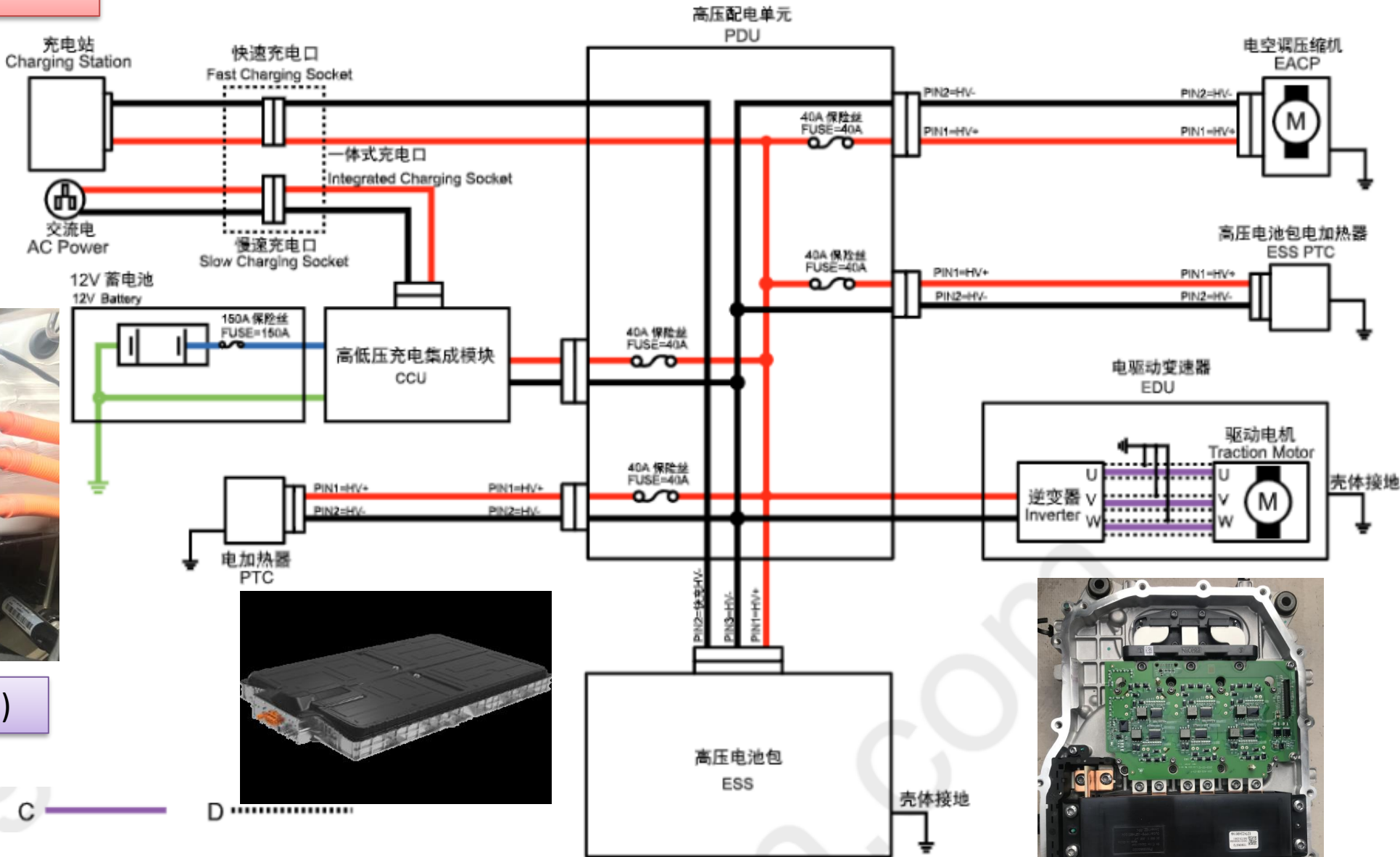
System Control Diagram :MG EP



Power distribution unit(PDU)

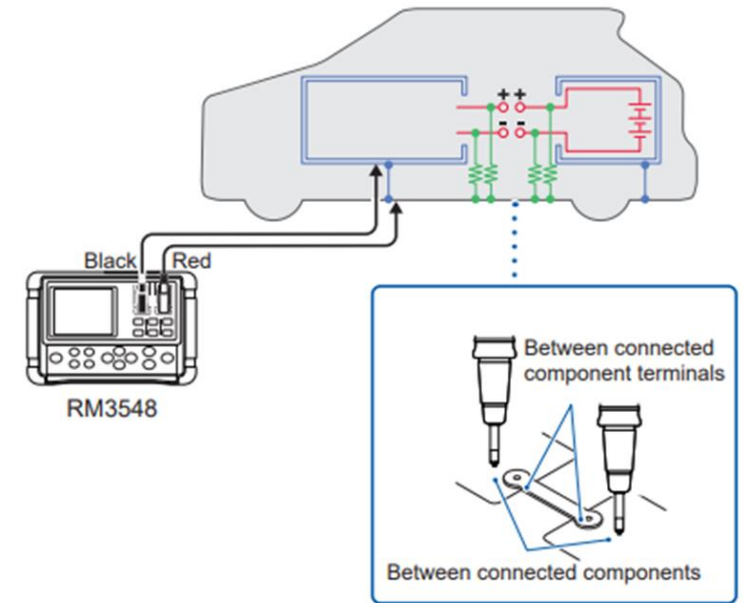
A  B  C  D 

A = Low Voltage DC,B = High Voltage DC , C = High Voltage AC,D = Shield



ระบบกราวด์ในรถยนต์ไฟฟ้า(Electric vehicle ground system)

- *Isolation ground or DC High voltage ground*
- *Chassis ground or DC Low voltage ground*
- *Alternating current(AC) ground or Protective earth (PE)*
- *IT Network system or Equipotential ground bonding*



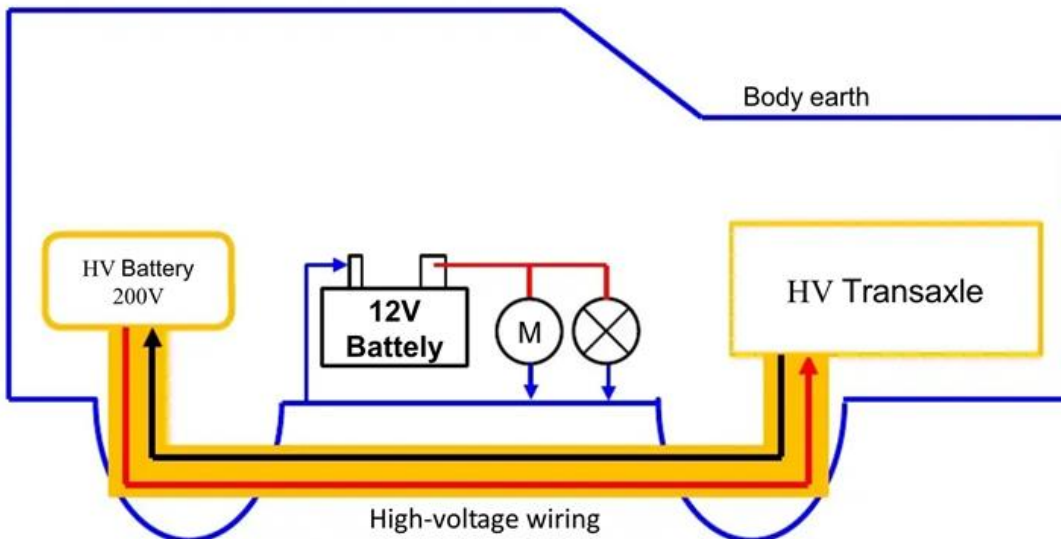
Example: Between connected components

Equipotential ground bonding



Alternating current(AC) ground

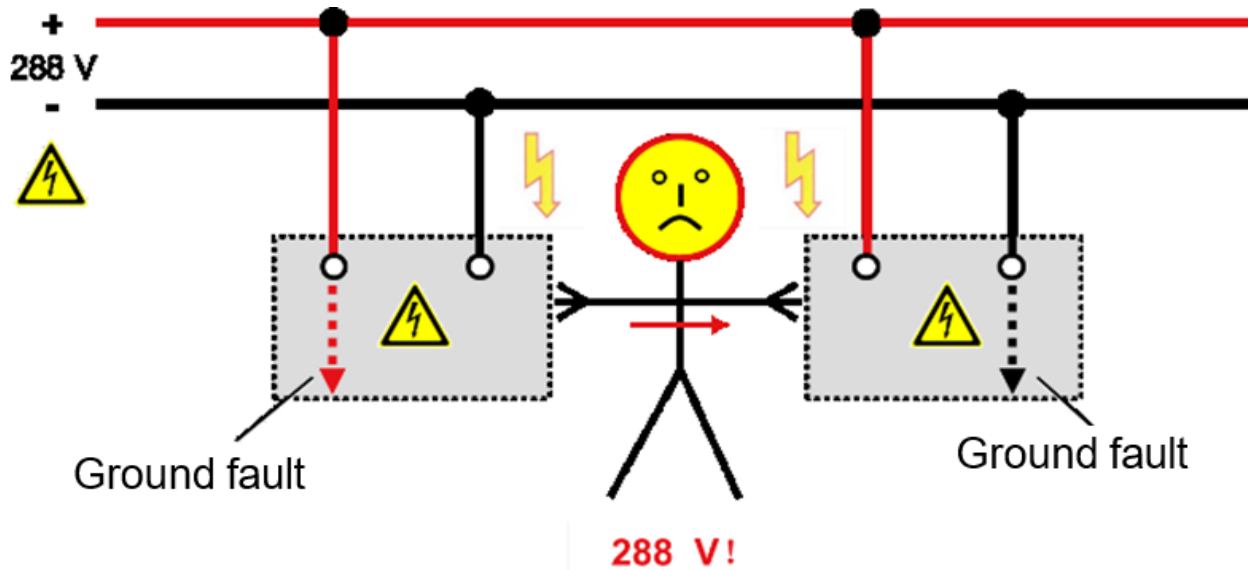
Isolation of the high-voltage system



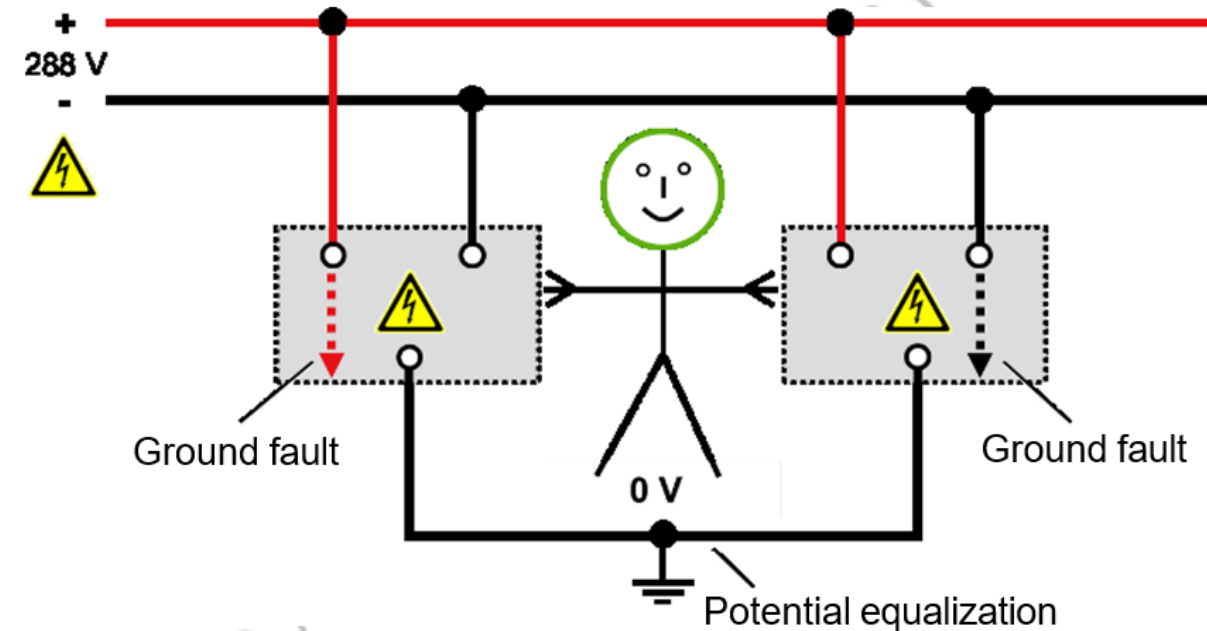
Safety devices : Equipotential ground bonding

The high-voltage network is an IT-like network; accordingly, both HV+ and HV- are galvanically isolated from the vehicle ground and all equipment bodies. IT "Isolé terre" (French) = isolated earth (the IT network is a non-earthed electrical supply)

Danger of a ground fault of high voltage components



Potential balance of high voltage components

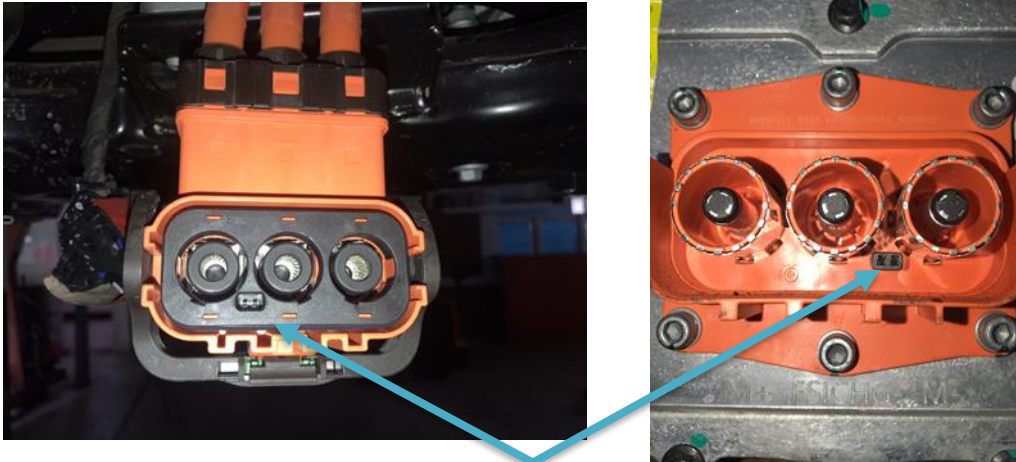


HVIL (High Voltage Interlock Loop) Monitor.

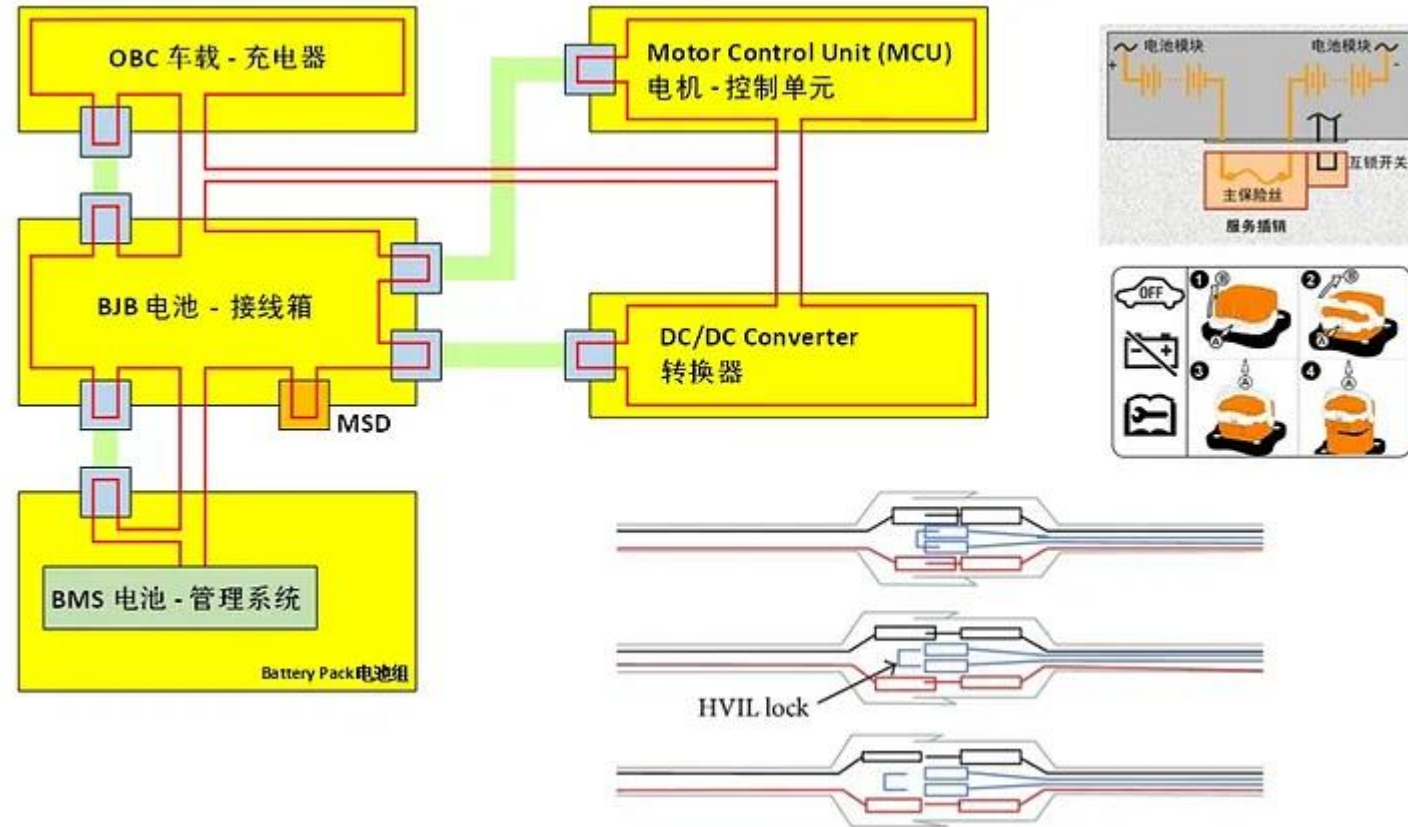
HVIL is a low voltage loop that routes through every high voltage (HV) component (OBC, BJB, MCU, DC/DC Converter, Battery Pack), HV connector and MSD in the vehicle.

Circuit Continuity Check

- The HVIL is a low-voltage circuit that runs in parallel with the high-voltage system.
- HVIL connects through all high-voltage connectors, components, and service disconnects.
- If any connector is loose or improperly seated, the loop is broken.



High Voltage Interlock Loop (HVIL)

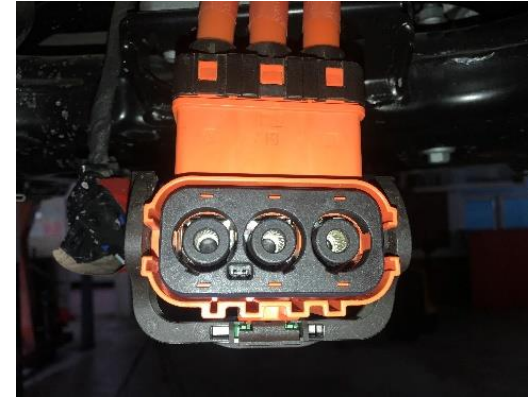


Safety system

1. Manual service disconnect(MSD)
2. High voltage inter lock loop (HVIL)



Manual service disconnect(MSD)



High voltage inter lock loop (HVIL)



Safety switch

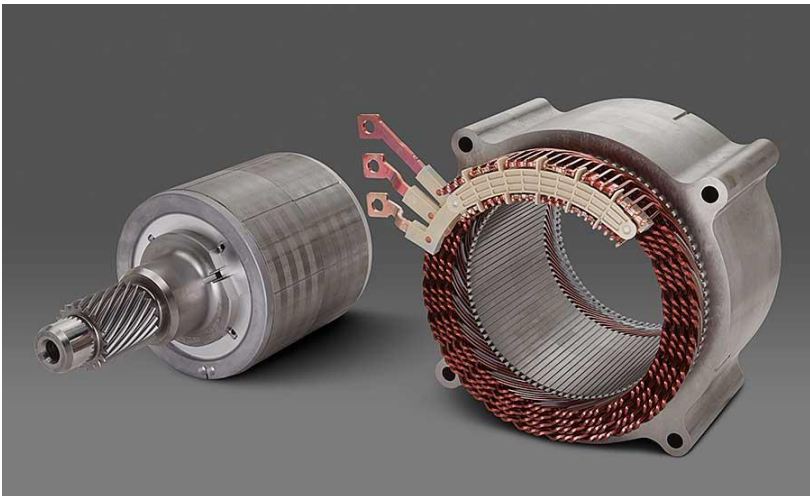
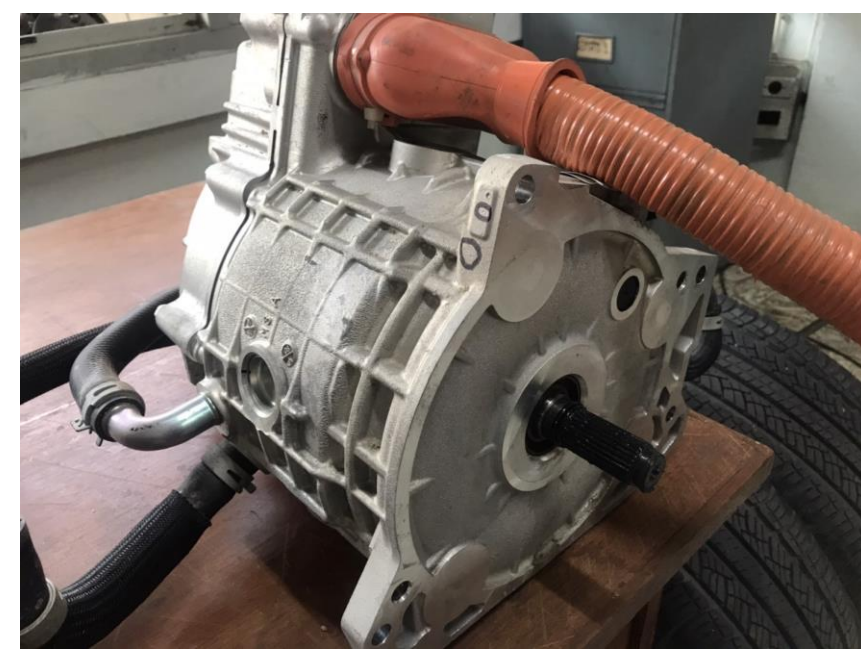
มอเตอร์ขับเคลื่อนรถยนต์ไฟฟ้า

ชนิดของมอเตอร์ขับเคลื่อนรถยนต์ไฟฟ้า(Traction motor)

ทำหน้าที่ขับเคลื่อนรถยนต์ไฟฟ้าให้เกิดการเคลื่อนที่โดยเปลี่ยนพลังงานไฟฟ้าเป็นพลังงานกลเพื่อส่งกำลังผ่านเกียร์ลดความเร็วรอบ(Single speed reduction gear)ไปยังล้อ

มอเตอร์ขับเคลื่อนรถยนต์ไฟฟ้า(Traction motor)ที่ใช้ในรถยนต์ มี 2 แบบ คือ

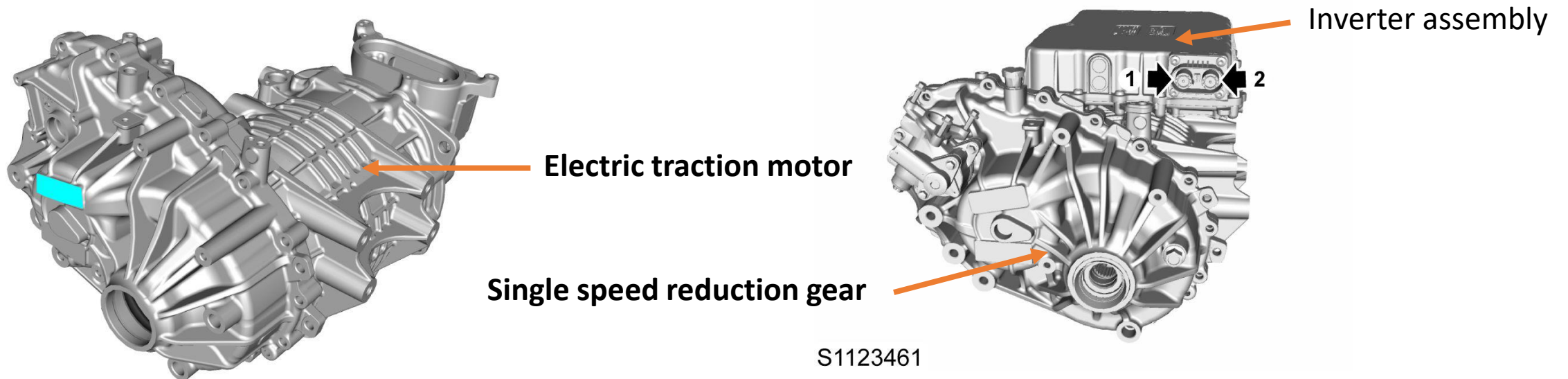
- Asynchronous motor ได้แก่ มอเตอร์แบบเหนี่ยวนำ(Induction motor) หรือ IM
- Synchronous motor ได้แก่ มอเตอร์ที่ใช้โรเตอร์แม่เหล็กถาวร(Permanent magnet synchronous motor) หรือ PMSM



ระบบขับเคลื่อนรถยนต์ไฟฟ้าและระบบส่งกำลัง(Electric drive transmission)

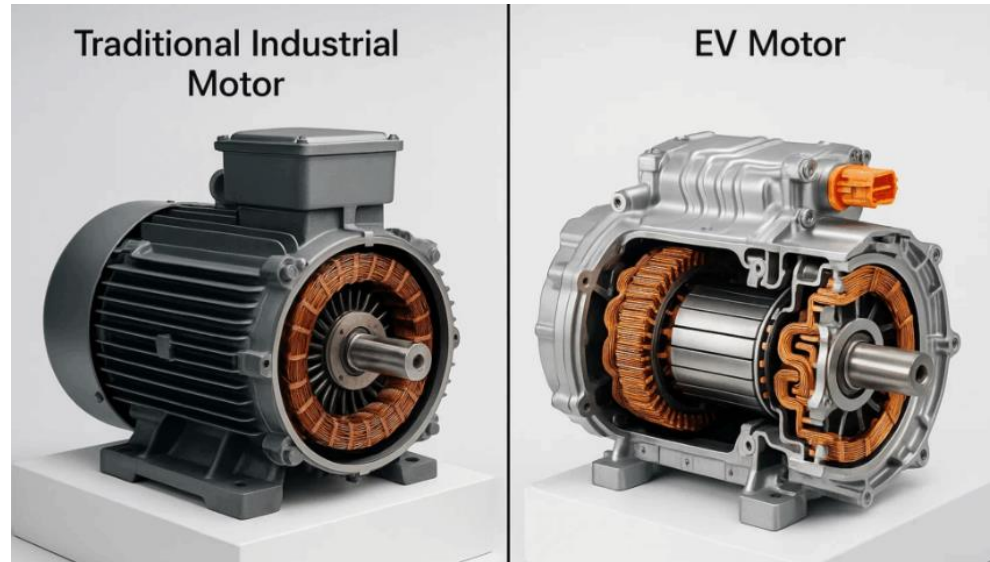
ประกอบด้วย

- มอเตอร์ไฟฟ้าขับเคลื่อน(Electric traction motor)
- ชุดอินเวอร์เตอร์(Inverter assembly) หรือ Power electronic box(PEB)
- ชุดเกียร์ลดความเร็วแบบหนึ่งความเร็ว(Single speed reduction gear)



ส่วนประกอบของมอเตอร์ขับเคลื่อนรถยนต์ไฟฟ้า

มอเตอร์ไฟฟ้าทำหน้าที่เปลี่ยนพลังงานไฟฟ้าเป็นกำลังงานกล เพื่อขับเคลื่อนรถยนต์ เป็นมอเตอร์ไฟฟ้ากระแสสลับ แบบ 3 เฟส



Basic Components

Stator: The stationary part of the motor, which contains U,V,W windings (coils) that generate a rotating magnetic field when energized with alternating current (AC).

Rotor: The rotating part of the motor, PMSM rotor which contains permanent magnets. These magnets create a constant magnetic field. IM or induction motor rotor contains conductor bars. This rotor call squarell cages rotor

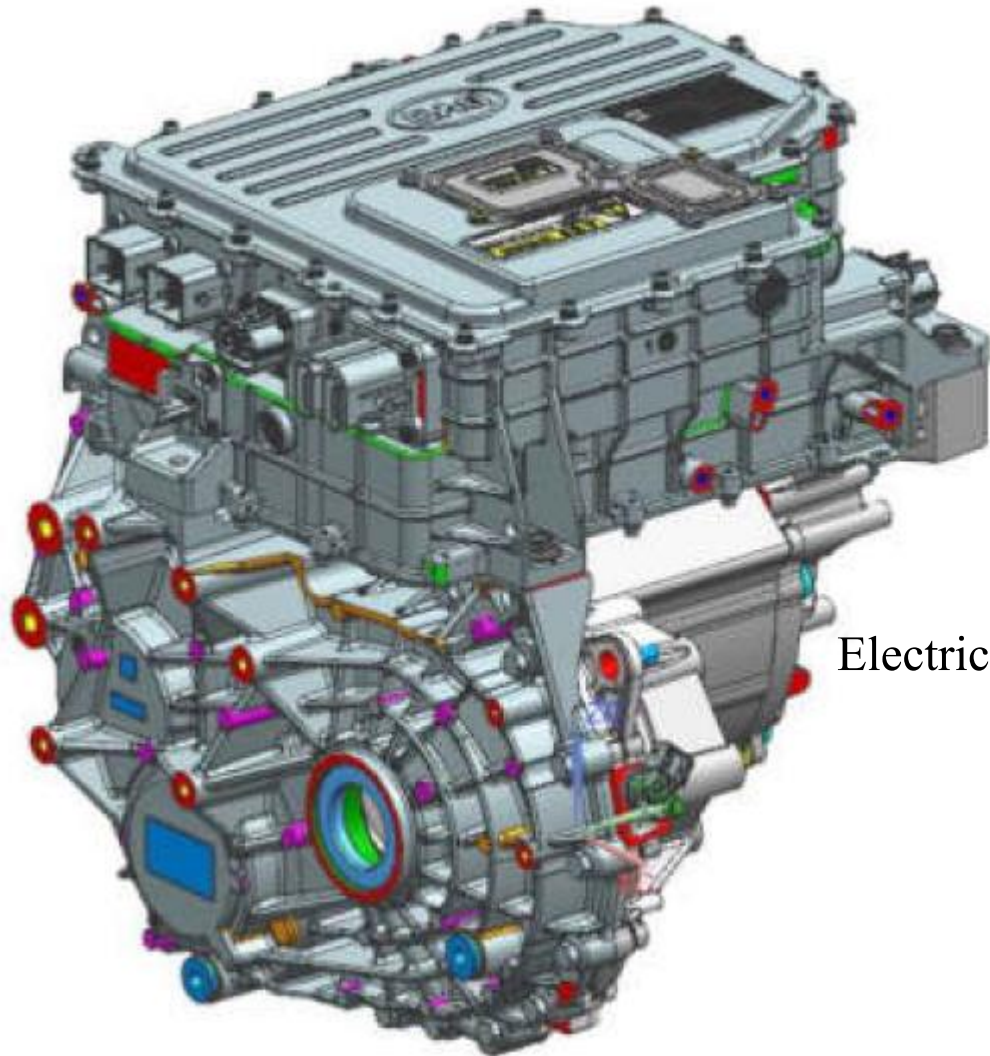
Shaft: Connected to the rotor, it transfers the mechanical power generated by the motor to the load.

Resolver: Position Sensor (Encoder/Resolver) – Used for precise rotor position detection (essential for control)

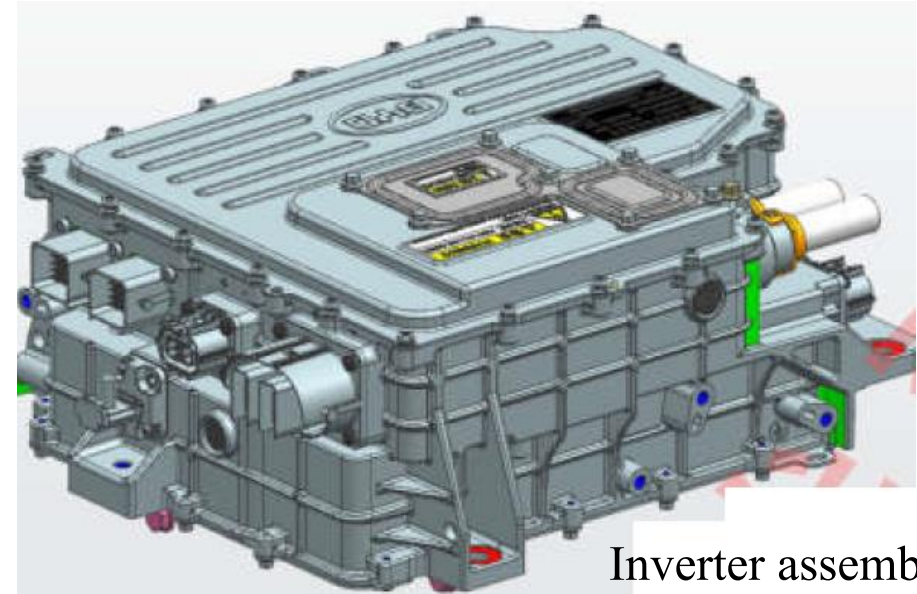
ชุดมอเตอร์ไฟฟ้าขับเคลื่อน : DEEPAL SO7



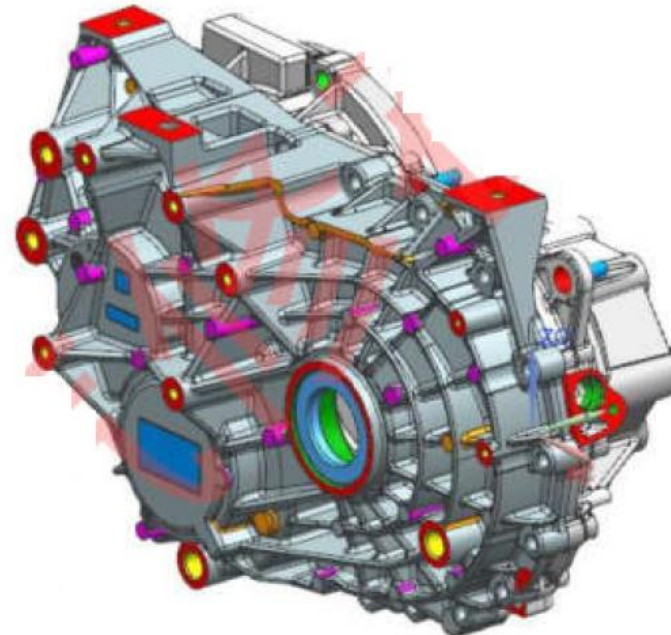
ชุดมอเตอร์ไฟฟ้าขับเคลื่อน : BYD Atto3



Electric motor

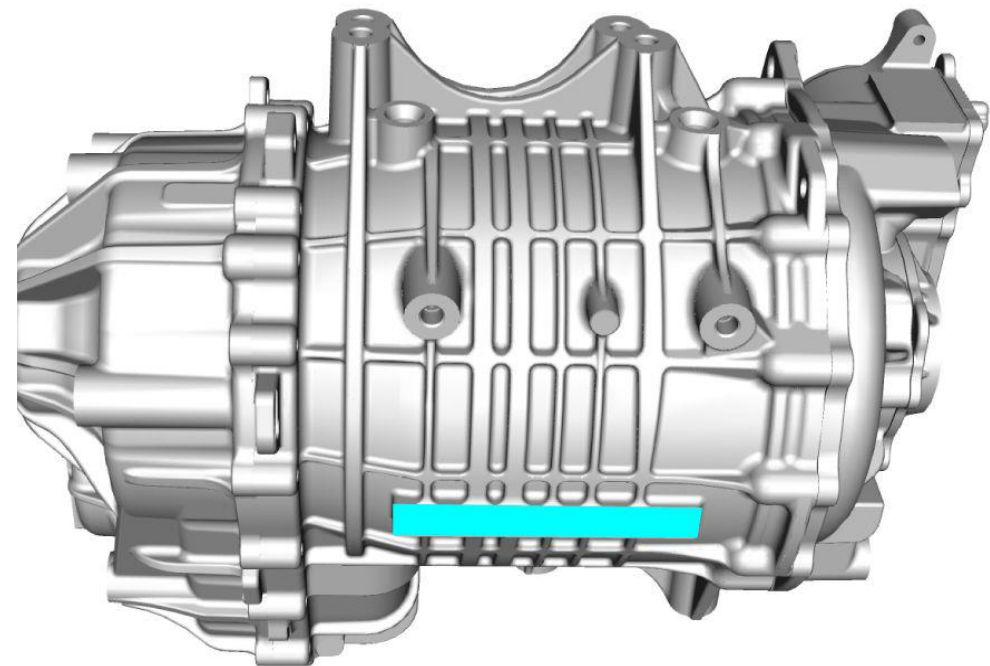
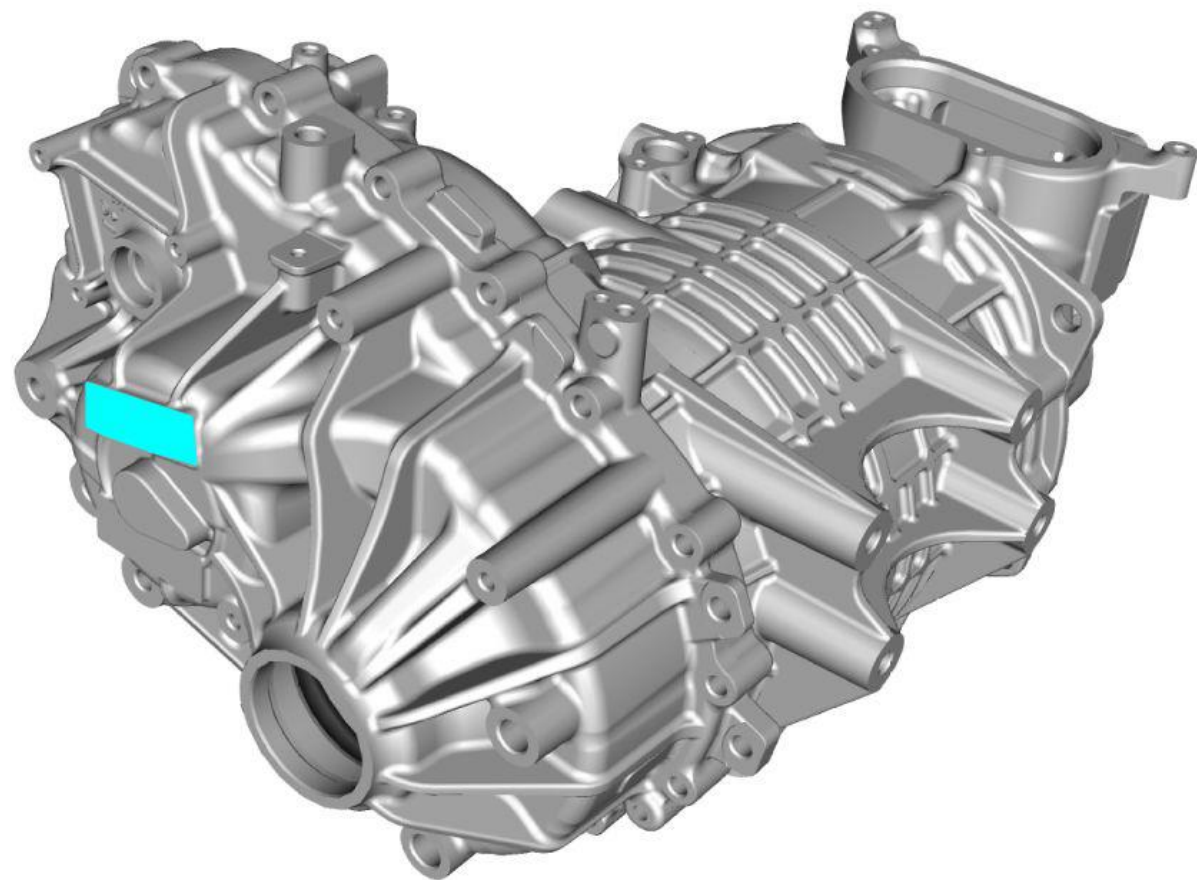


Inverter assembly

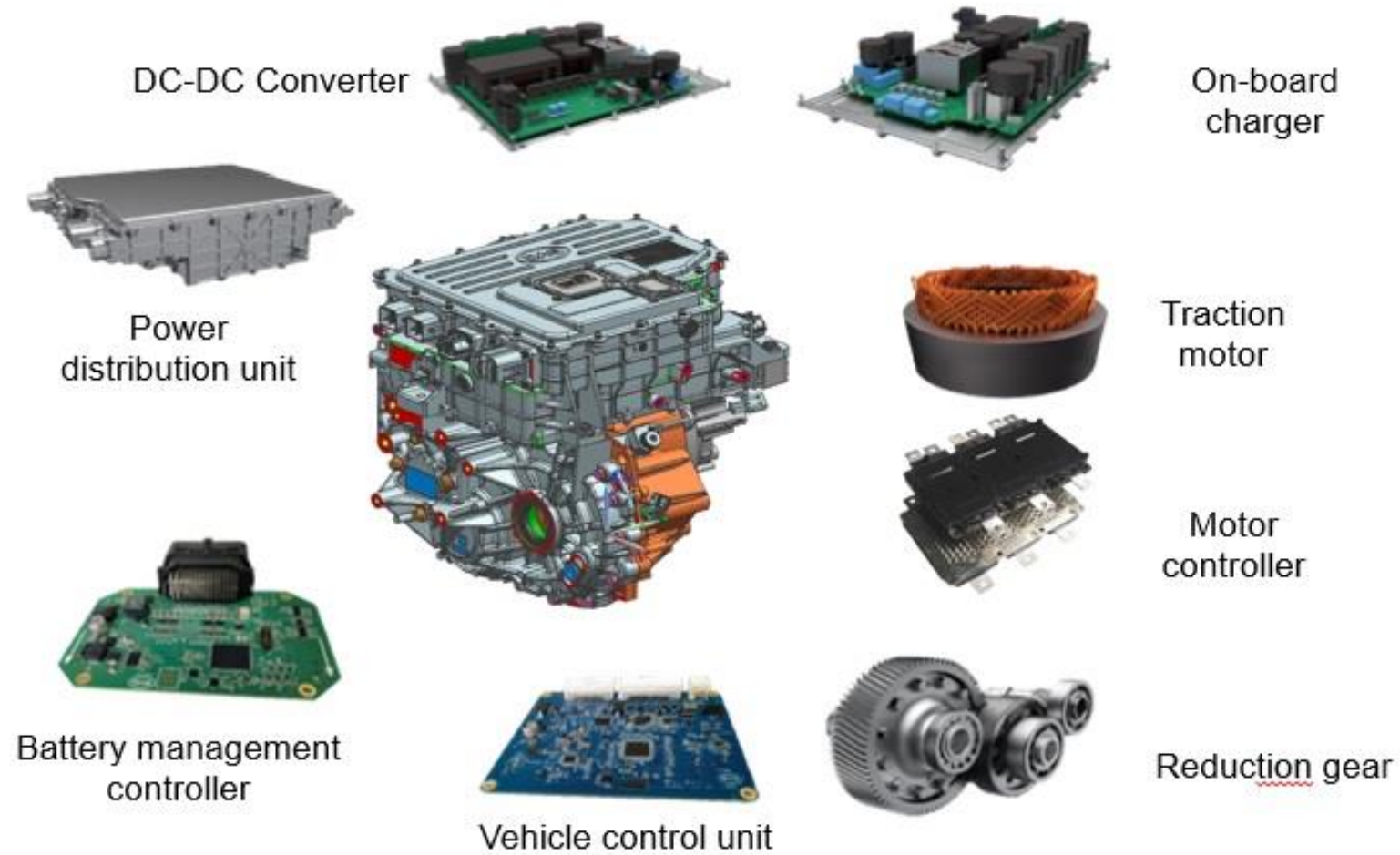


Single speed
reduction gear

ชุดมอเตอร์ไฟฟ้าขับเคลื่อน : MG EP



Powertrain domain controller: BYD



ผู้ช่วยศาสตราจารย์สุวิทย์ วงษ์เย็น

ส่วนประกอบภายในของมอเตอร์ไฟฟ้า (Electric motor components)

มอเตอร์ไฟฟ้าสำหรับขับเคลื่อน เป็นมอเตอร์ไฟฟ้ากระแสสลับ 3 เฟส มีส่วนประกอบหลักได้แก่ ขดลวด
มอเตอร์ (Stator Coil) โรเตอร์ (Rotor) และเบร็กเพลาโรเตอร์



Single speed reduction gear

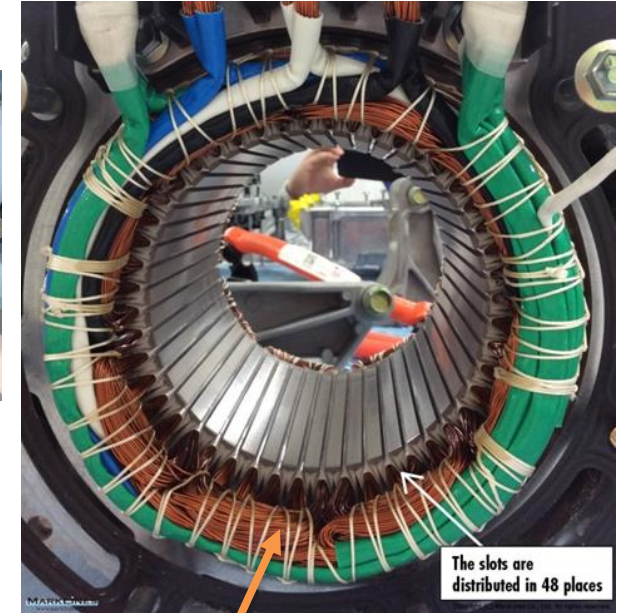


เบร็กเพลาโรเตอร์

PM Rotor



เฟส U,V,W ของขดลวดมอเตอร์
ทำมุม 120 องศา

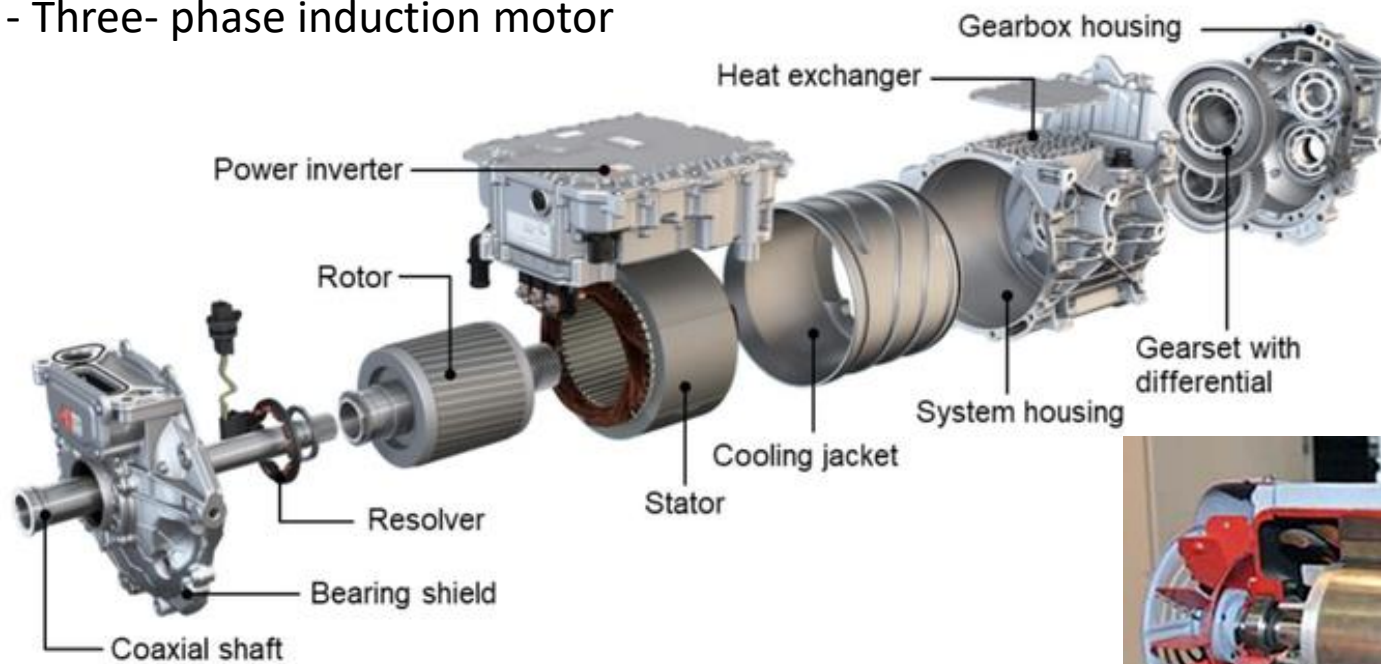


Stator Coil

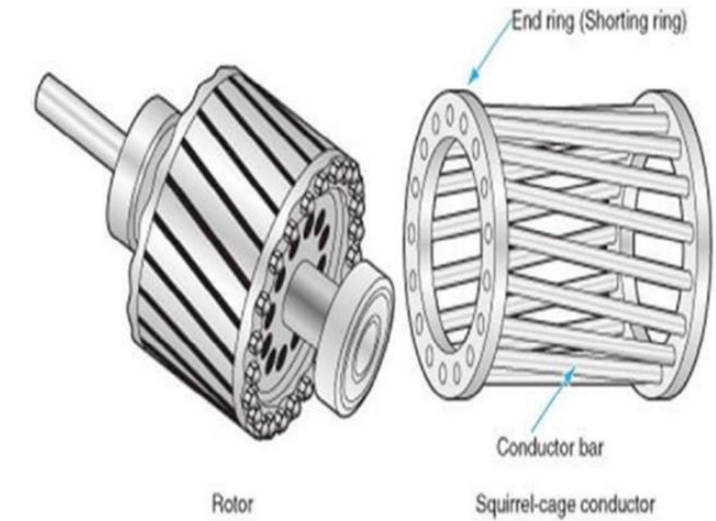
1. Asynchronous motor components

Asynchronous motor or Induction motor

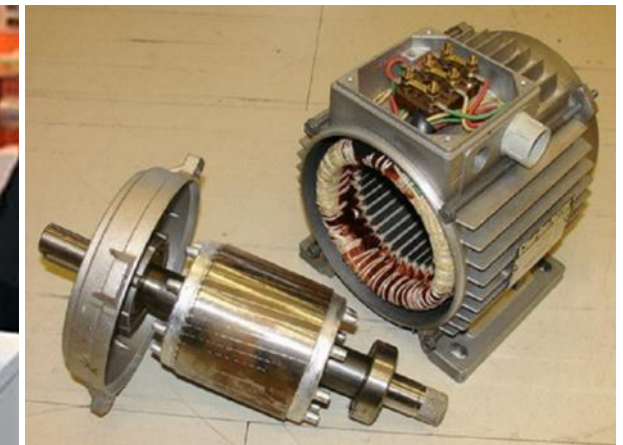
- Single-phase induction motor
- Three- phase induction motor



Squirrel Cage Rotor



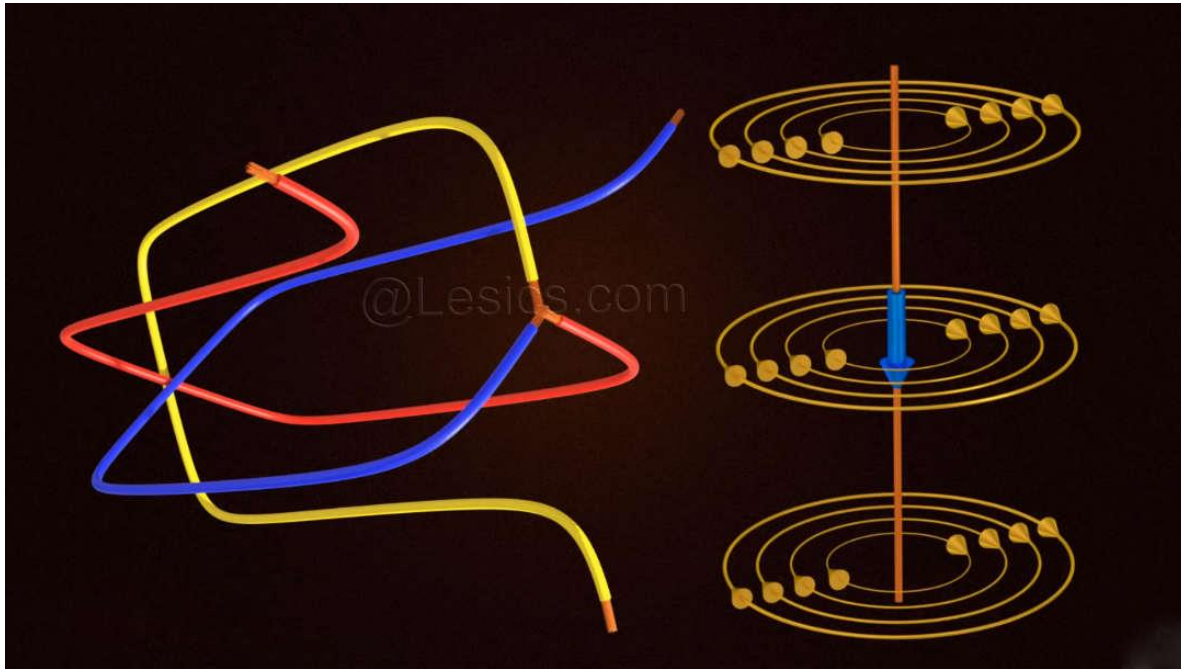
AC Motor : Induction Motor Drive And Components



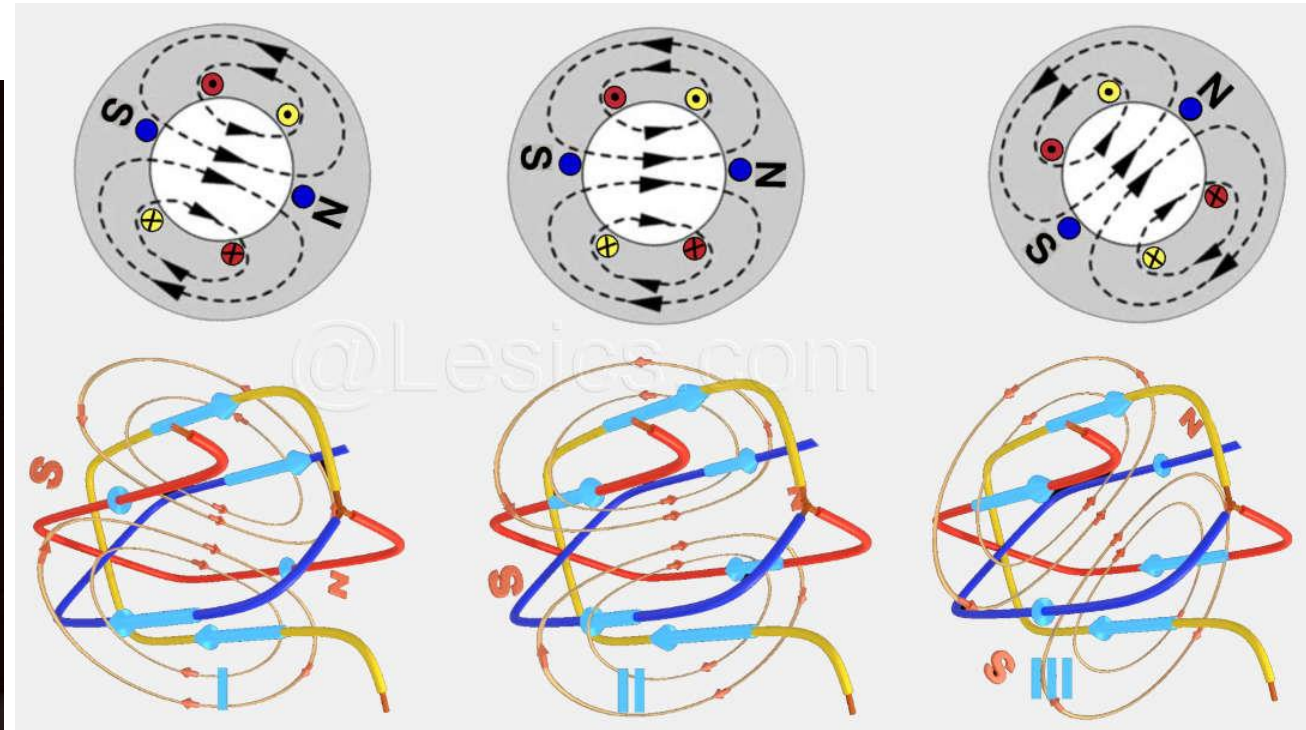
The concept of a Rotating Magnetic Field (RMF)

To understand the phenomenon of a rotating magnetic field, it is much better to consider a simplified 3 phase winding with just 3 coils. A wire carrying current produces a magnetic field around coils. Now for this special arrangement, the magnetic field produced by 3 phase A.C current will be as shown at a particular instant.

The components of A.C current will vary with time. Two more instances are shown in the following figure, where due to the variation in the A.C current, the magnetic field also varies.



A Magnetic field is produced around the single wire



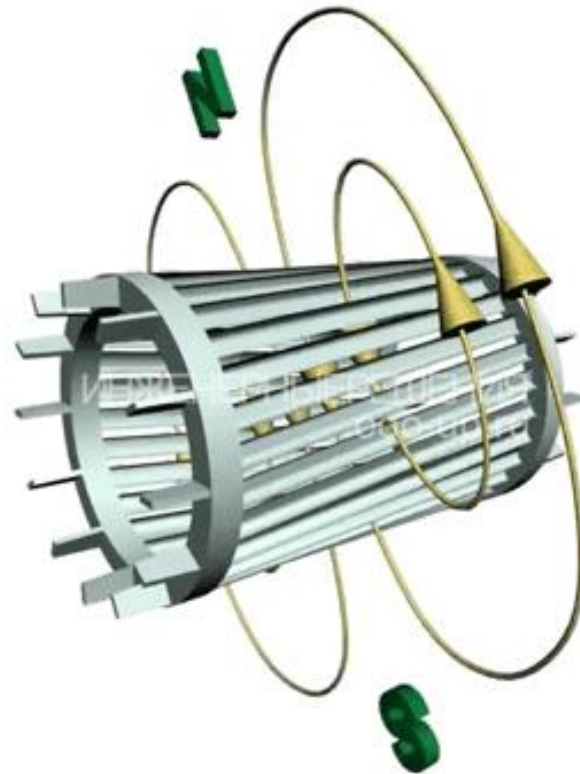
Concept of a Rotating Magnetic Field is illustrated here

หลักการทำงานของมอเตอร์แบบเหนี่ยวนำ(Induction motor)

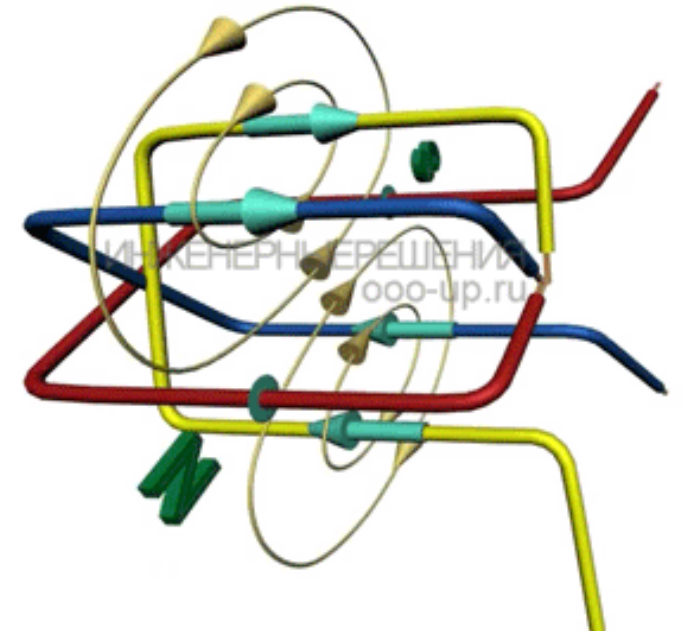
The concept of a Rotating Magnetic Field (RMF)



Stator coil:Magnetic field

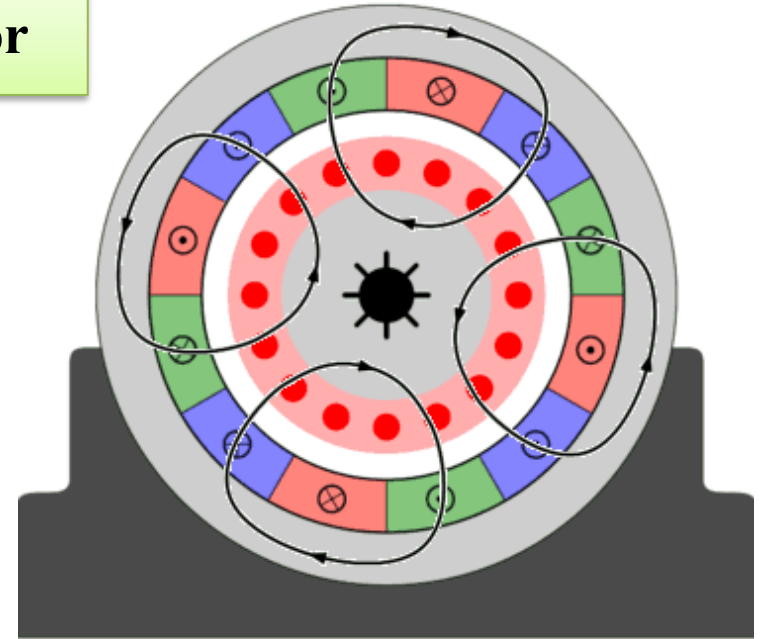
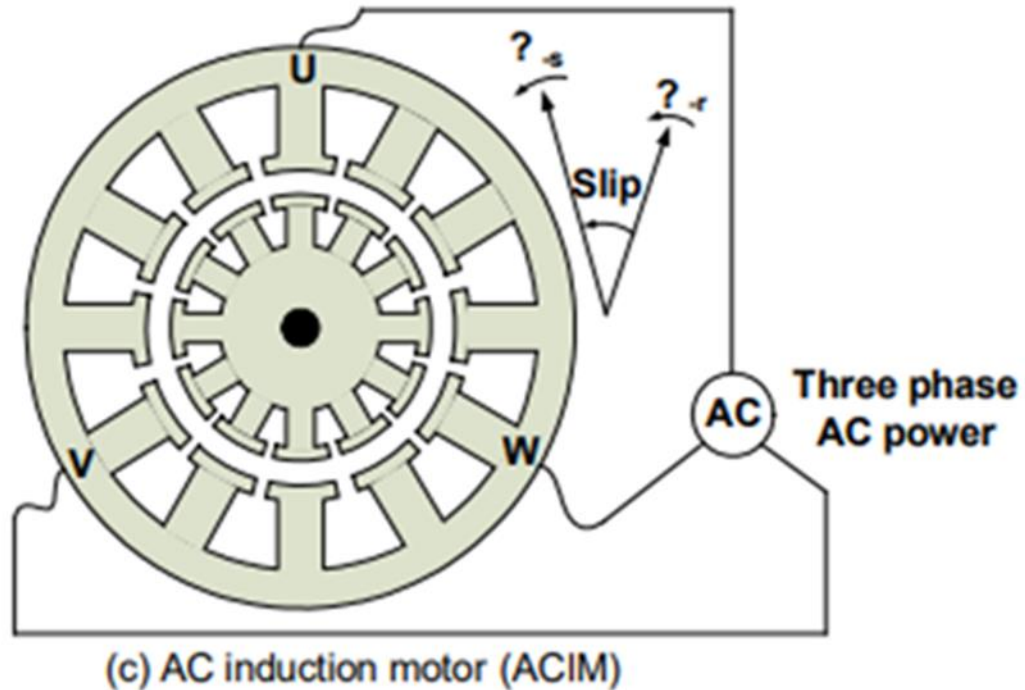


Squirrel case rotor



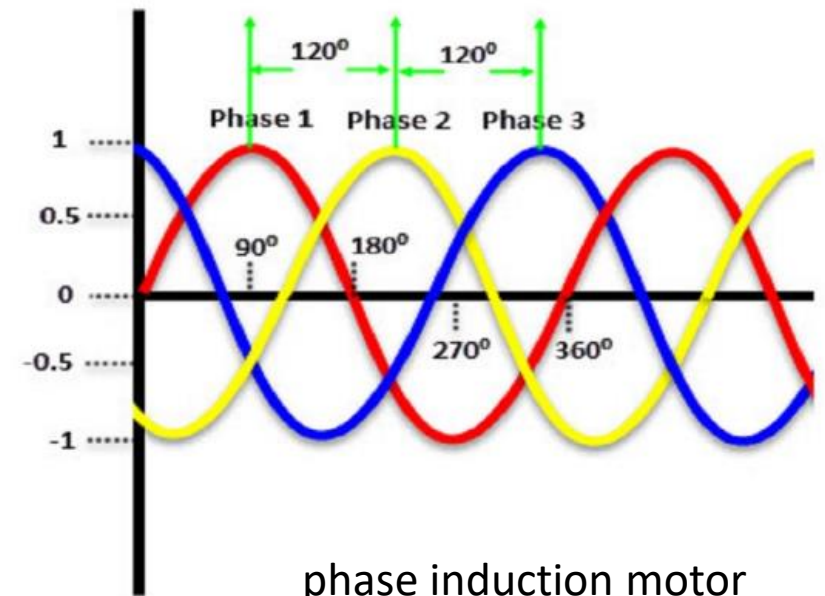
Induction current

การควบคุมการทำงานของ Induction Motor



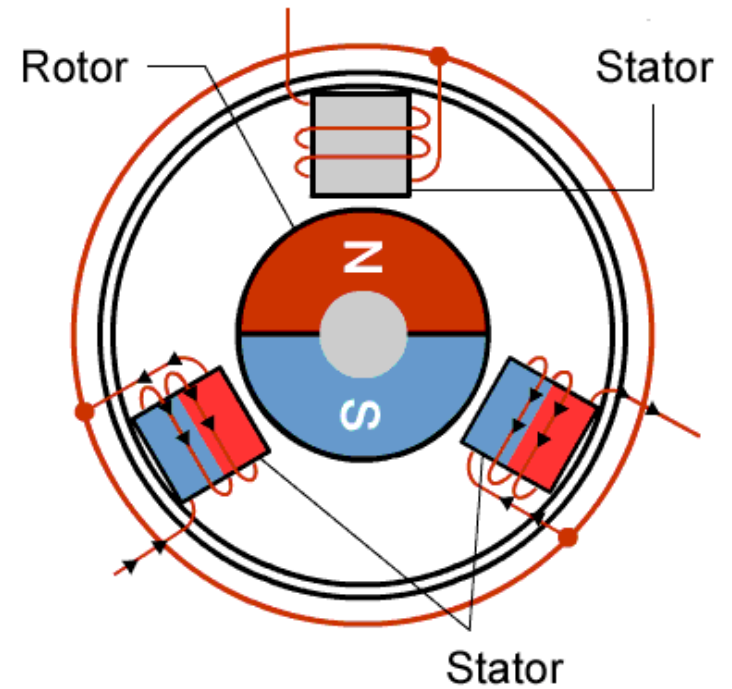
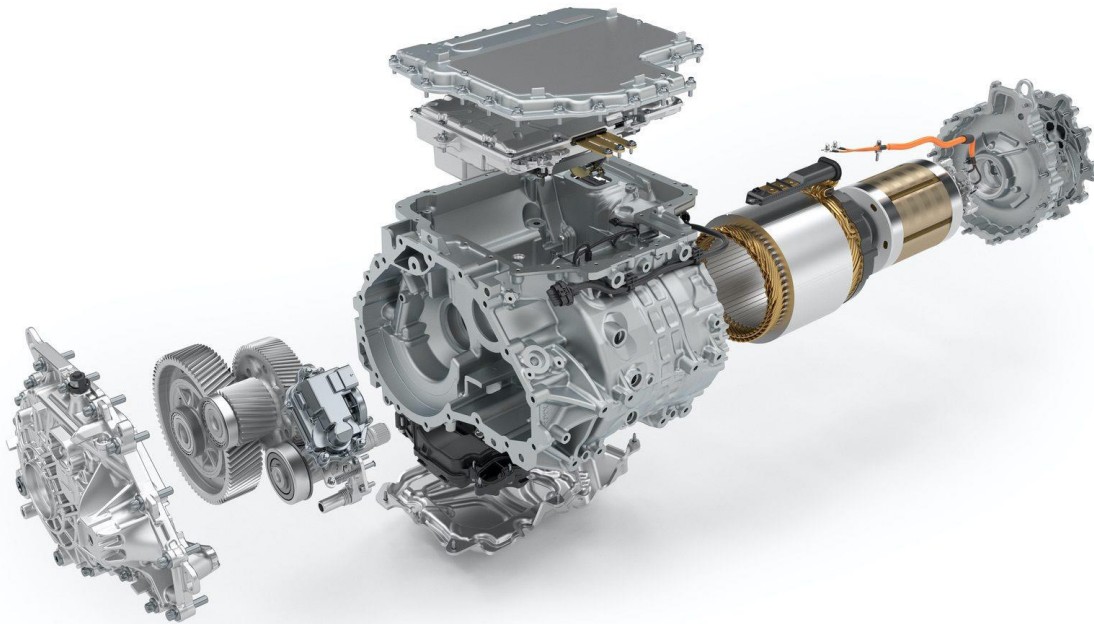
Type Of Induction Motor

- Single phase induction motors are not a self-starting motor.
- Three phase induction motor are a self-starting motor.



2. Permanent magnet Synchronous Motor

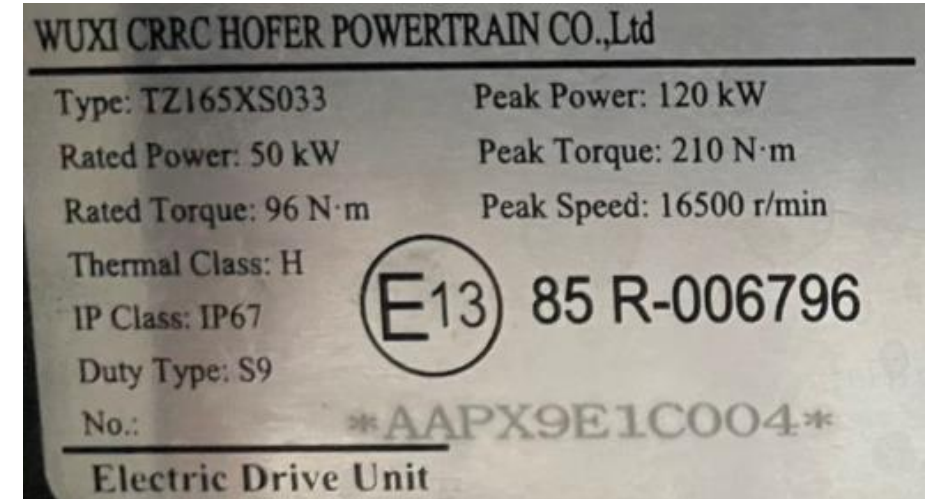
ส่วนประกอบของมอเตอร์แบบ PMSM



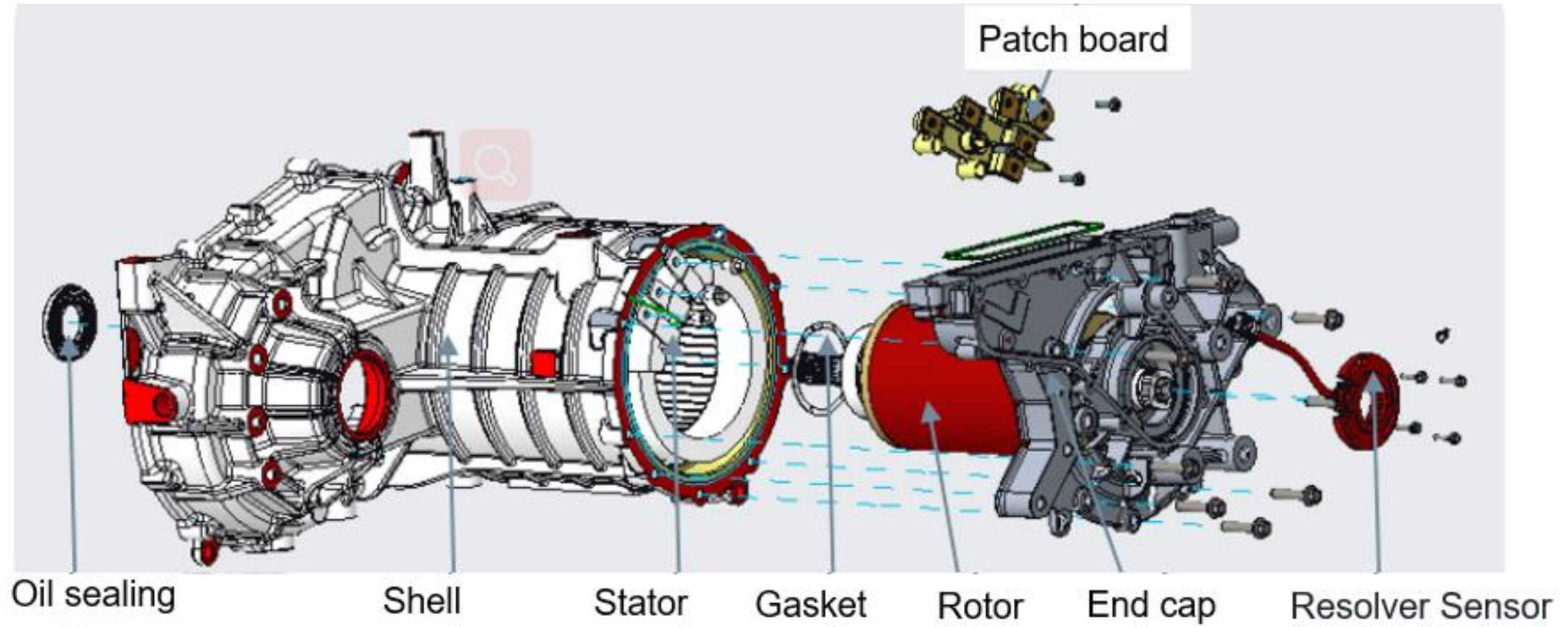
Drive Motor System Introduction specification :NETA

➤ System Parameters

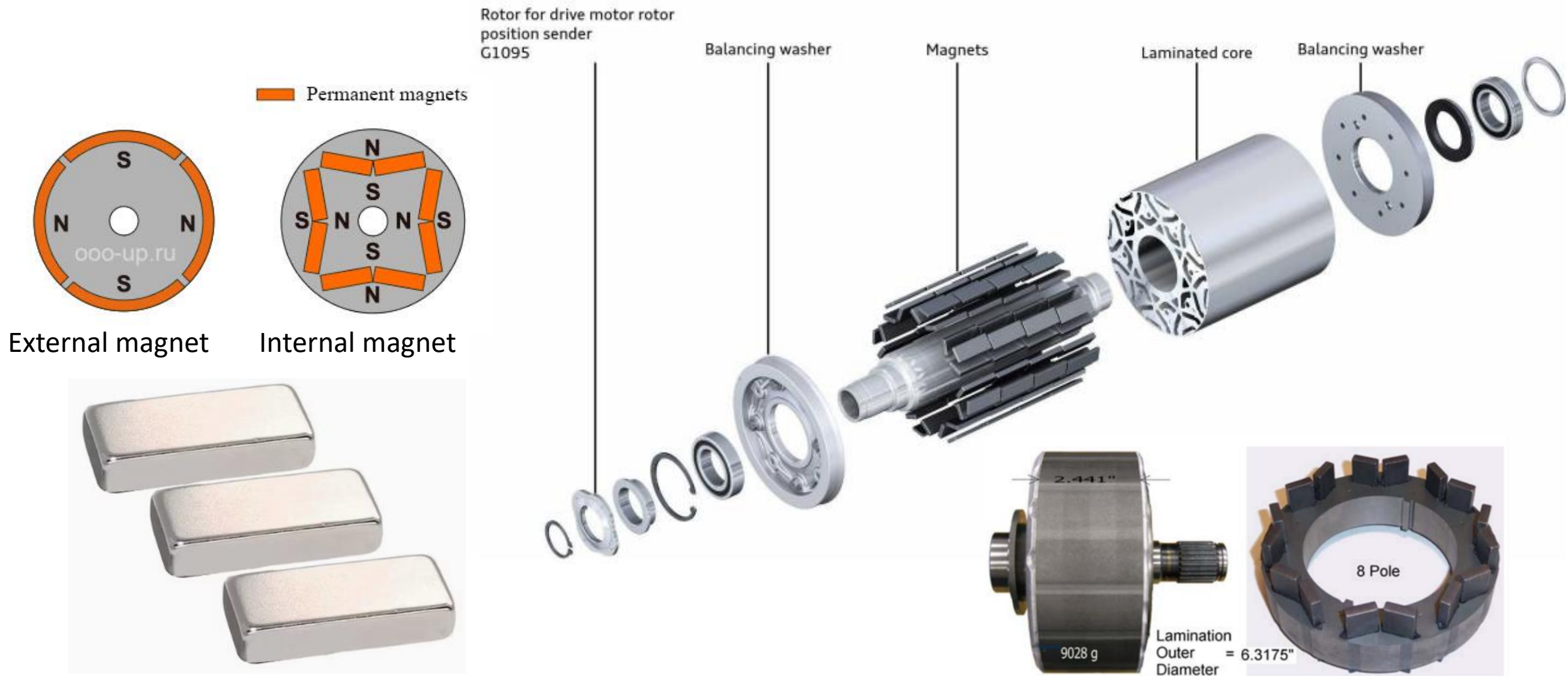
Motor Controller		Drive Motor	
Name	Parameter	Name	Parameter
Manufacturer	CRRC	Peak power	120 kW
Model	P24180C12101	Peak torque	210 N·m
Internal name	Electric drive assembly	Peak speed	16500 rpm
Type	Permanent magnet synchronous	Rated power	50 kW
Combination mode	Three-in-one	Rated torque	96 N·m
IP Class	IP67	Rated speed	4980 rpm
Cooling type	Water cooled	DC rated voltage (VDC)	360
Coolant flow	≥ 8 L/min	DC voltage range (VDC)	270 - 450
Battery voltage level	12 V	Reduction gear ratio	12.33



Drive Motor System Structure



Permanent magnet Rotor And Components



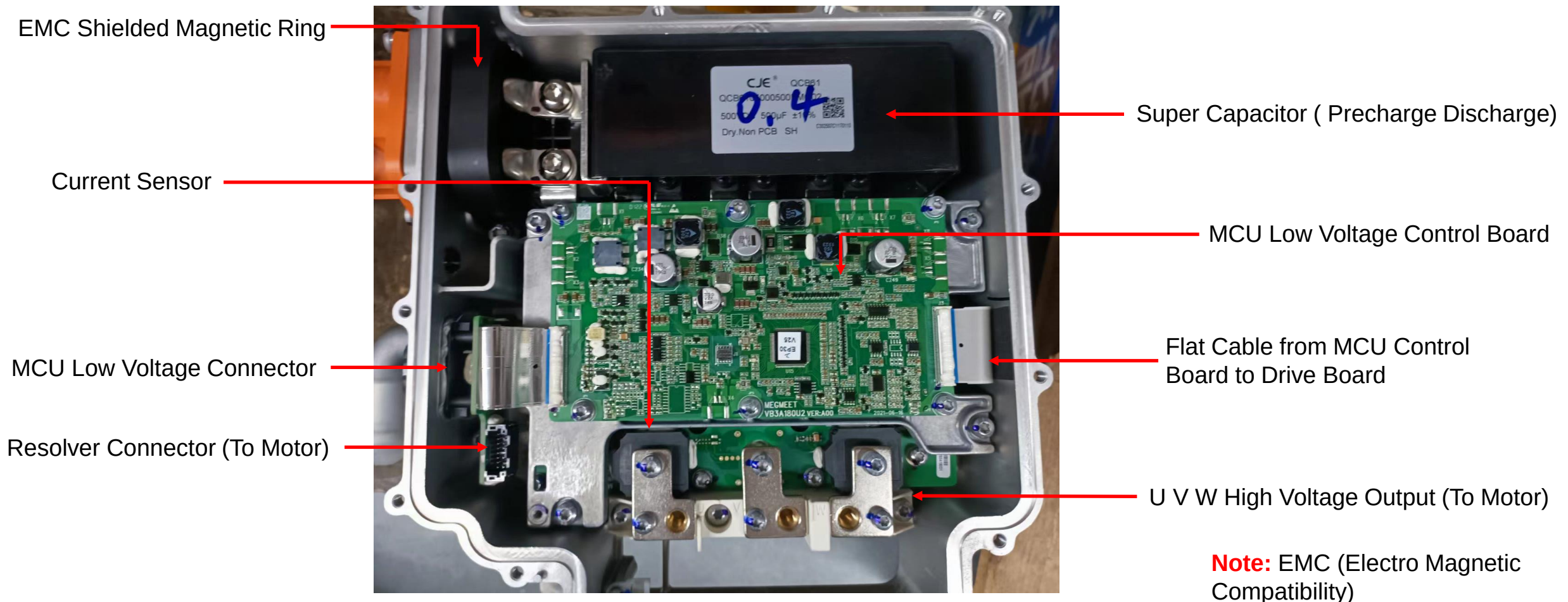
Details of the Camry rotor assembly and the “V” orientation of the neodymium iron boron (NdFeB) interior PM

Camry rotor assembled (left) and partially disassembled (right).

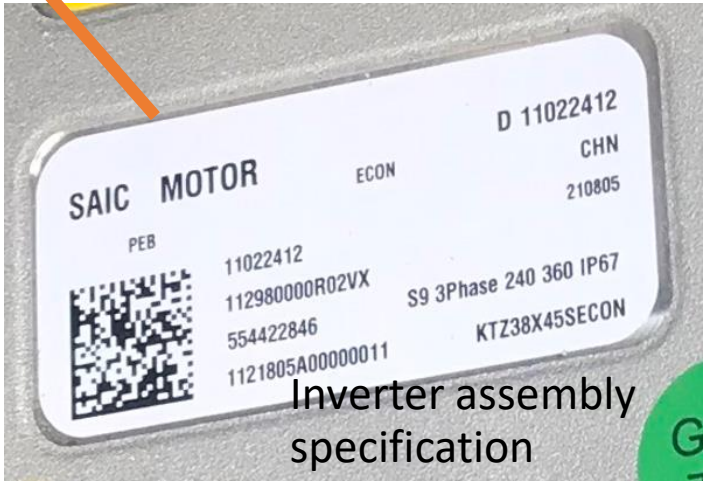
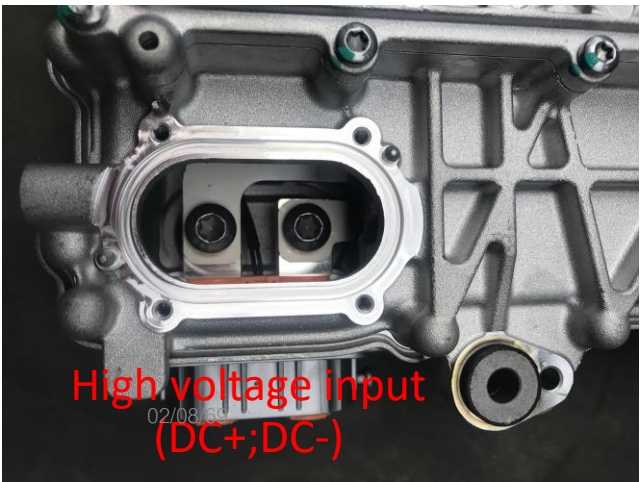
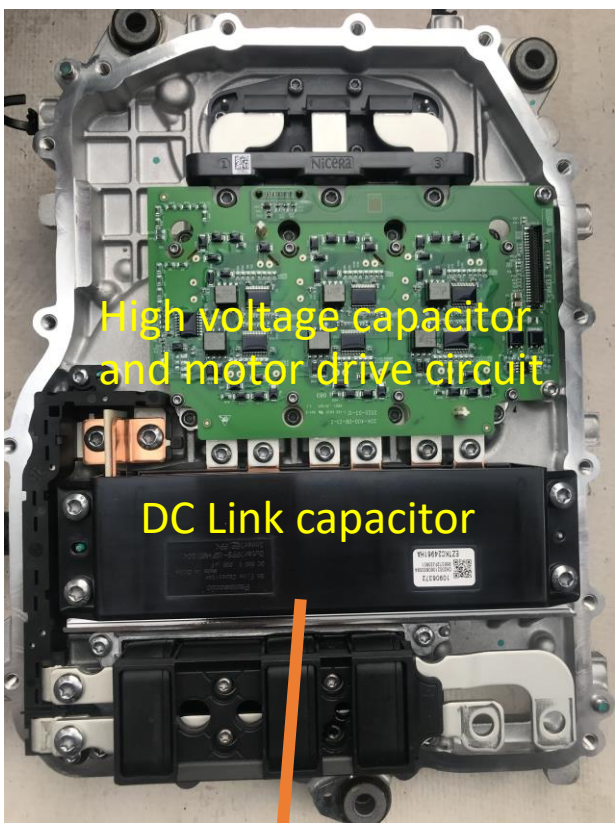
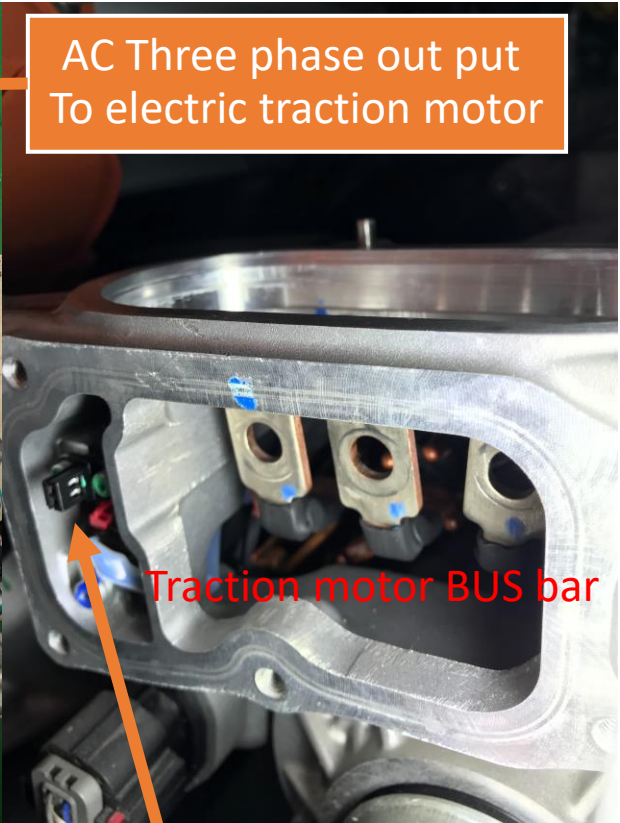
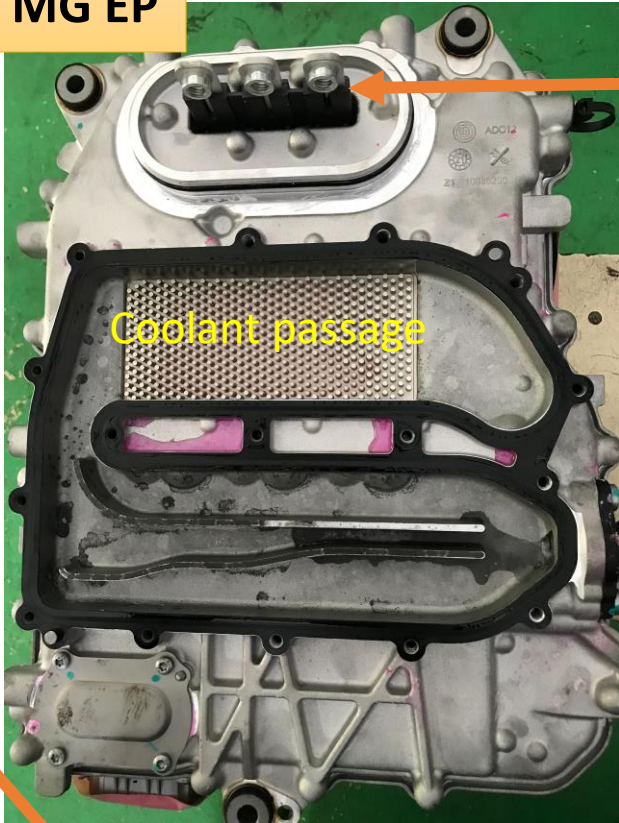
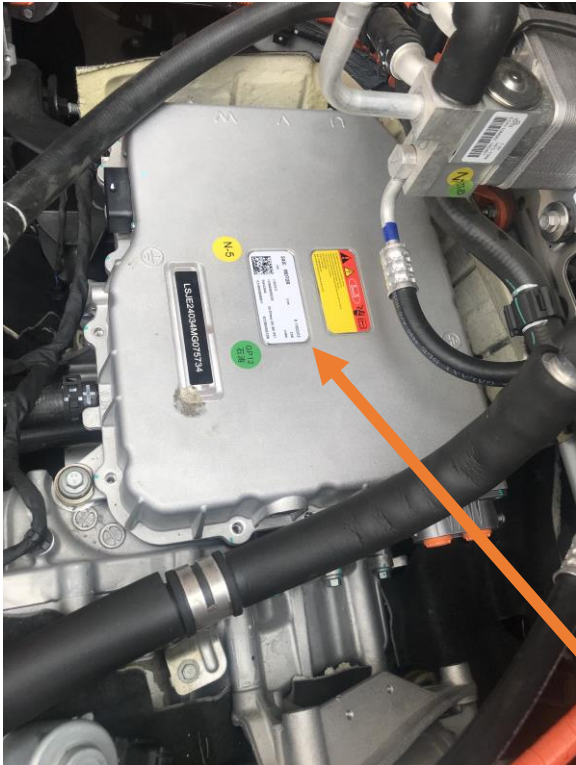
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Drive Motor System Structure

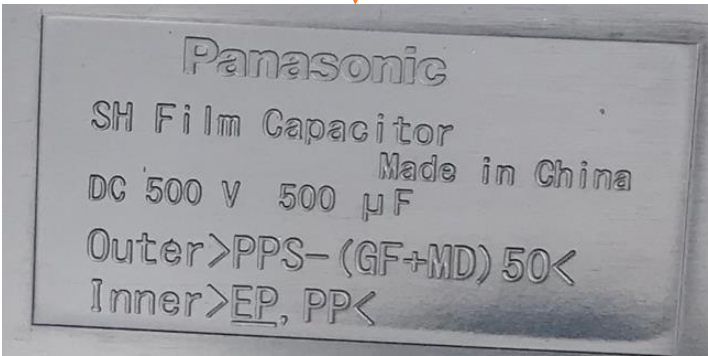
➤ Motor Controller Structure



Example Inverter Assembly : MG EP

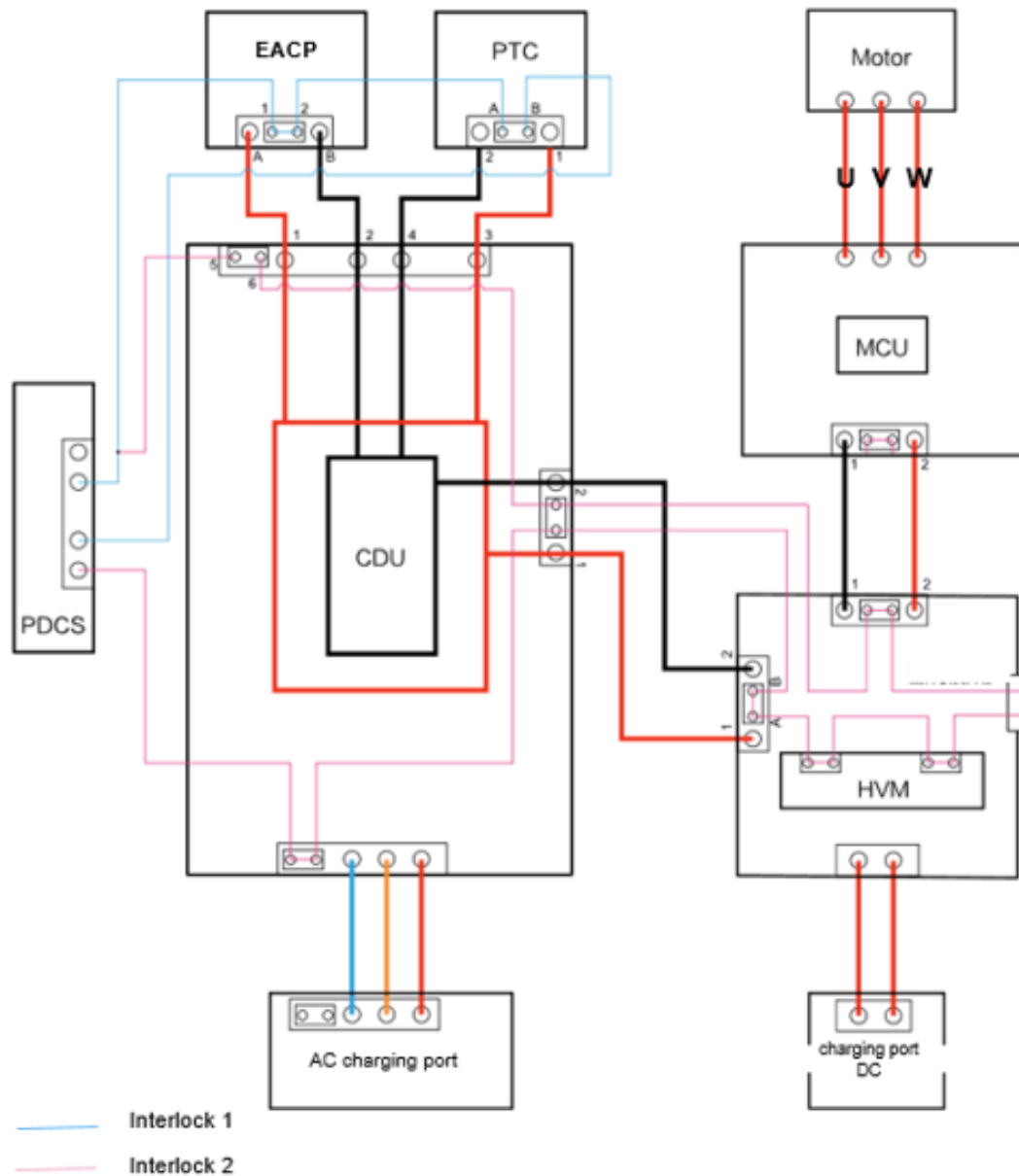


High voltage inter lock loop

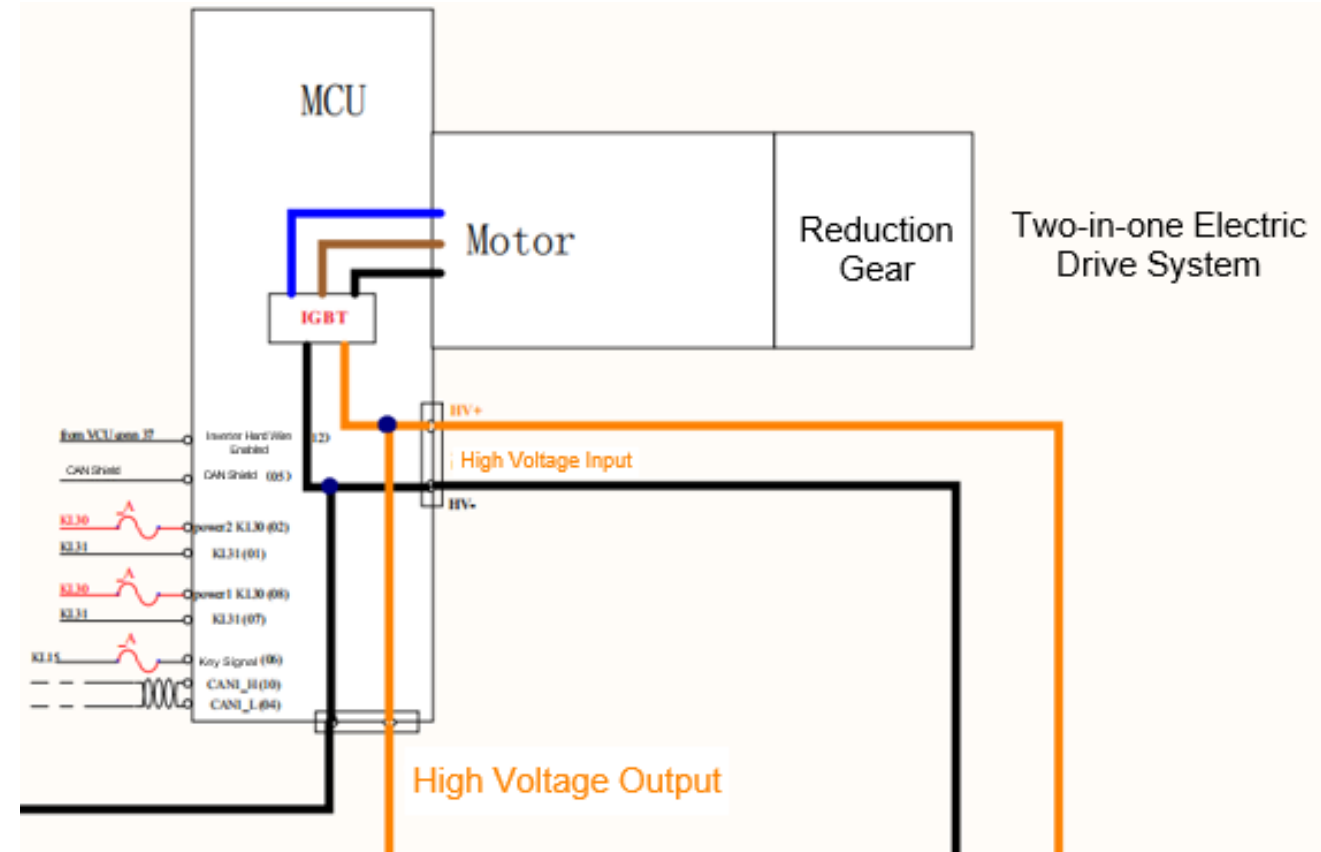


Capacitor DC500V 500 uF

Drive Motor System Operation

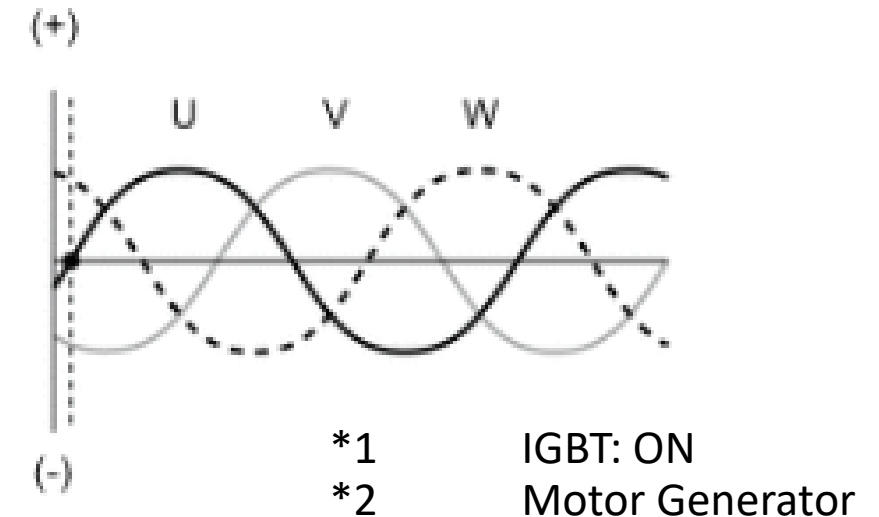
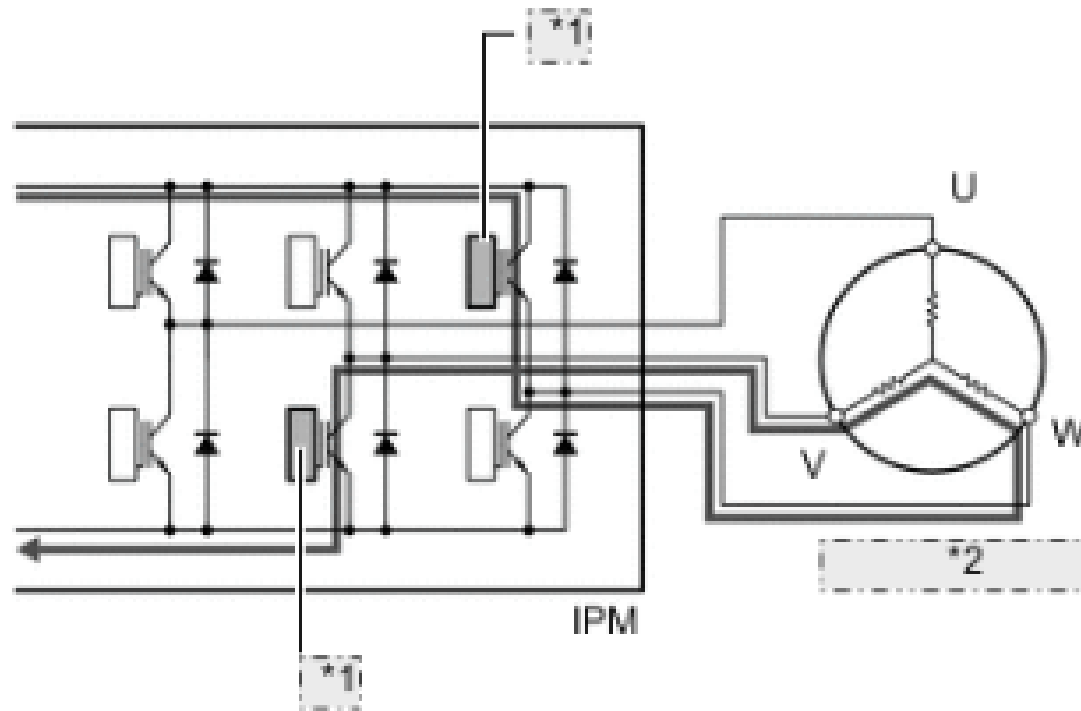


➤ Drive Motor Principle Structure



Schematic Diagram of Motor Controller System Peripheral High and Low Voltage

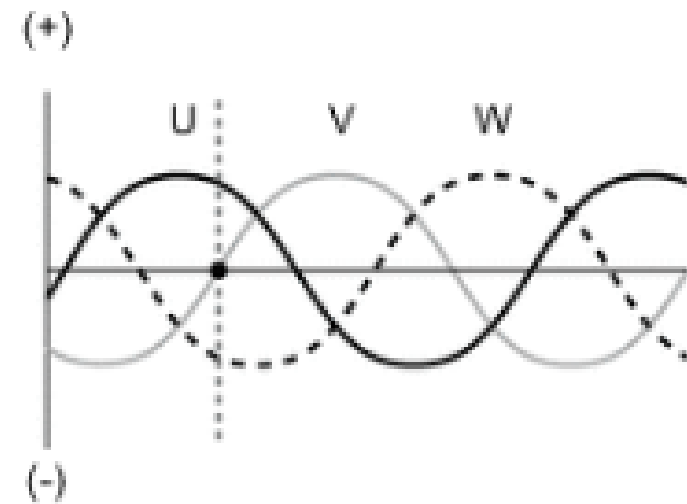
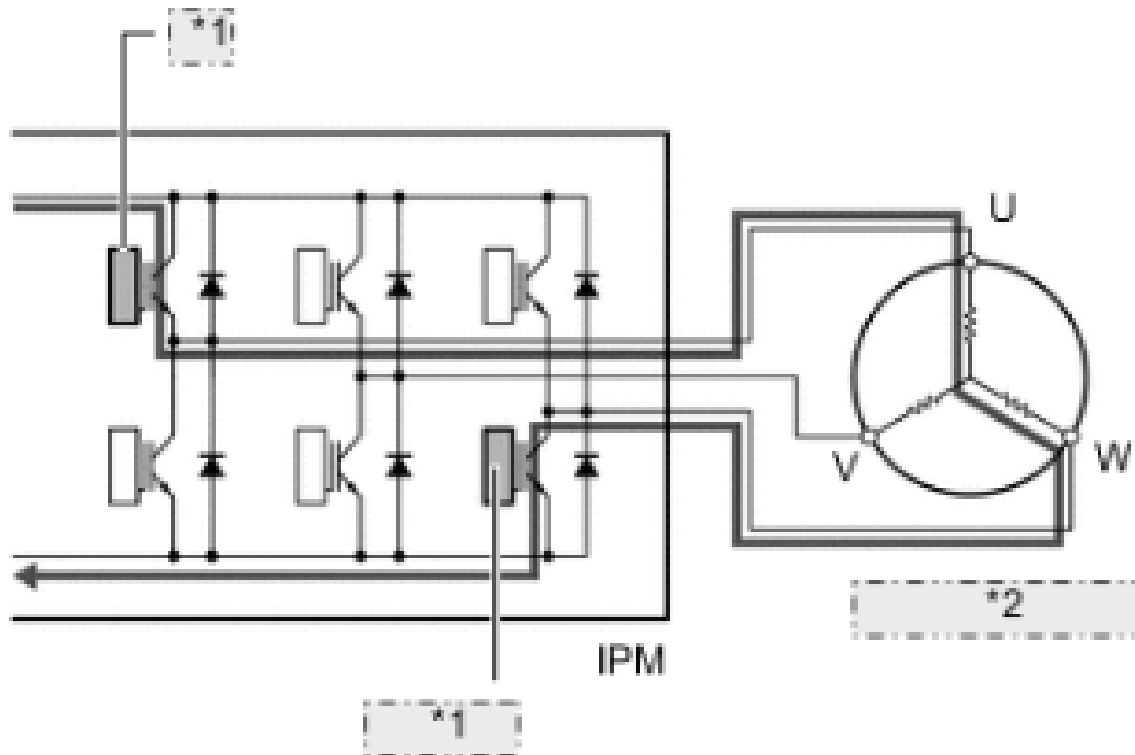
W-V Phase Electric motor control



The MG ECU controls the Insulated Gate Bipolar Transistors (IGBTs) in the Intelligent Power Module (IPM) based on the signals received from the hybrid vehicle control ECU. The IGBTs are used for switching for the U, V, and W phases of each motor generator. There are 6 IGBTs that switch on and off to control each individual motor generator in accordance with operation as either a motor or as a generator.

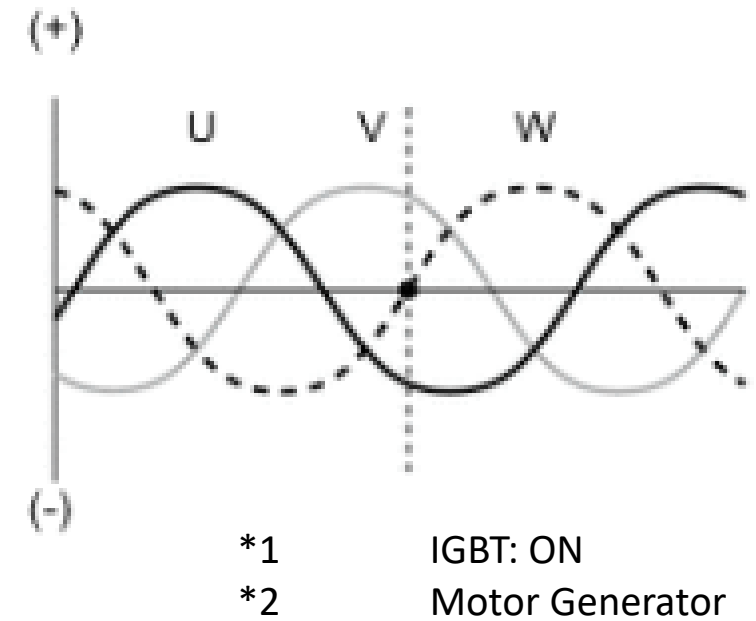
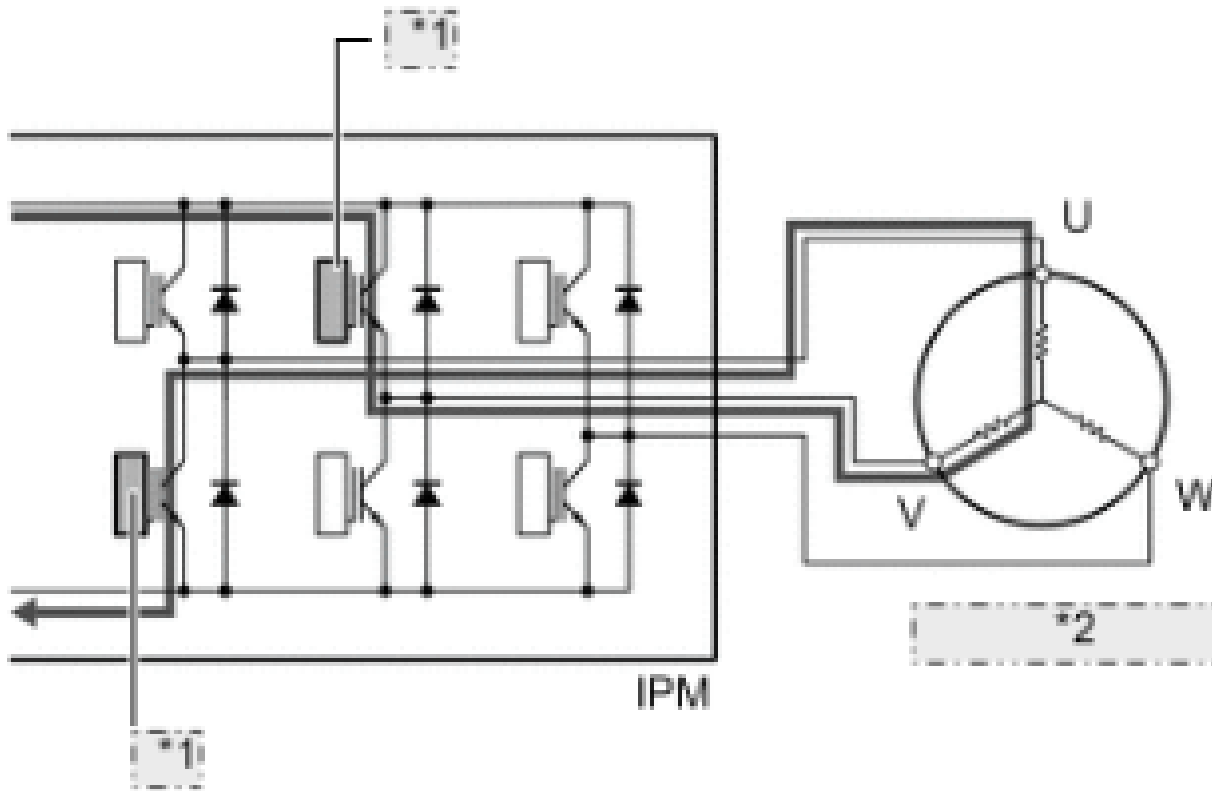
The illustration below describes the basic control used when the motor generator functions as a motor. The IGBTs in the IPM switch on and off to supply 3-phase alternating current to the motor generator. In order to create the motive force required of the motor generator as calculated by the hybrid vehicle control ECU, the MG ECU switches the IGBTs on and off in order to control the speed of the motor generator.

U-W Phase Electric motor control



*1 IGBT: ON
*2 Motor Generator

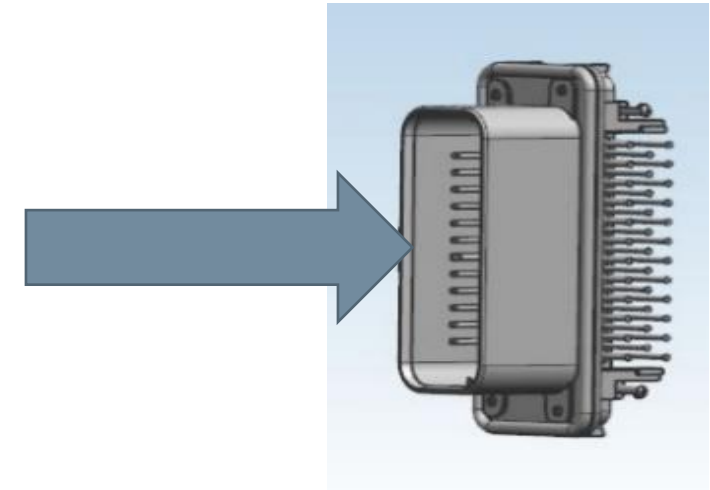
V-U Phase Electric motor control



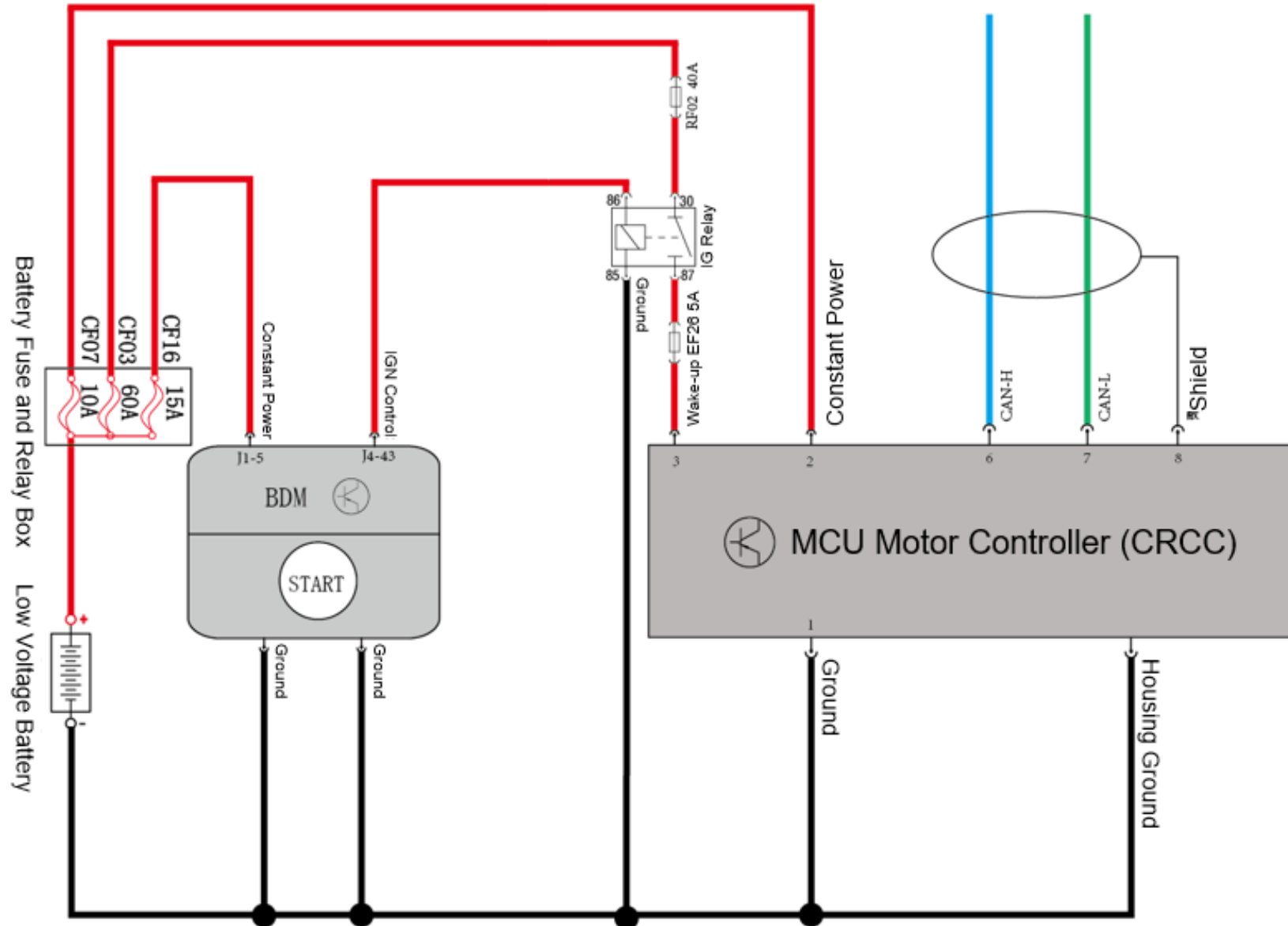
Drive Motor System Operation

➤ MCU Controller Low Voltage Pin Definition

Pin	Signal Description		Pin	Signal Description	
1	KL31	Low voltage ground	5	Null	
2	KL30	Battery power supply positive	6	CAN-H	CAN high
3	KL15	Ignition signal	7	CAN-L	CAN low
4	Null		8	GND_shield	Shield ground



Drive Motor System Operation

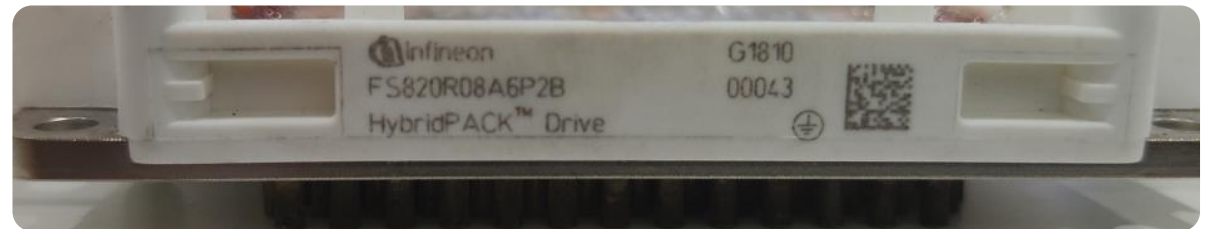
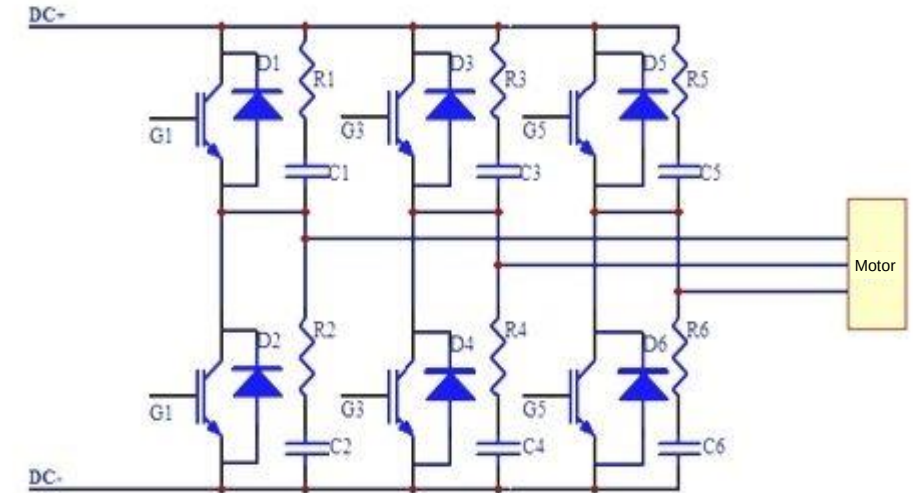
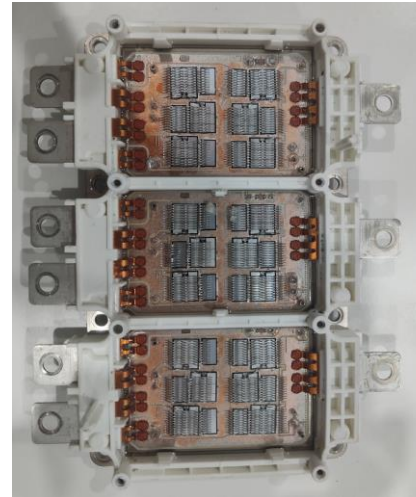


MCU Circuit Schematic Diagram

Drive Motor System Operation

➤ IGBT Introduction

- ◆ IGBT is the abbreviation of **Insulated Gate Bipolar Transistor**. It is a power electronic integrated circuit board with high power, mainly used for inverter circuit and other inverting circuits to invert DC voltage into AC with adjustable frequency. So it is called the "CPU" of power electronic device
- ◆ As a core component of electric vehicle electronic control technology, IGBT is mainly used to control the conversion of DC and AC of vehicle drive system, which can directly determine the key indexes that affect vehicle performance, such as torque, maximum output power, etc.

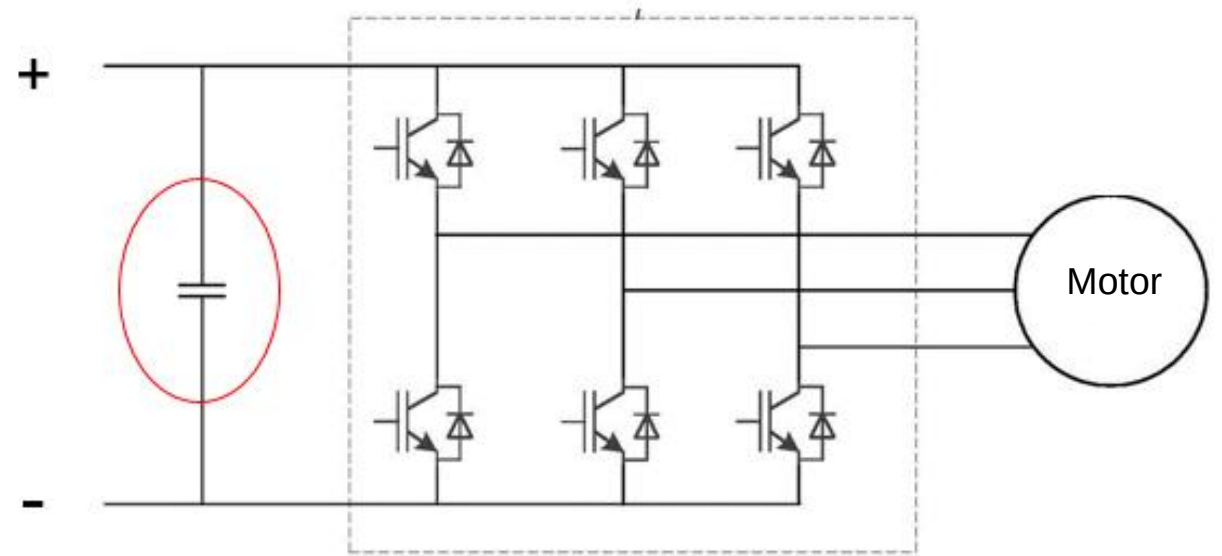


IGBT Set - Infineon

Drive Motor System Operation

➤ MCU Capacitor

- ◆ The internal of MCU is equipped with super capacitor, which has the function of keeping voltage stable;
- ◆ Super capacitor is generally located under the IGBT heat sink. The capacitor takes part in the pre charging when power on and discharging when power off;
- ◆ Measured value of super capacitor: Approximately 560 uF (factory test part, for reference only)



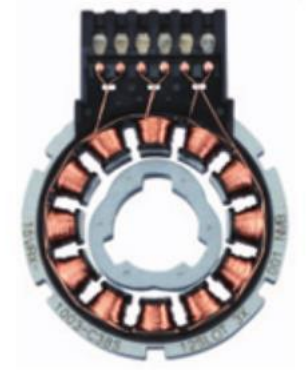
Drive Motor System Operation

➤ Resolver Sensor

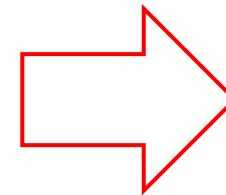
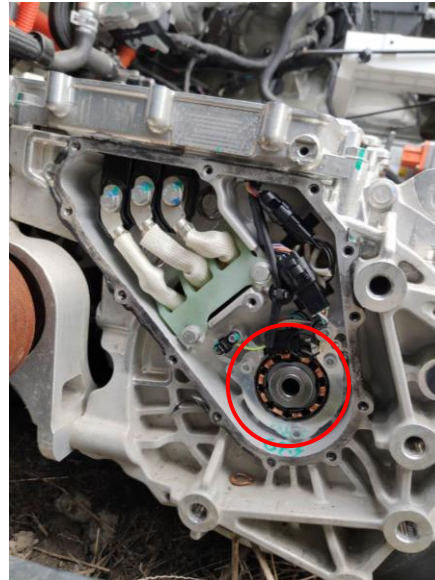
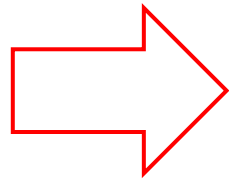
Function: Monitor the angle speed, position and rotation direction of motor rotor and feed them back to motor controller

Structure: Sensor coil is fixed on the housing and signal ring gear is fixed on the motor rotor (it is similar to the installation relationship between generator crankshaft position sensor and crankshaft)

Circuit: Directly connected to motor controller. Three circuit coils (excitation, sine and cosine) form a 6-wire sensor, and shielded wire ground point is also provided



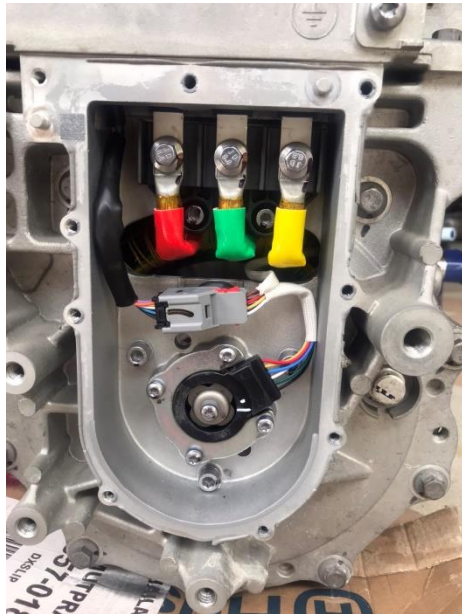
Resolver Sensor



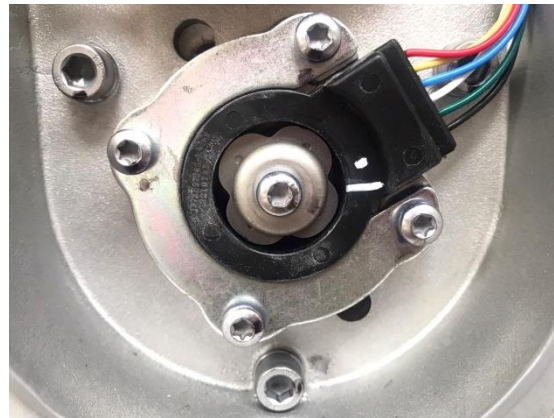
Resolver and Rotor Signal

A resolver is an extremely reliable and compact sensor that precisely detects the magnetic pole position. Knowing the precise position of the magnetic poles of the rotor. Detect the rotating position of the rotor and control the timing of current application to the coil. The stator of the resolver contains 3 types of coils: excitation coil A, detection coil S and detection coil C.

The rotor of the resolver is oval, the distance of the gap between the stator and the rotor varies with the rotation of the rotor



NETA Traction motor and resolver



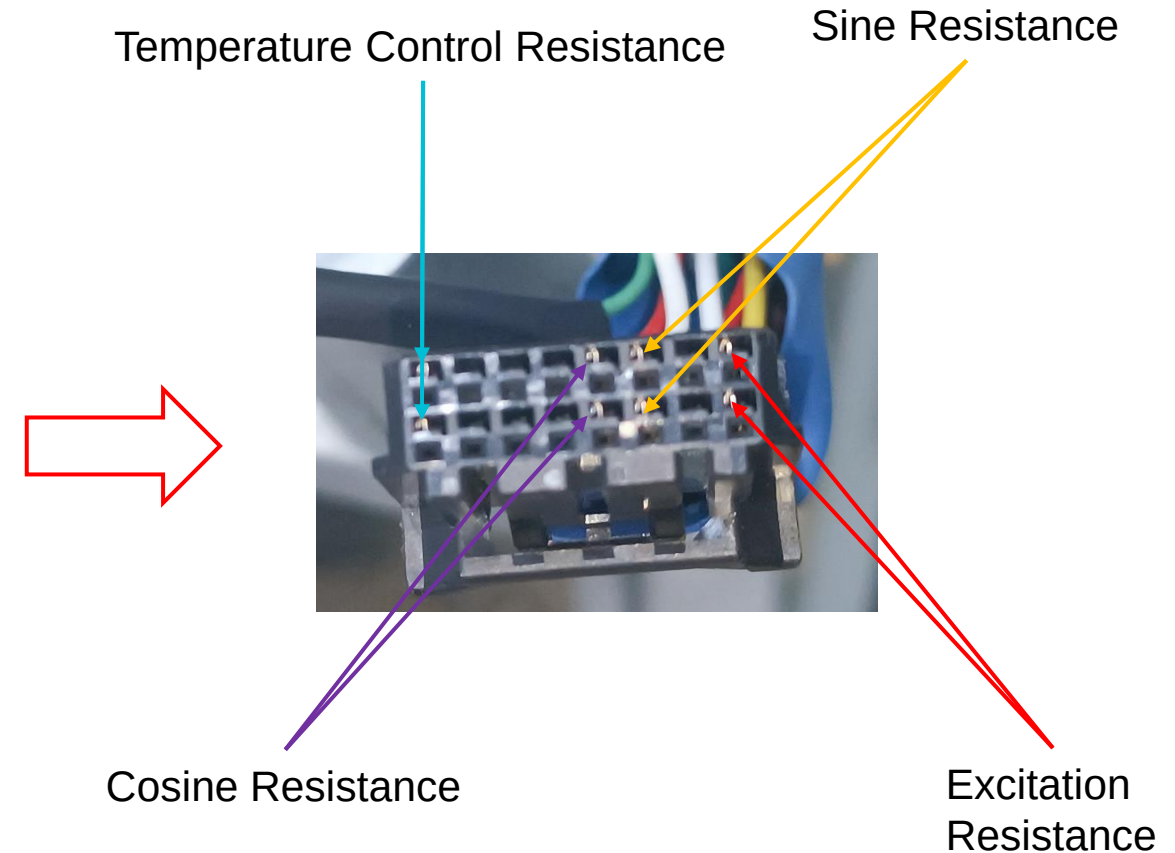
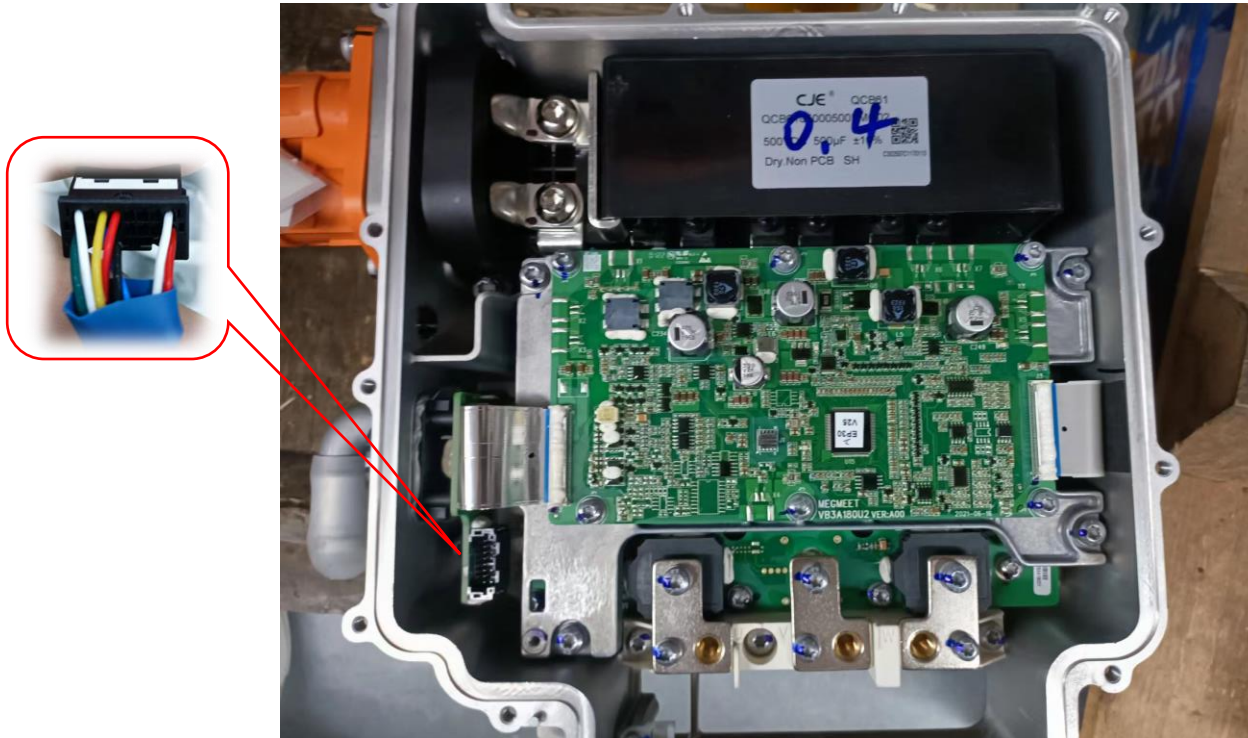
Resolver and rotor



Toyota Hybrid

Drive Motor System Operation

➤ Resolver Connector Location and Definition (CRRC 120 kW)

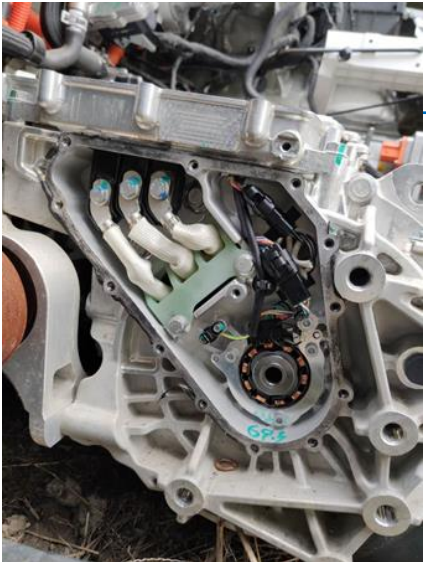
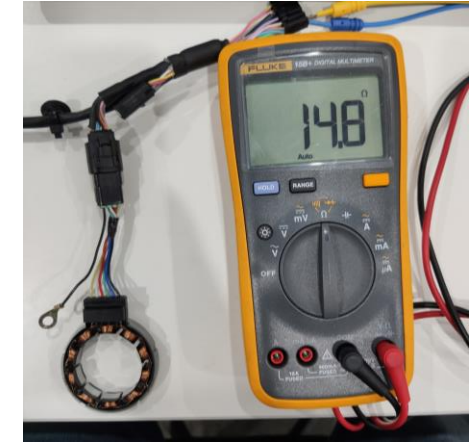


Drive Motor System Operation

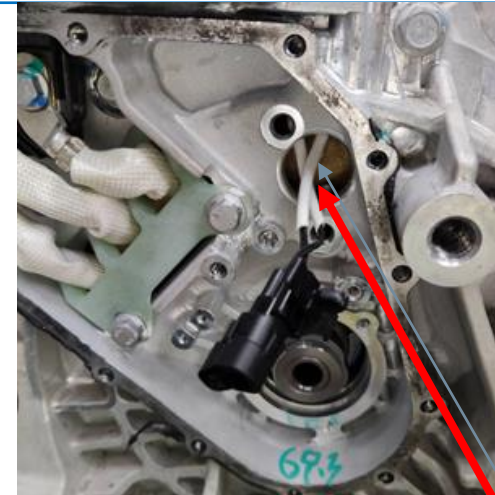
Resolver Sensor Measurement

- ◆ Sine winding resistance: 42.6 Ω
- ◆ Cosine winding resistance: 36.7 Ω
- ◆ Excitation winding resistance: 14.8 Ω

(Note: The measured values below are for factory test parts, for reference only)



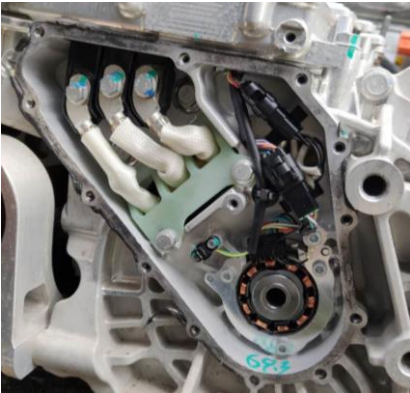
Resolver sensor



Motor temperature sensor 1: 1.078 K Ω

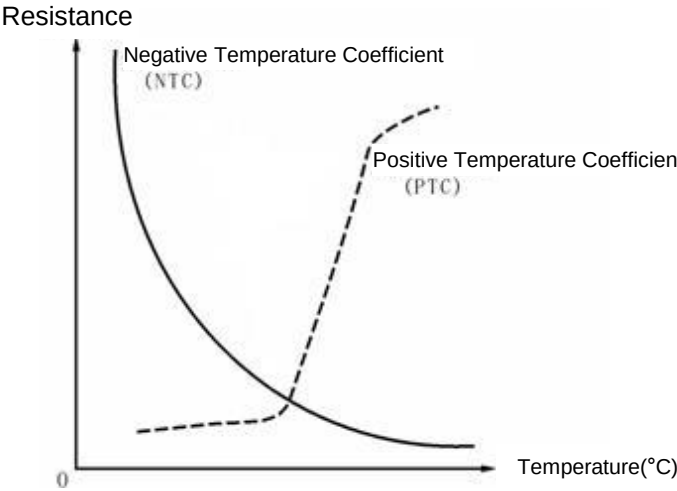
Drive Motor System Operation

➤ Temperature Sensor



Measurement value of temperature sensor: 106.8 KΩ

Temperature (°C)	-20	-15	-10	-5	0	5	10	15	20	25	30	35	40
Resistance (KΩ)	0.932	0.942	0.962	0.981	1	1.019	1.039	1.058	1.077	1.096	1.039	1.058	1.154
Temperature	45	50	55	60	65	70	75	80	85	90	95	100	105
Resistance (KΩ)	1.173	1.193	1.212	1.231	1.25	1.27	1.289	1.308	1.327	1.347	1.366	1.385	1.44



Drive Motor System Operation

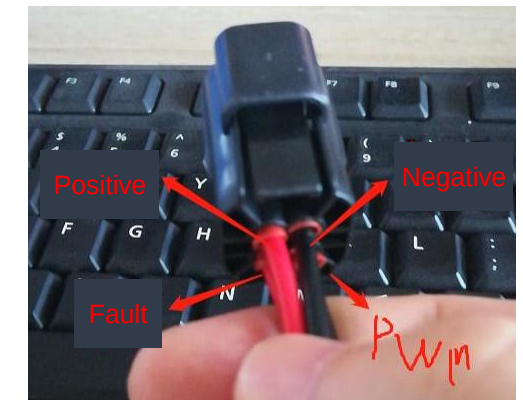
➤ Cooling System Control Strategy

Items		Temperature Valve Opening Value	Temperature Valve Closing Value	Note
Low speed fan	Drive motor	75℃	70℃	When motor stalls and compressor operates
	Motor controller	70℃	65℃	
	DC/DC controller	70℃	65℃	
	Charger	60℃	55℃	
High speed fan	Motor	85℃	70℃	When medium voltage switch is closed
	Motor controller	80℃	70℃	
	Charger	65℃	70℃	
Cooling water pump	Drive motor	45℃	< 45℃	EP31 is the VCU output duty ratio control, which realizes the PWM adjustment of pump speed/coasting or the pump operates at full speed when the motor stalls
	Motor controller	40℃	< 40℃	
	DC/DC	45℃	40℃	
	Charger	45℃	40℃	

Common System DTCs and Cases

➤ Motor System Malfunction Cases

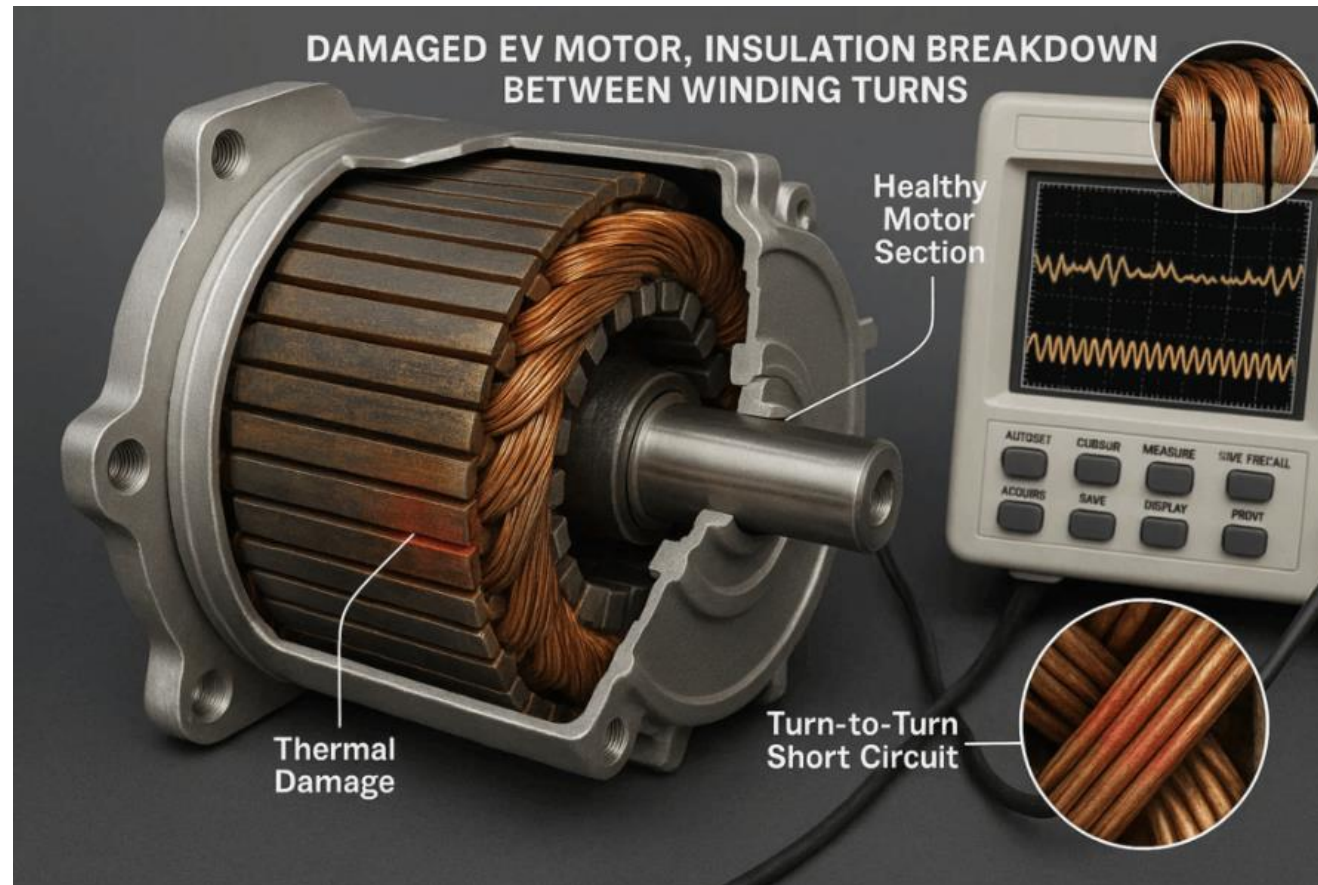
- ◆ **Malfunction symptom:** Motor system fault is displayed on instrument cluster, and vehicle cannot be accelerated
- ◆ **Malfunction classification:** Drive motor system malfunction
- ◆ **Malfunction description:** Drive motor fault is displayed on instrument cluster suddenly while driving normally, then vehicle cannot be accelerated and there is no response when depressing accelerator pedal, vehicle speed is reduced to 20 km/h, but electrical equipment in vehicle operates normally
- ◆ **Malfunction repair:** Motor fault is displayed on instrument cluster when vehicle speed is limited in 20 km/h. It can be preliminarily judged that the motor is in limp mode and output power is limited. It is a power limitation strategy applied to motor, mainly including motor overtemperature and MCU overtemperature
 1. Check front compartment and find that the fan is operating at a high speed
 2. Motor water pump does not operate
 3. Motor body is hot
 4. Read data stream and it displays that the temperature of MCU main board is 97°C
- ◆ **Troubleshooting:** Confirm that the fault is caused by water pump not operating. Its connector is 4-wire (refer to Cooling System Control for details). In addition to the power ground wire, there is also PWM input control, fault feedback. Check if water pump power supply ground circuit is normal. Water pump resumes operating after disconnecting PWM wire harness terminal. Confirm that VCU controller PWM signal output is normal. The fault is solved after replacing an electronic water pump.



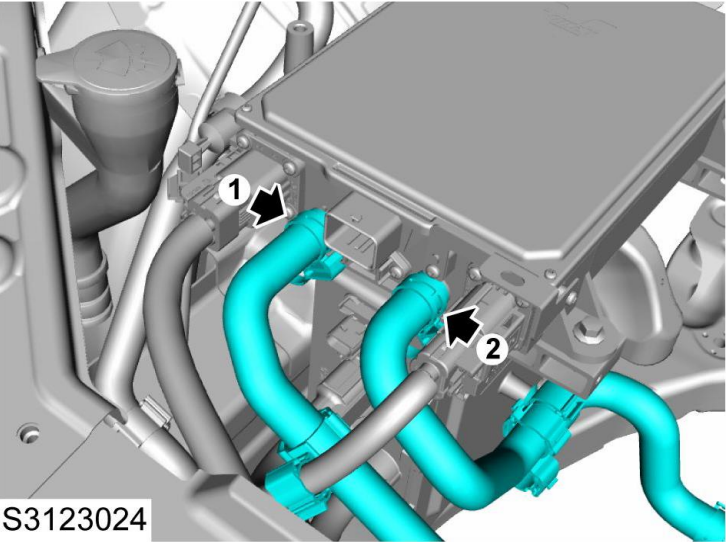
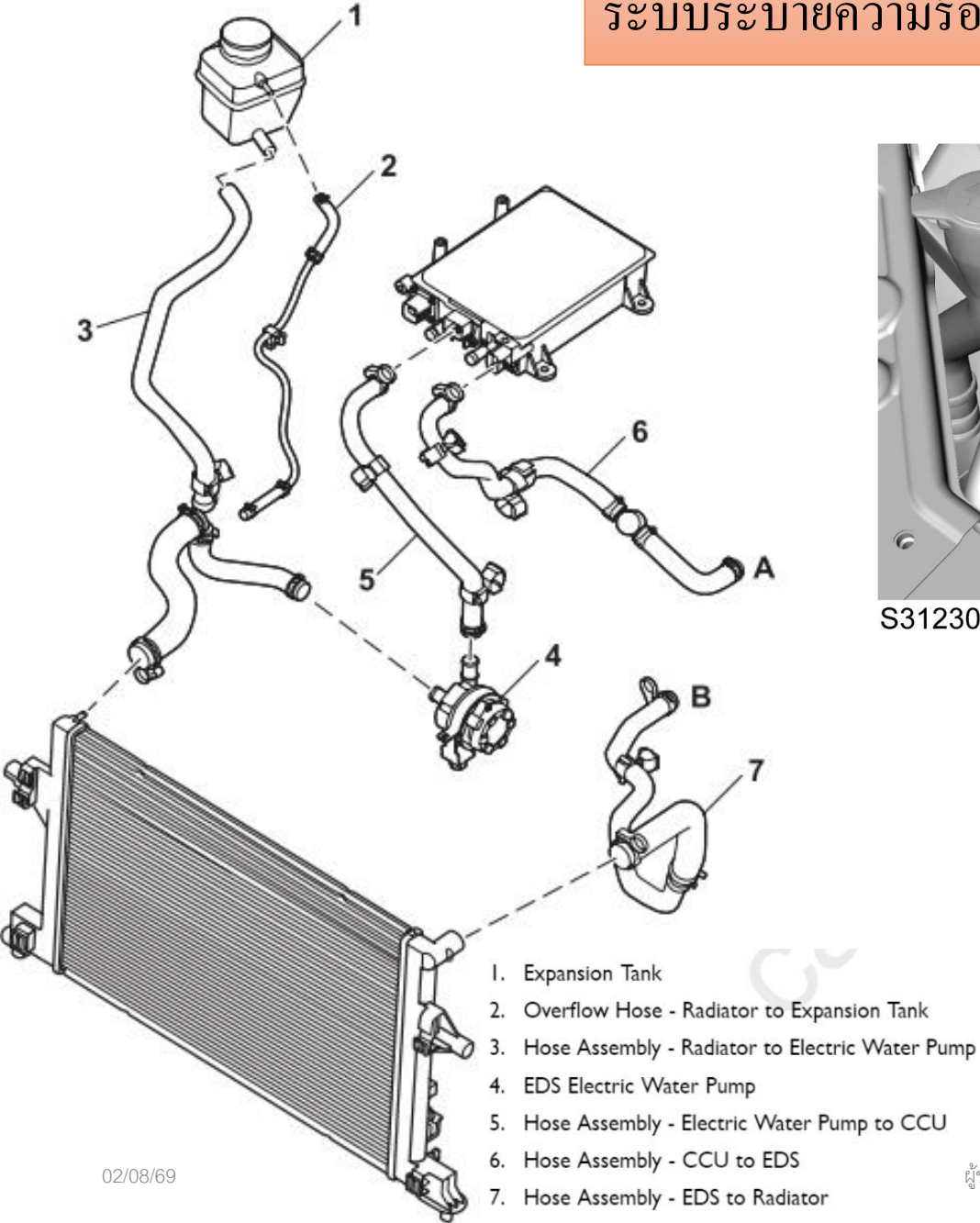
Common EV Motor Failure Modes and Detection

Winding Insulation Failures

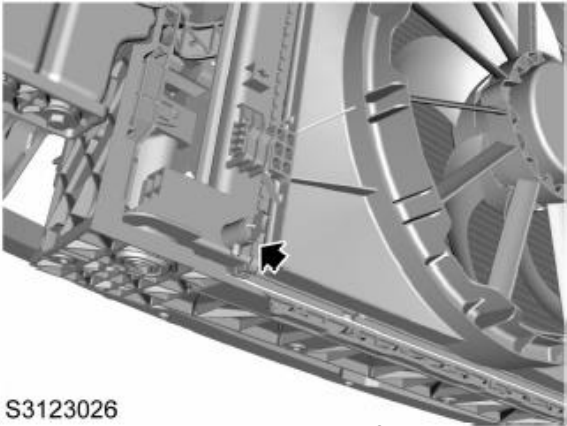
- Turn-to-turn shorts
- Phase-to-phase failures
- Ground faults



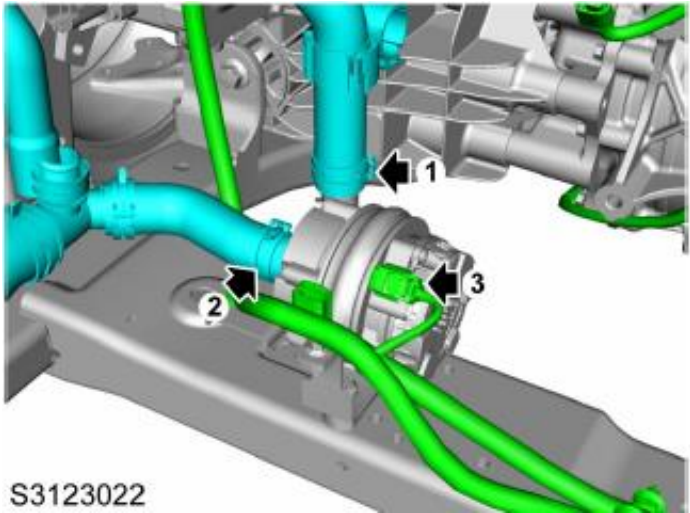
ระบบระบายความร้อนระบบขับเคลื่อนและระบบชาร์จ



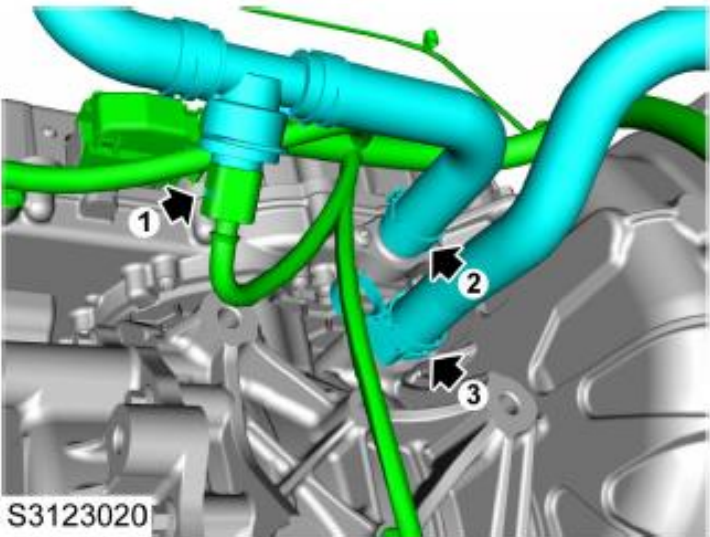
Charger cooling hose



Drain valve

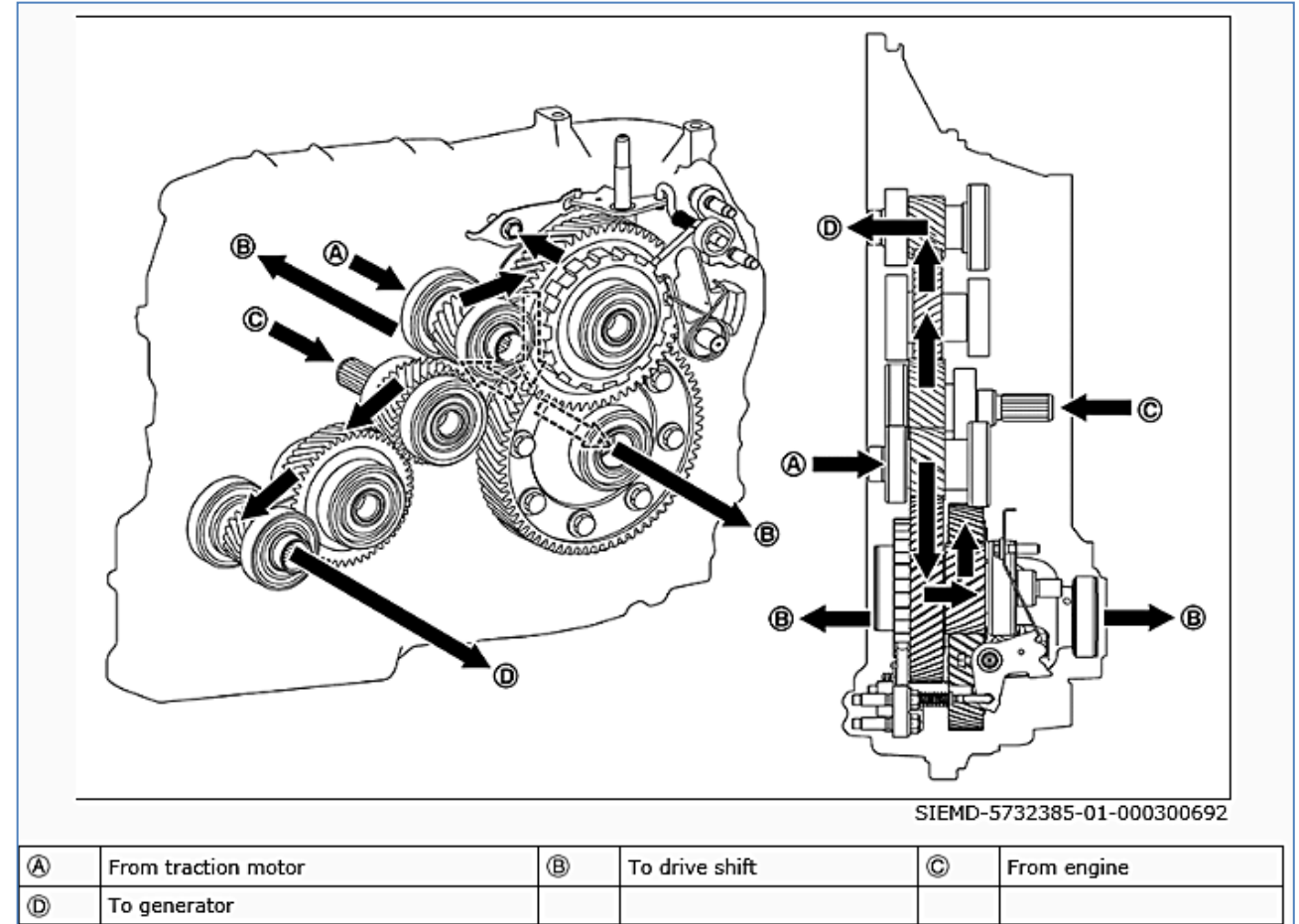
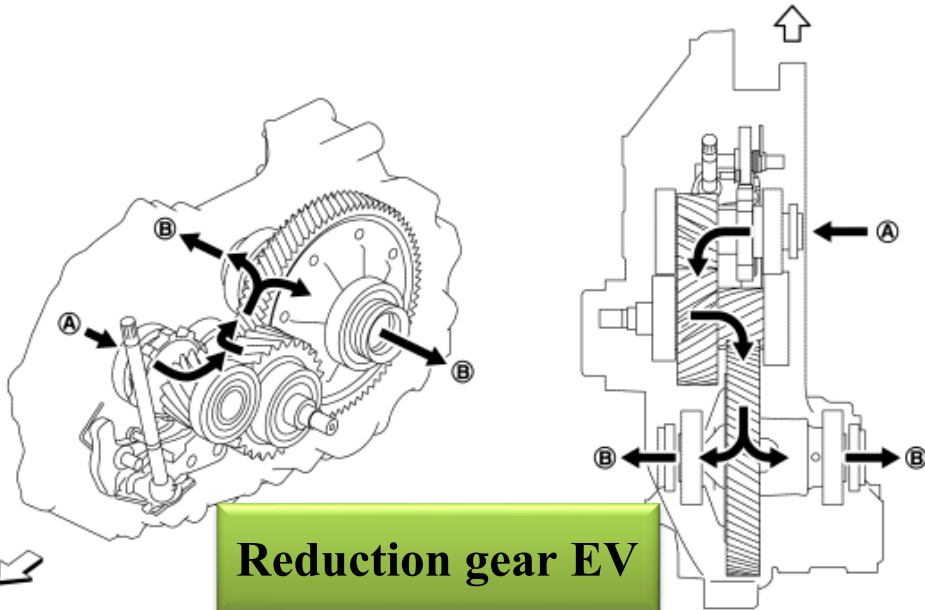
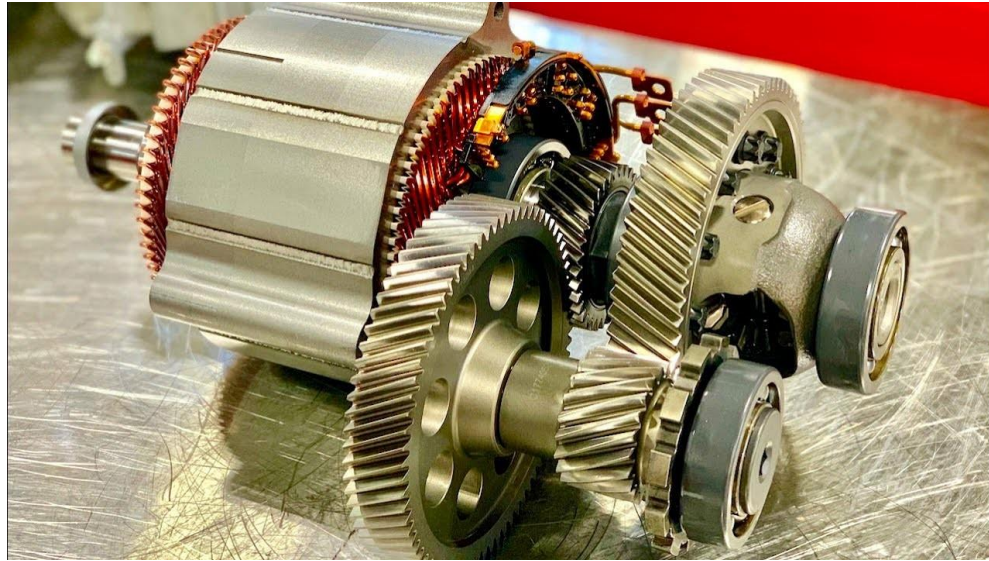


Electric water pump



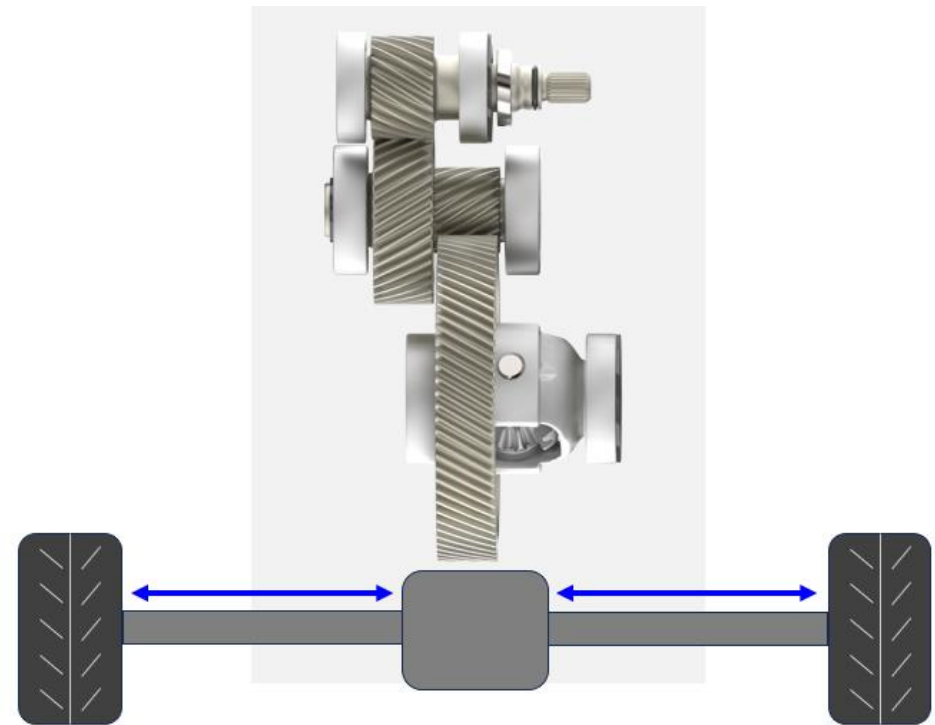
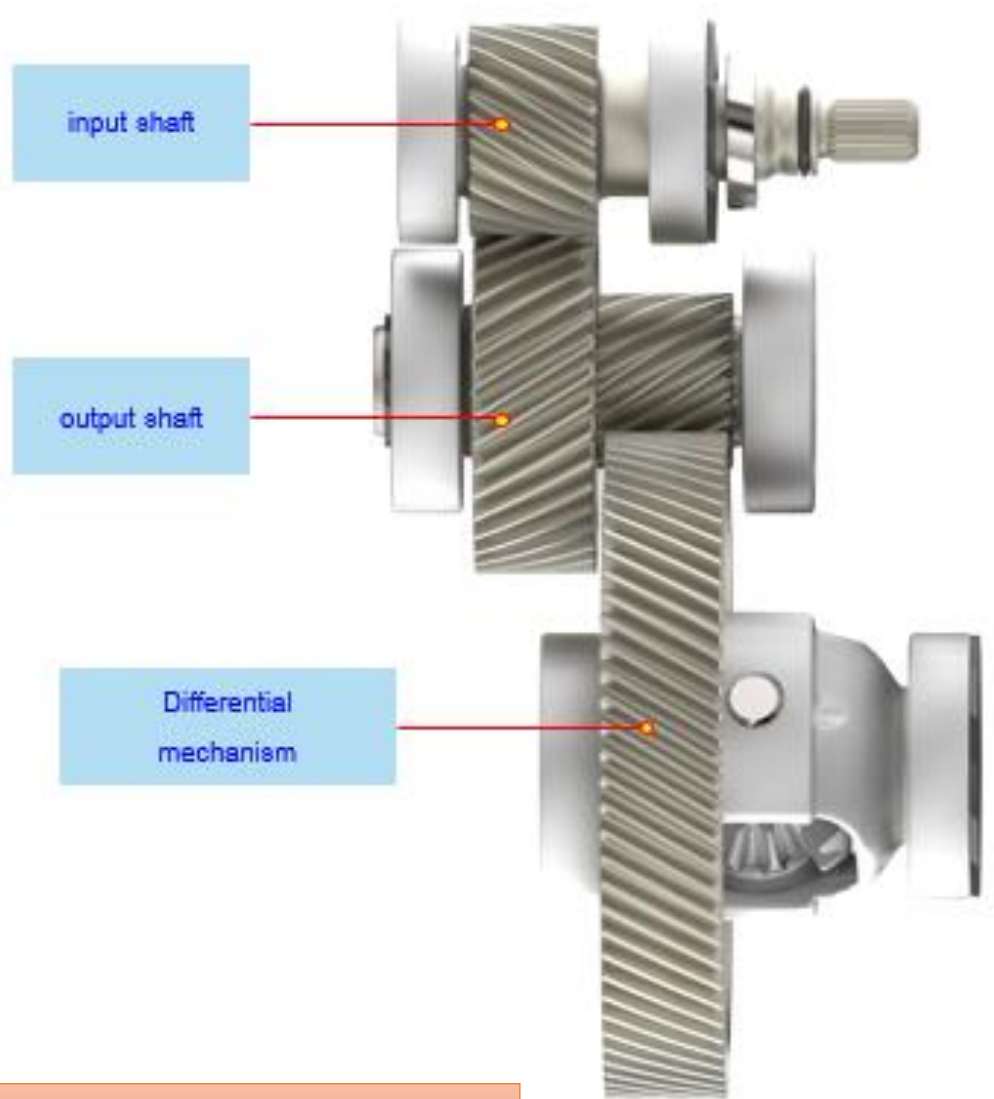
Temp.sensor and EDS cooling

BEV Transmission System : Single speed reduction gear



Reduction and multiply gear HEV

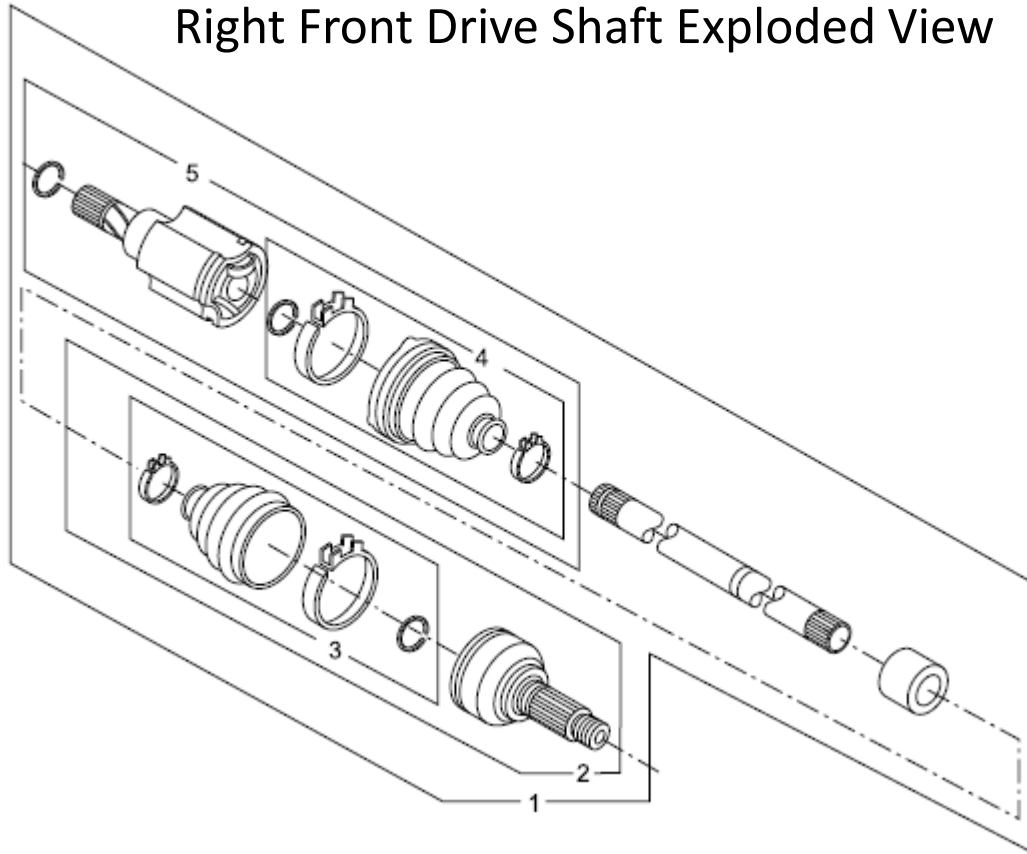
BEV Transmission System : Single speed reduction gear



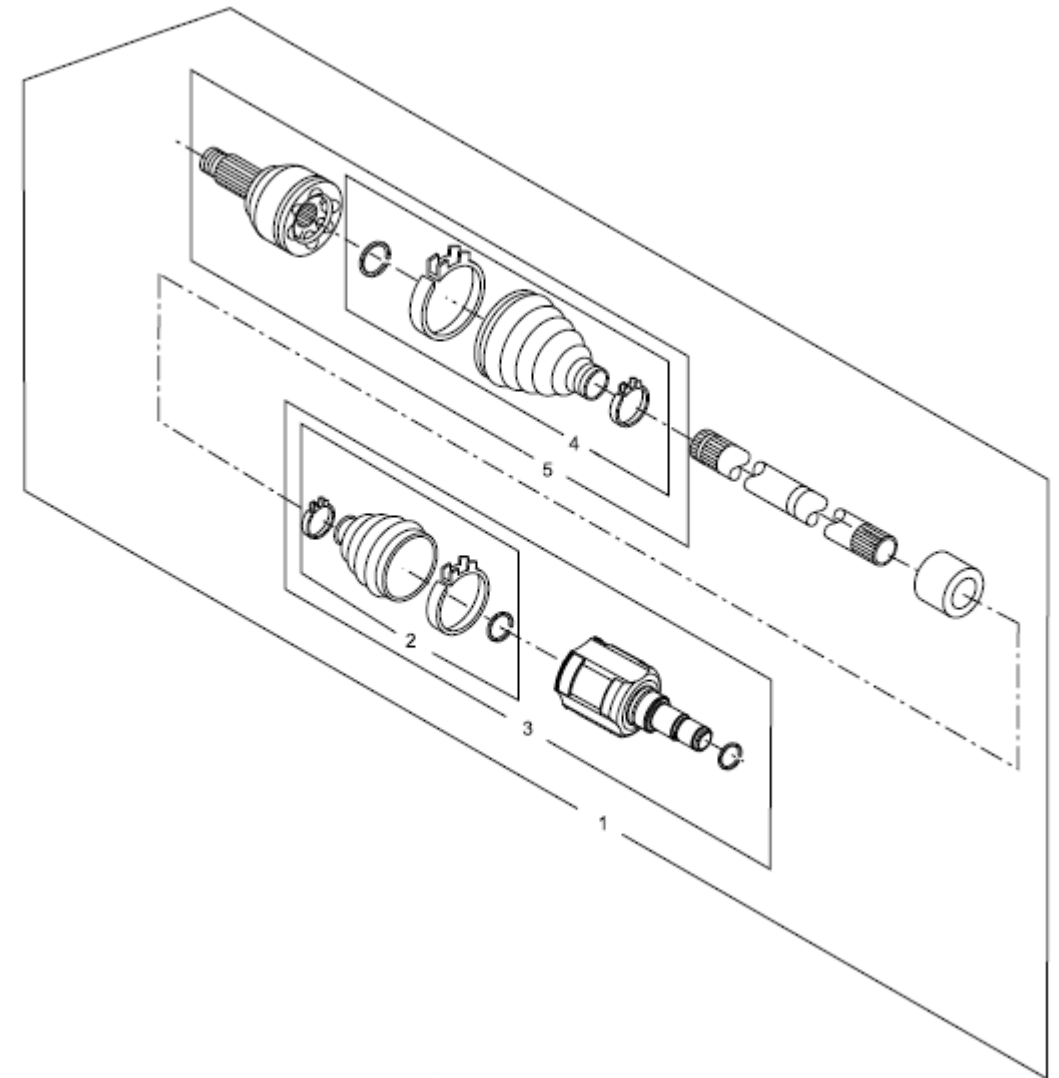
Single speed reduction gear

Driveline/Axle

System Layout
Right Front Drive Shaft Exploded View

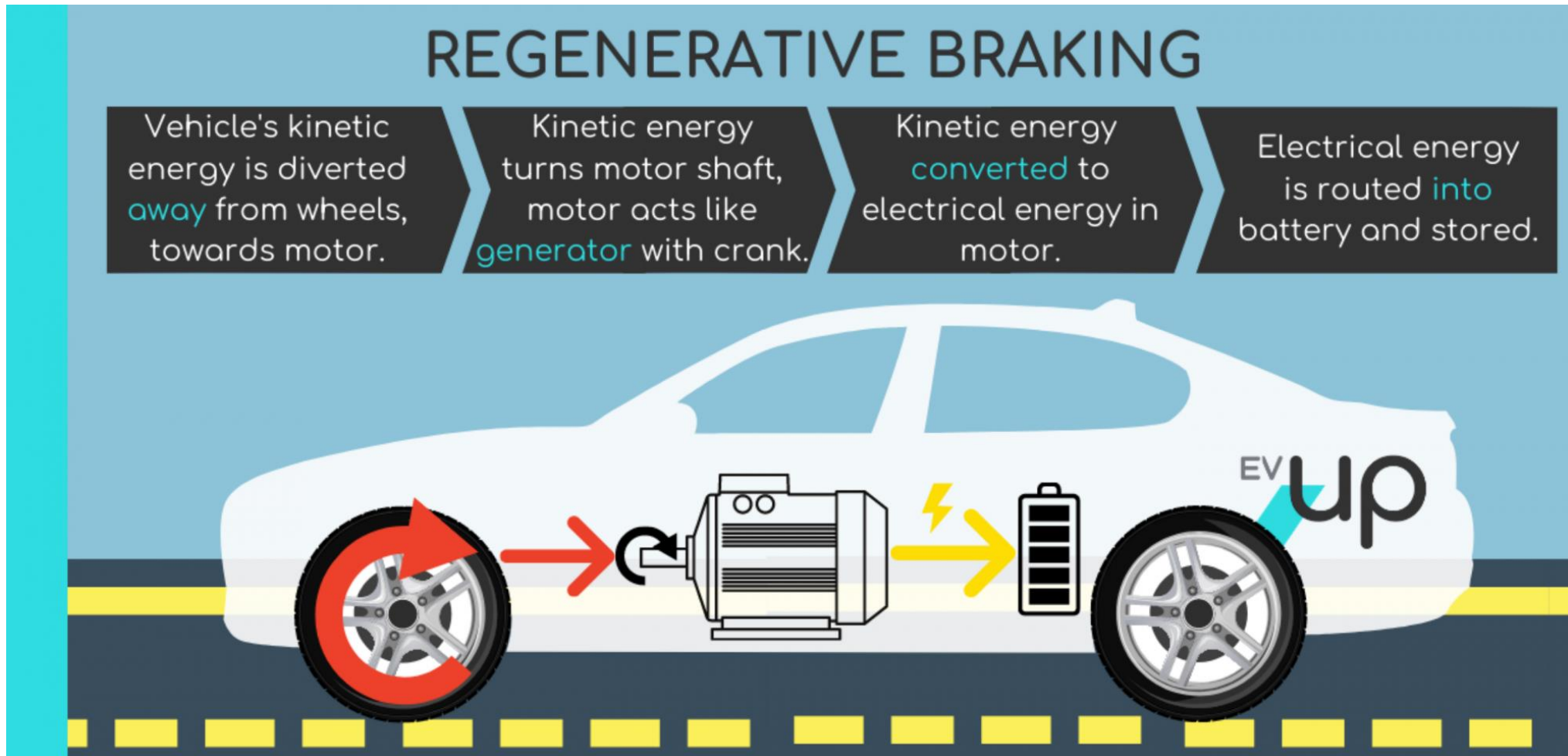


1. Left Drive Shaft Assembly - Front Wheel
2. Left Outer Joint Assembly Service Kit - Front Wheel
3. Left Outer Joint Sleeve Service Kit - Front Wheel
4. Left Inner Joint Sleeve Service Kit - Front Wheel
5. Left Inner Joint Assembly Service Kit - Front Wheel



1. Right Drive Shaft Assembly - Front Wheel
2. Right Inner Joint Sleeve Service Kit - Front Wheel
3. Right Inner Joint Assembly Service Kit - Front Wheel
4. Right Outer Joint Sleeve Service Kit - Front Wheel
5. Right Outer Joint Assembly Service Kit - Front Wheel

Regenerative Braking



<https://www.evup.com.au/about-evup/ev-news/how-does-regenerative-braking-work>

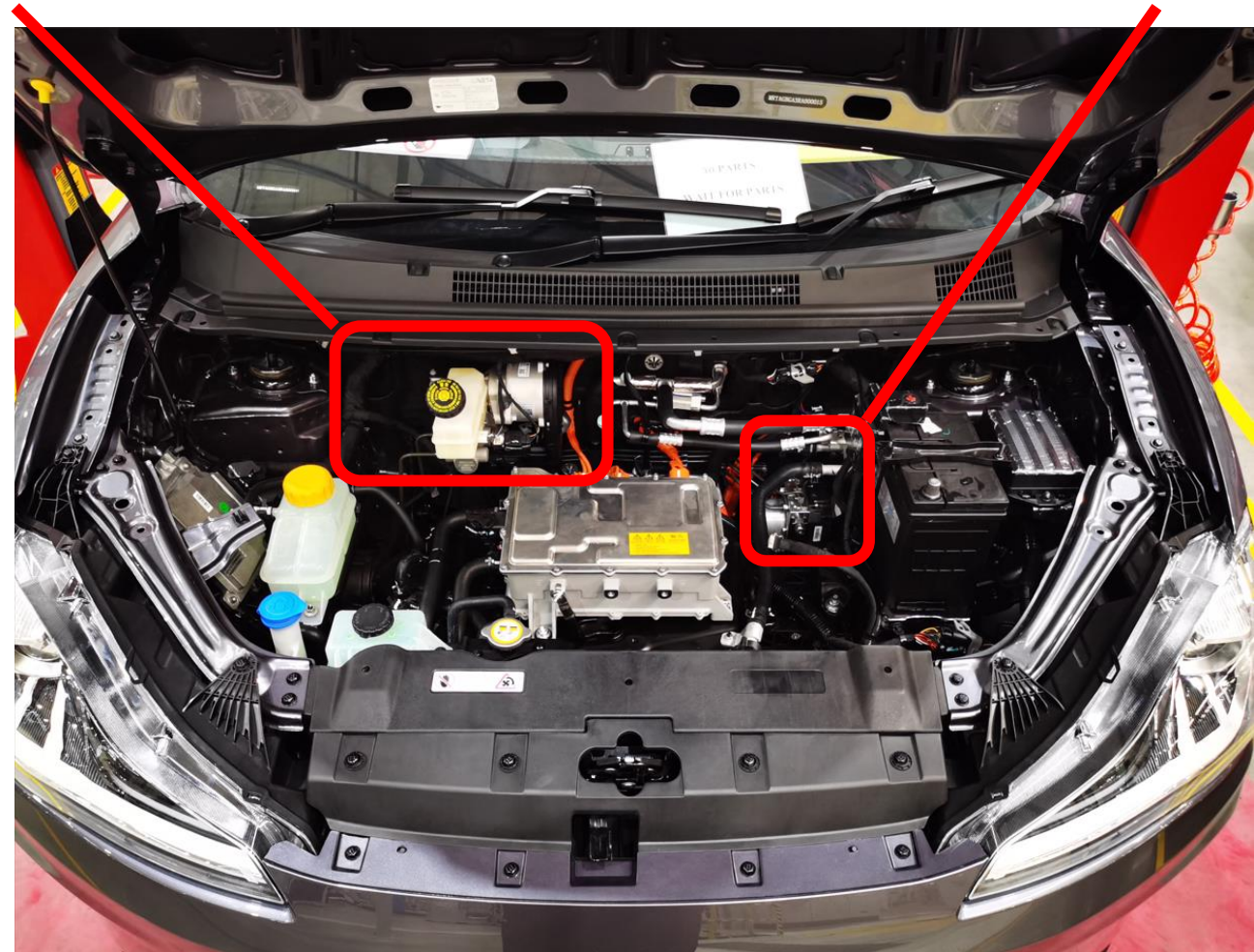
Brake system : NETA V

Introduction to brake system

- The main function of the brake system is to slow down or even stop a moving vehicle, to keep the speed of a downhill vehicle stable and to keep a stopped vehicle stationary. It is the most important active safety system in a vehicle. According to the function, it is divided into parking part and driving part.
- Neta V-II adopts the X-type dual-circuit hydraulic brake system, with front and rear disc brakes, EPBi, EHB and ESC.

EHB-EPBi

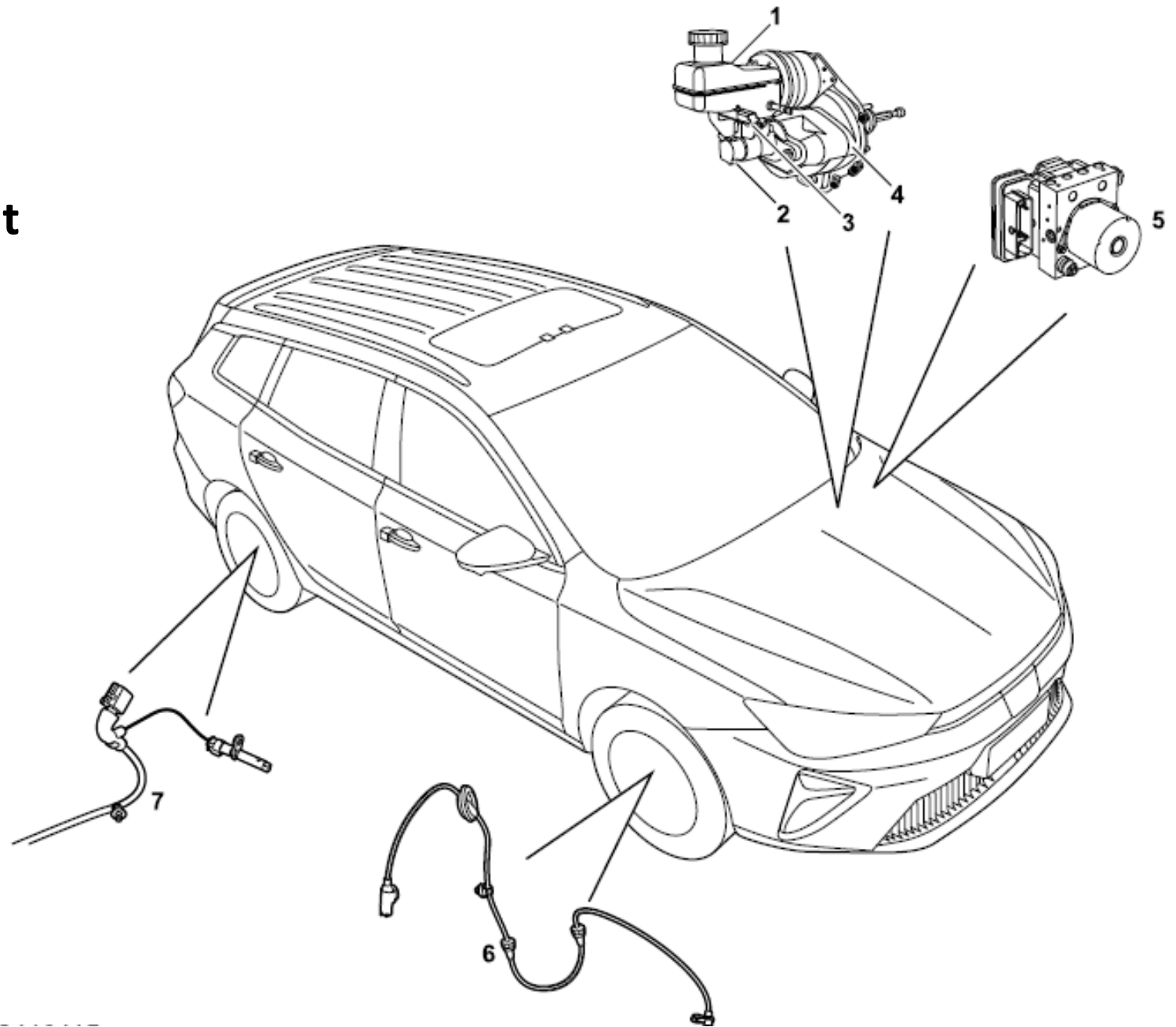
ESC



Brake System : MG EP

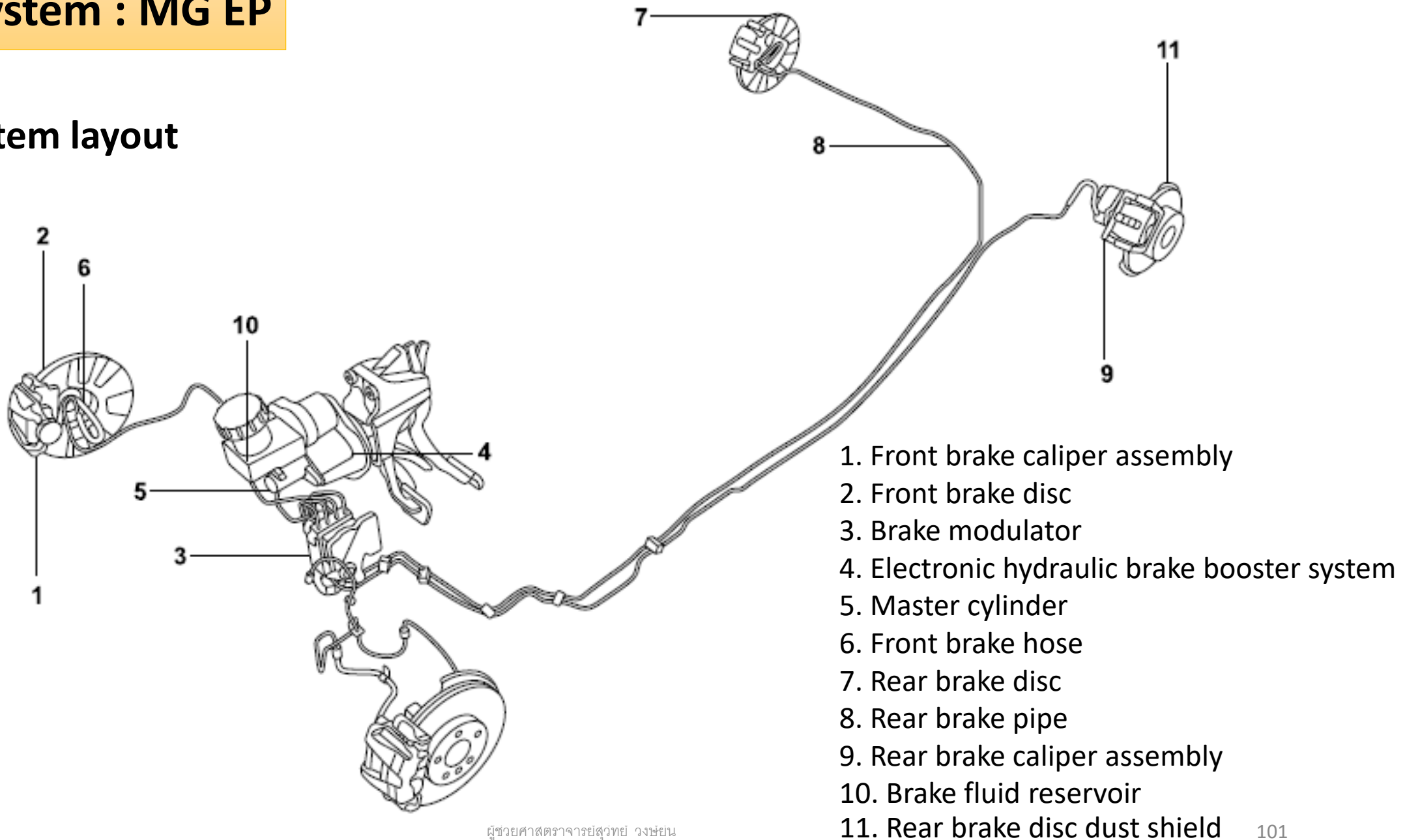
Brake Modulator System Layout

1. Brake fluid reservoir
2. Master cylinder
3. Brake fluid level switch
4. Electronic hydraulic brake booster
5. Brake modulator
6. Front wheel speed sensor
7. Rear wheel speed sensor



Brake System : MG EP

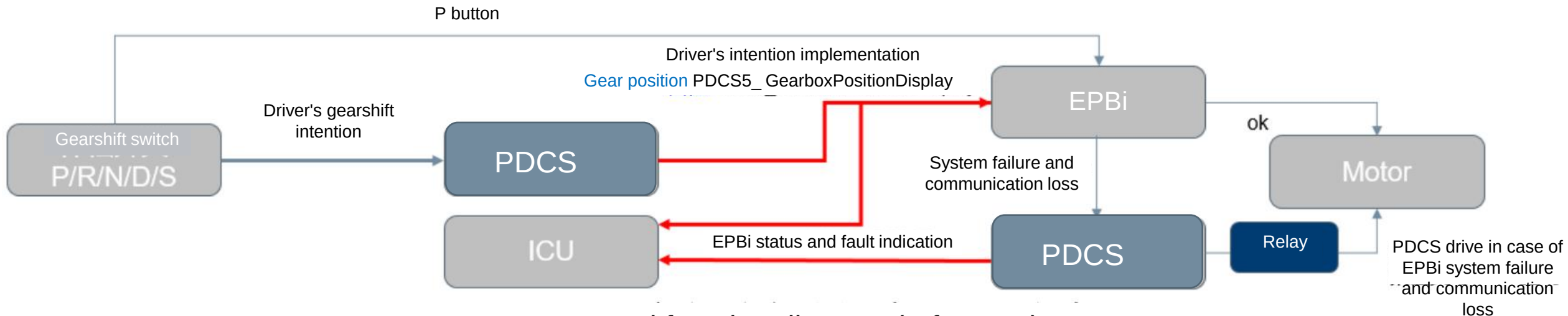
Brake system layout



Brake system: EPBi

EPBi basic function principle

- The driver sends the gearshift intention to PDCS through the gearshift switch, and PDCS sends the gear position signal to EPBi through CAN communication; at the same time, EPBi also directly receives the P position signal from the gearshift switch.
- EPBi self-test is normal and fault-free, and EPBi drives the EPBi motor normally. At the same time, the EPBi status signal is fed back to ICU for interactive display.
- If any fault is found in EPBi self-test, it is fed back to the PDCS via CAN communication, and PDCS drives the EPBi motor via the backup parking brake relay. At the same time, EPBi and MCU will feed the relevant fault information back to ICU to illuminate the fault indicator light.

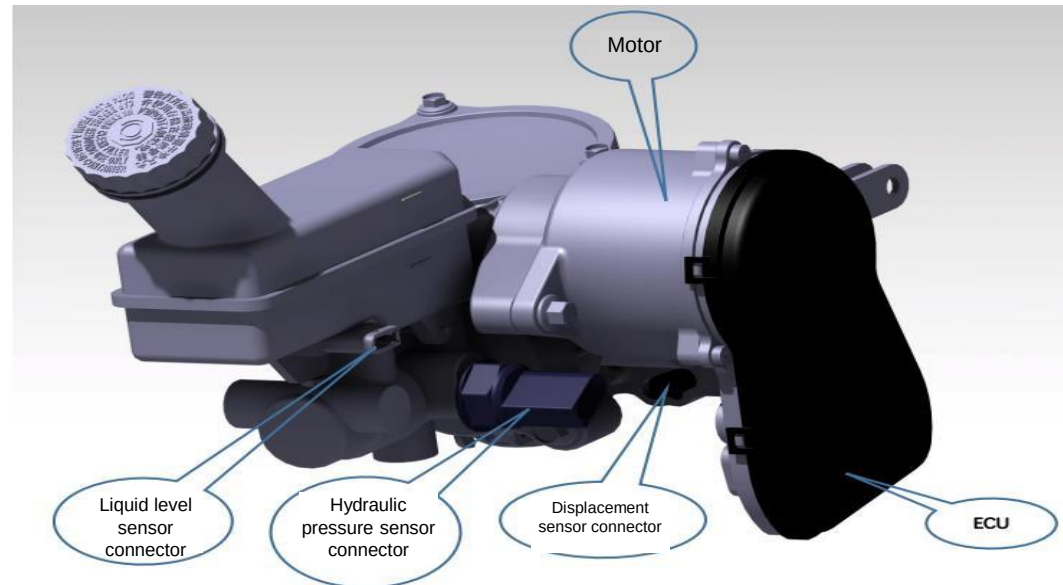
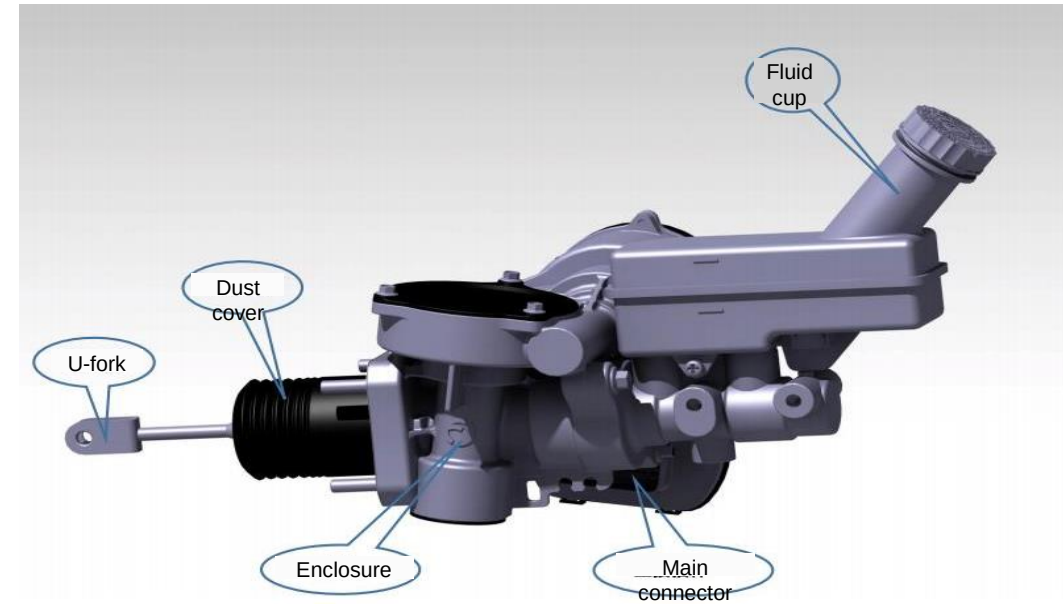


EPBi function diagram (reference)

Brake system: EHB

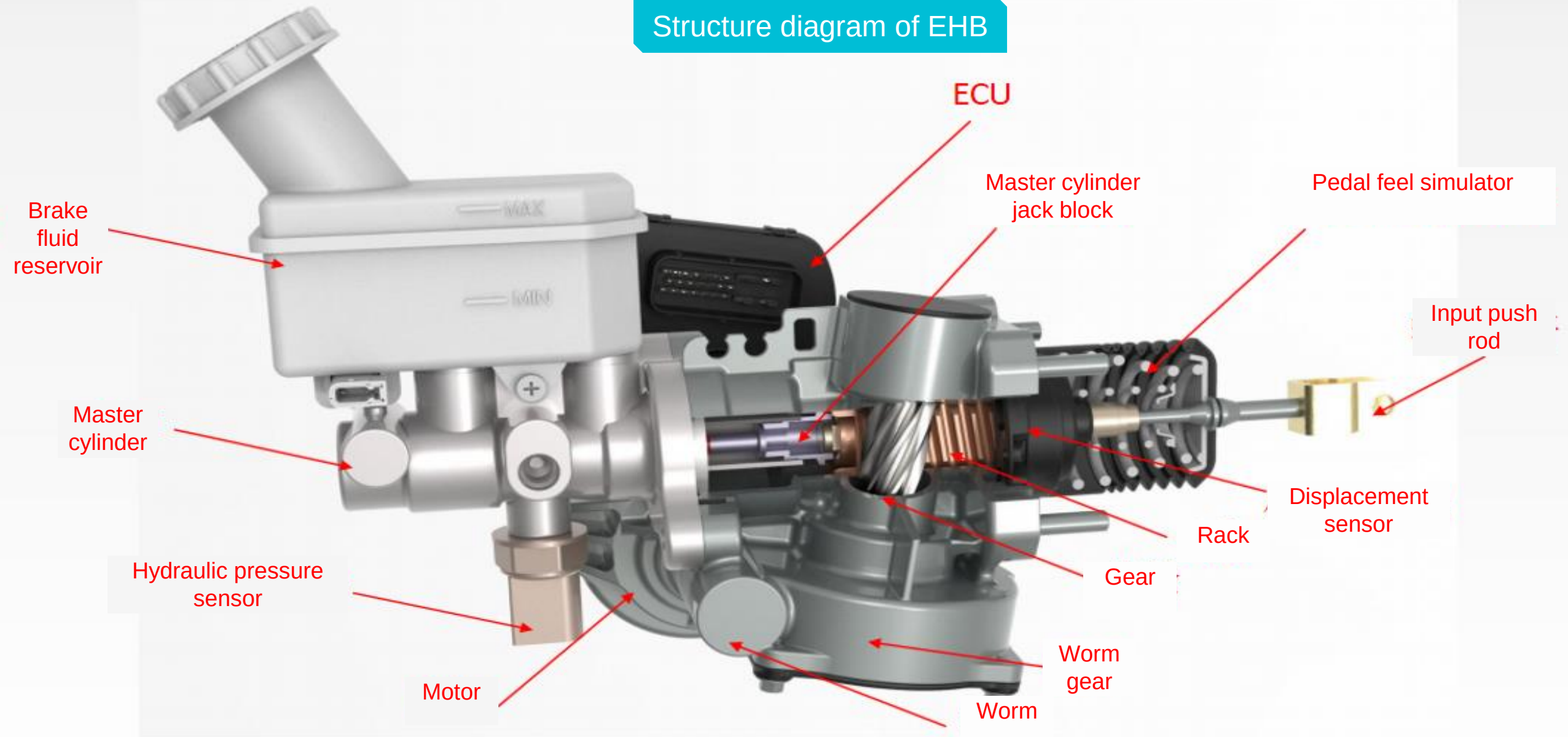
Electro-hydraulic brake (EHB) system

- ◆ The electro-hydraulic brake (EHB) system (referred to as EHB for short) adopts some hydraulic braking components of the conventional hydraulic brake system and has an additional set of electronic control units to maximize the performance of the brake system. It replaces the conventional vacuum-based brake booster with motor-driven booster by replacing some mechanical components in the traditional brake system with electronic components.
- ◆ EHB includes the pedal feel simulator assembly, motor and controller assembly, displacement sensor, liquid level sensor, hydraulic pressure sensor, transmission assembly and brake master cylinder. EHB adopts a full decoupling scheme to achieve the brake-by-wire function and adjustable pedal feel function.
- ◆ At present, the whole series of Neta V adopt the EHB.



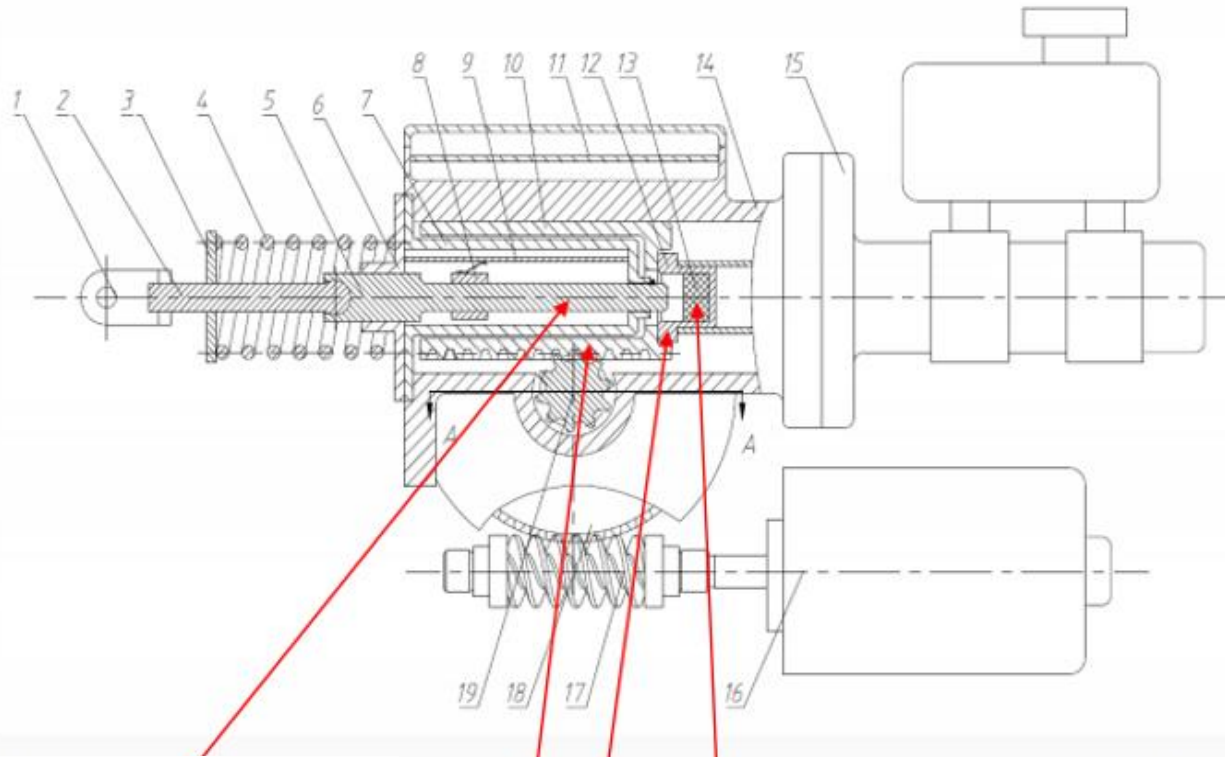
Brake system: EHB

Structure diagram of EHB



Brake system: EHB

Basic structure diagram of EHB

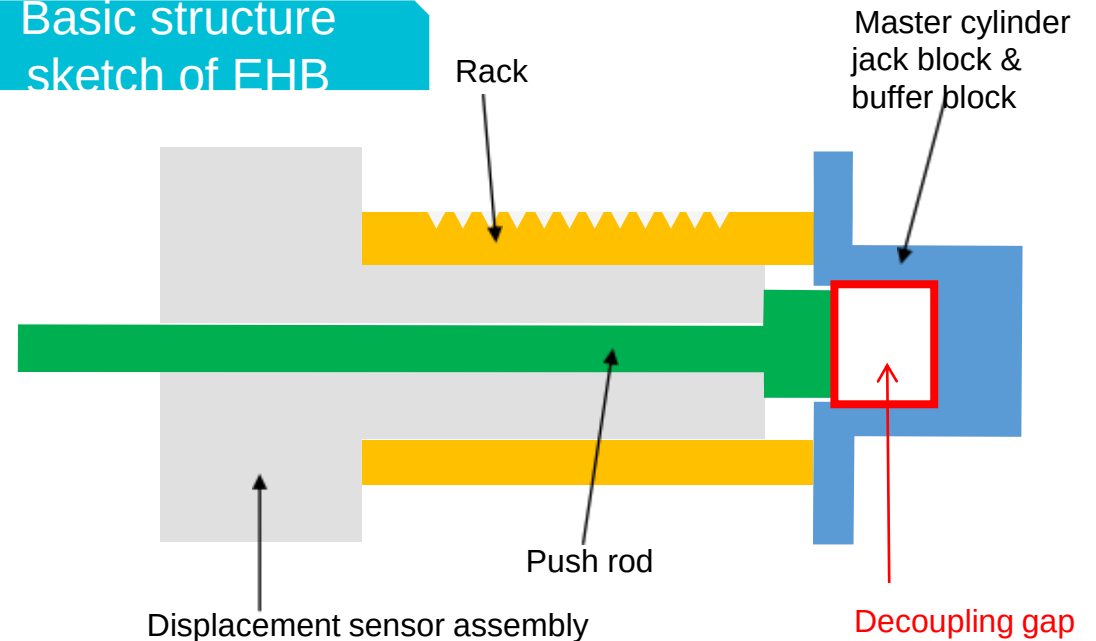


- 1. U-fork
- 2. Ball joint push rod
- 3. Spring seat
- 4. Pedal feel spring
- 5. Push rod
- 6. Sensor mount
- 7. Sensor shell
- 8. Sensor slider

- 9. Sensor PCB
- 10. Rack
- 11. ECU PCB
- 12. Master cylinder jack block
- 13. Decoupling chamber buffer block
- 14. Enclosure
- 15. Brake master cylinder
- 16. Motor

- 17. Worm
- 18. Worm gear
- 19. Gear shaft

Basic structure sketch of EHB

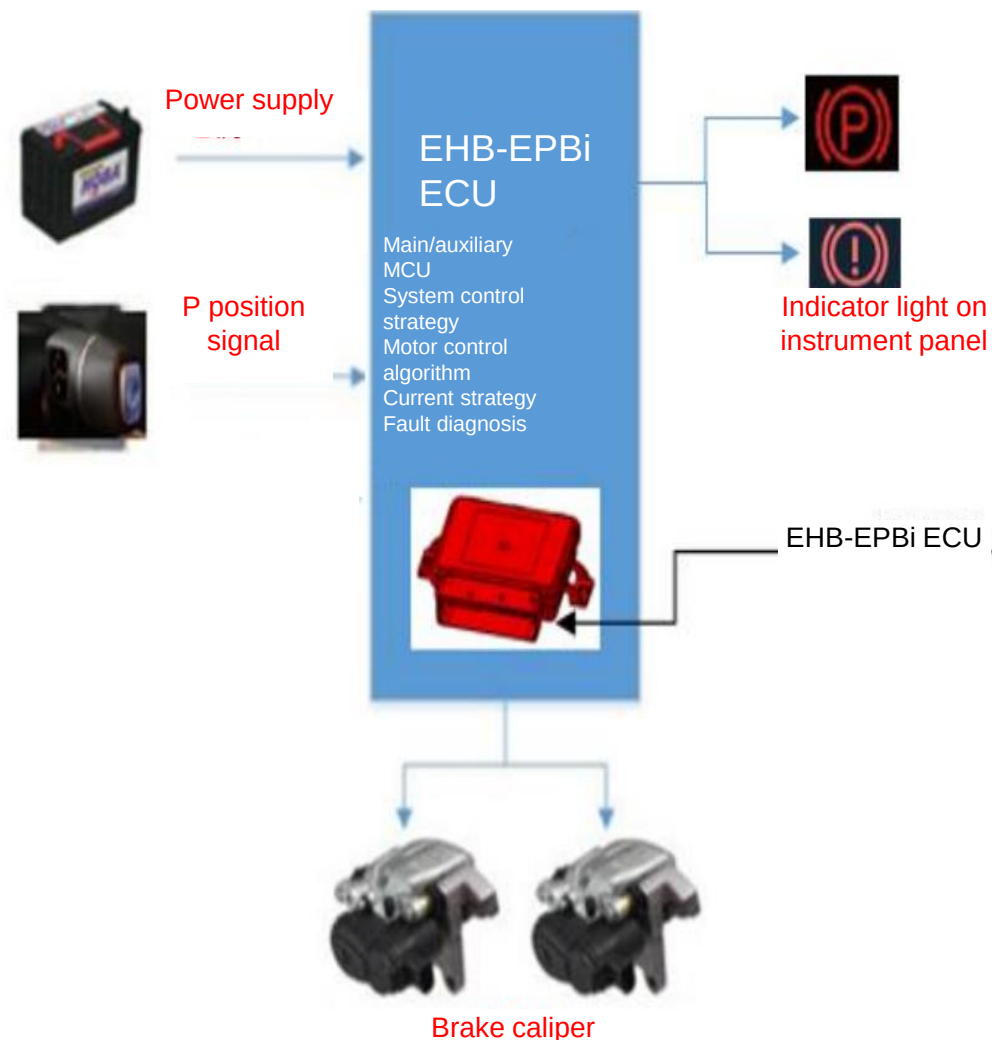


- ◆ Push rod: to transfer the brake pedal action
- ◆ Rack: to transfer the motor action
- ◆ Master cylinder jack block: to transfer the push rod/rack force to the master cylinder piston
- ◆ Decoupling chamber buffer block: to buffer the impact between the push rod and master cylinder jack block

Brake system: EPBi

Electric parking brake (EHB-EPBi)

- The EPBi of Neta V-II mainly consists of EPBi button, EPBi ECU and rear brake caliper with EPBi motor.
- According to the EPBi button hardwired signal or gear position and pedal position signals, the caliper motor is rotated forward or backward under the control of EPBi ECU to apply/release the brake. The vehicle battery is used as its power supply, and brake application/releasing can be realized only by controlling the amplitude and direction of caliper motor current/voltage without complicated control mechanism.
- According to "GB 21670-2008 Technical requirements and test methods for passenger car braking system", the brake system must provide a second set of braking device, by which the vehicle can be parked on a slope of 8% when fully loaded, in addition to the service brake. This function is known as "backup parking brake" which in Neta V-II is achieved by using PDCS module to control the right rear wheel.



The EPBi ECU is integrated in EHB

Brake system: EPBi

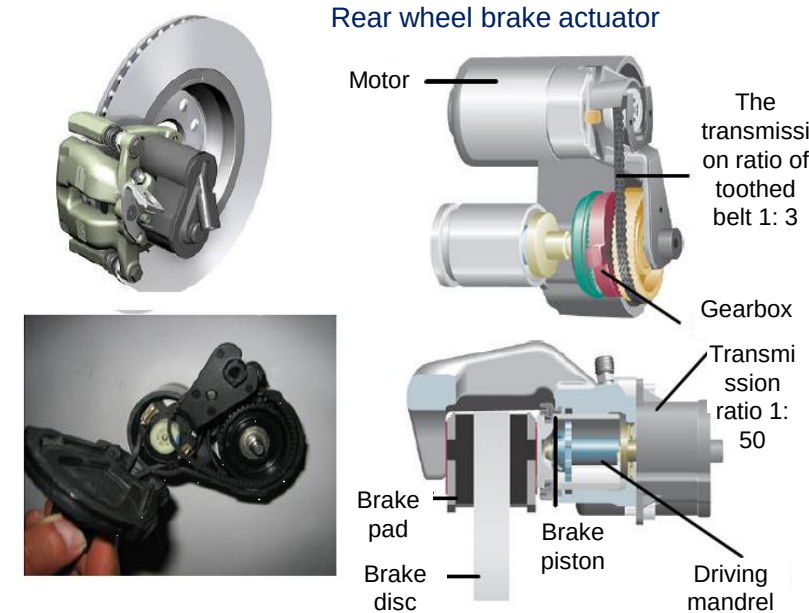
EPBi-related components



- The EPBi button of Neta V is integrated into the gearshift switch. Pressing the P button can trigger the EPBi signal.
- Under normal system conditions, shifting to another gear position will release the parking brake.
- A virtual EPBi button is provided in the drop-down menu on the center console display for interaction via CAN signals.



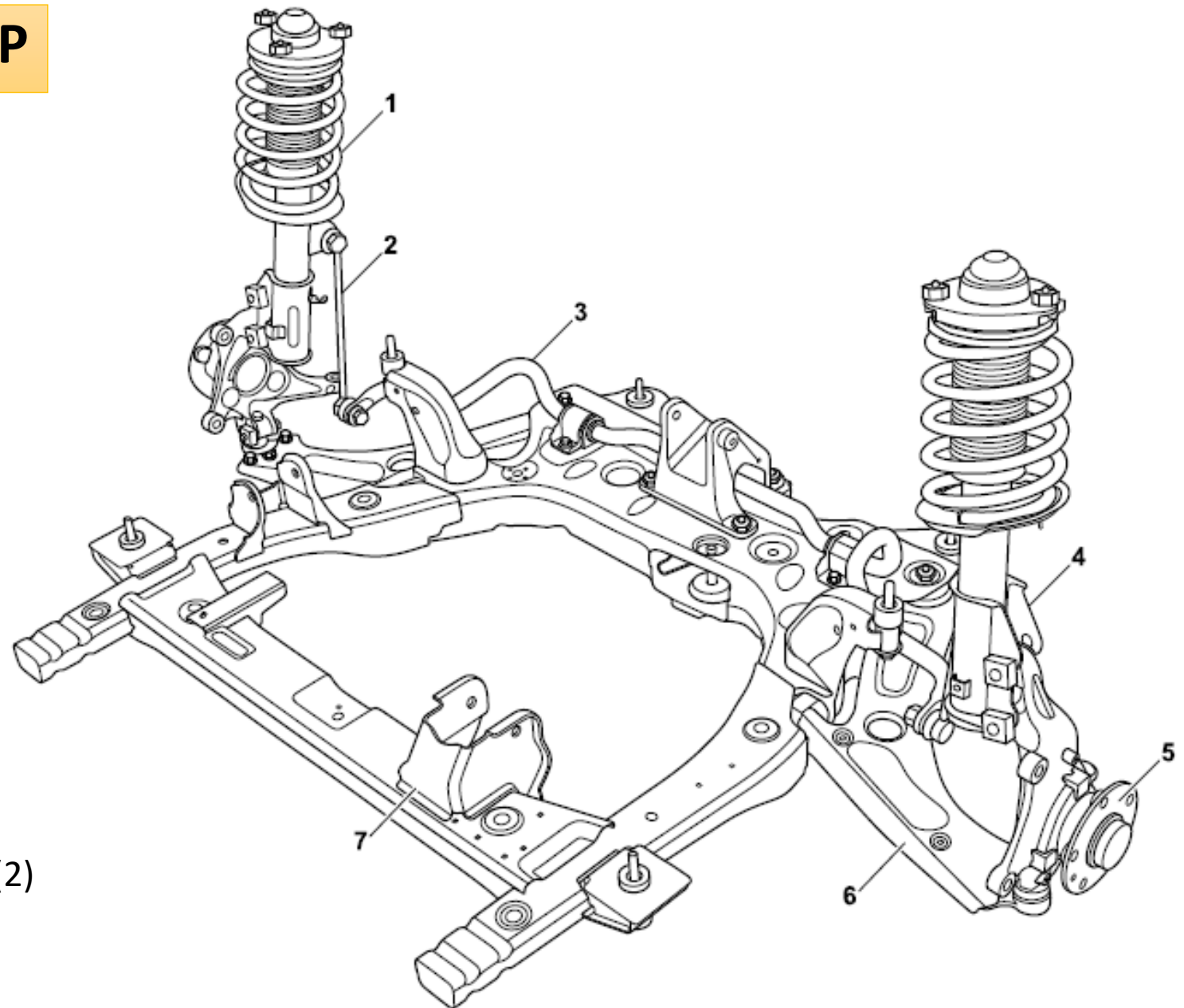
- When EPBi is applied, it will monitor the slope information and determine the clamping force required by the EPBi motor.
- EPBi has a built-in slope sensor. To ensure the accuracy of slope sensor, the EPBi should be installed parallel to the vehicle floor, so that the X-axis of the controller is perpendicular or parallel to the vehicle floor plane.
- **The EPBi ECU is integrated in EHB**



- The motor decelerates and increases torque through large and small gears, belts and gearbox, driving the mandrel to rotate, ejecting the brake piston, pushing the brake pad and clamping the brake disc to achieve the parking brake.
- When the brake disc is clamped, the driving mandrel stops rotating, which causes the motor to be unable to rotate and to enter the locked state, and the motor current becomes higher. When the EPBi ECU identifies the corresponding high current, it determines that the brake disc is clamped and stops the power output to the motor.

Suspension System :MG EP

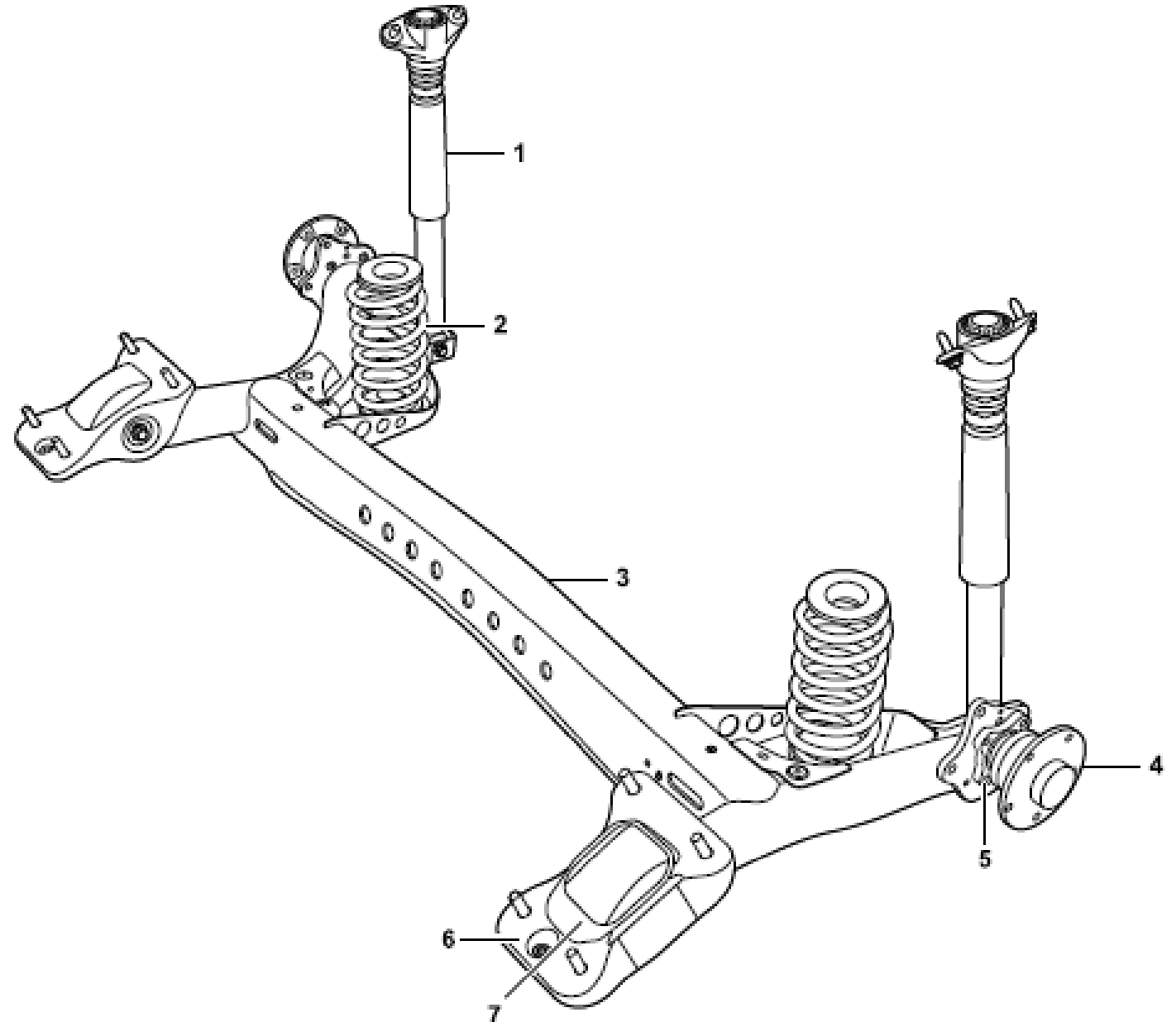
Front Suspension Layout



1. Front Shock Absorber Assembly (2)
2. Front Anti Roll Bar Link (2)
3. Front Anti Roll Bar
4. Front Swivel Hub Assembly (2)
5. Drive Flange and Hub Bearing Assembly (2)
6. Front Lower Arm Assembly (2)
7. Front Subframe

Suspension System :MG EP

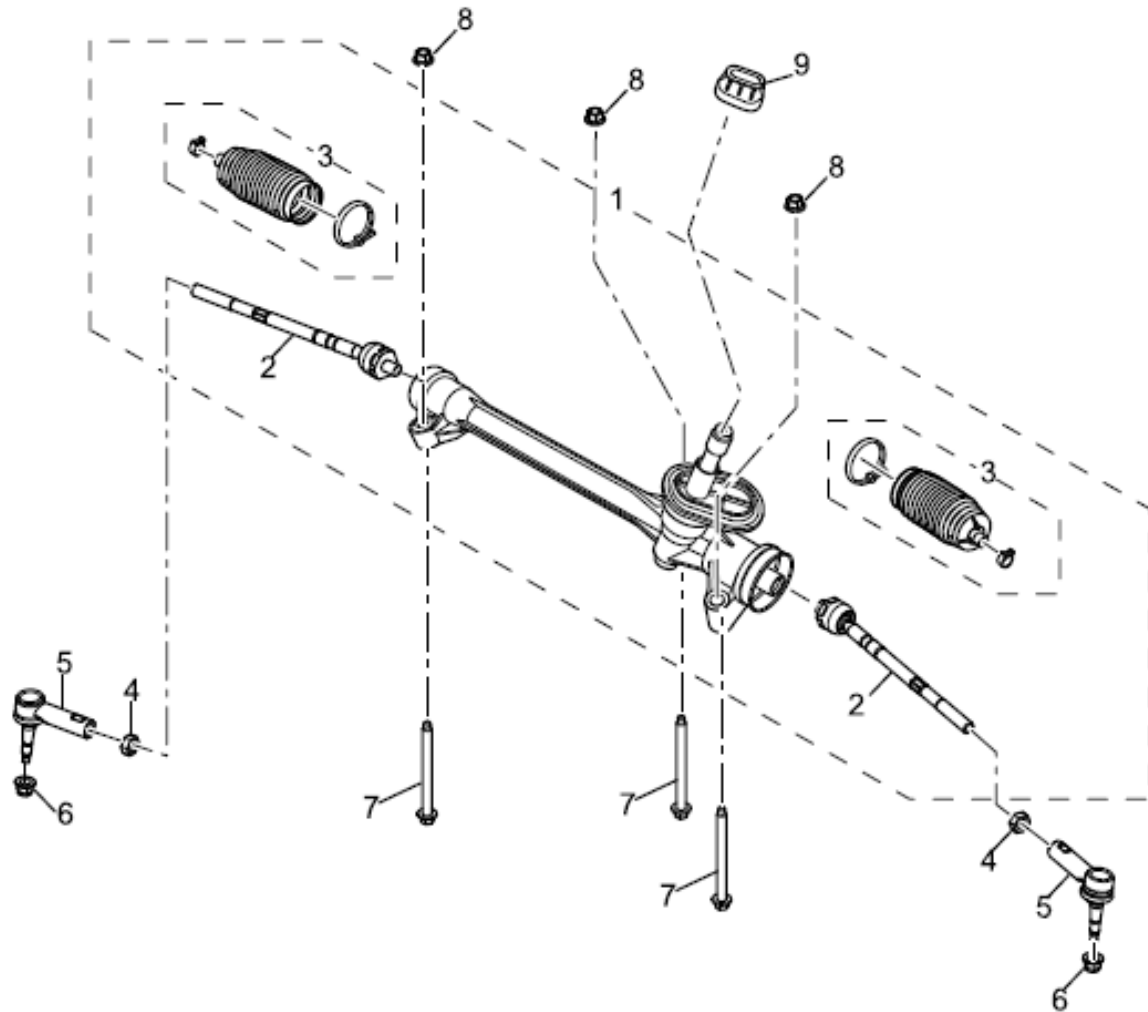
Rear Suspension Layout — Rear Shock Absorber



1. Rear Shock Absorber Assembly (2)
2. Rear Coil Spring Assembly (2)
3. Rear Torsion Beam Assembly
4. Rear Wheel Hub Bearing Assembly (2)
5. Rear Wheel Carrier (2)
6. Rear Torsion Beam Bracket (2)
7. Rear Torsion Beam Bracket Dust Panel (2)

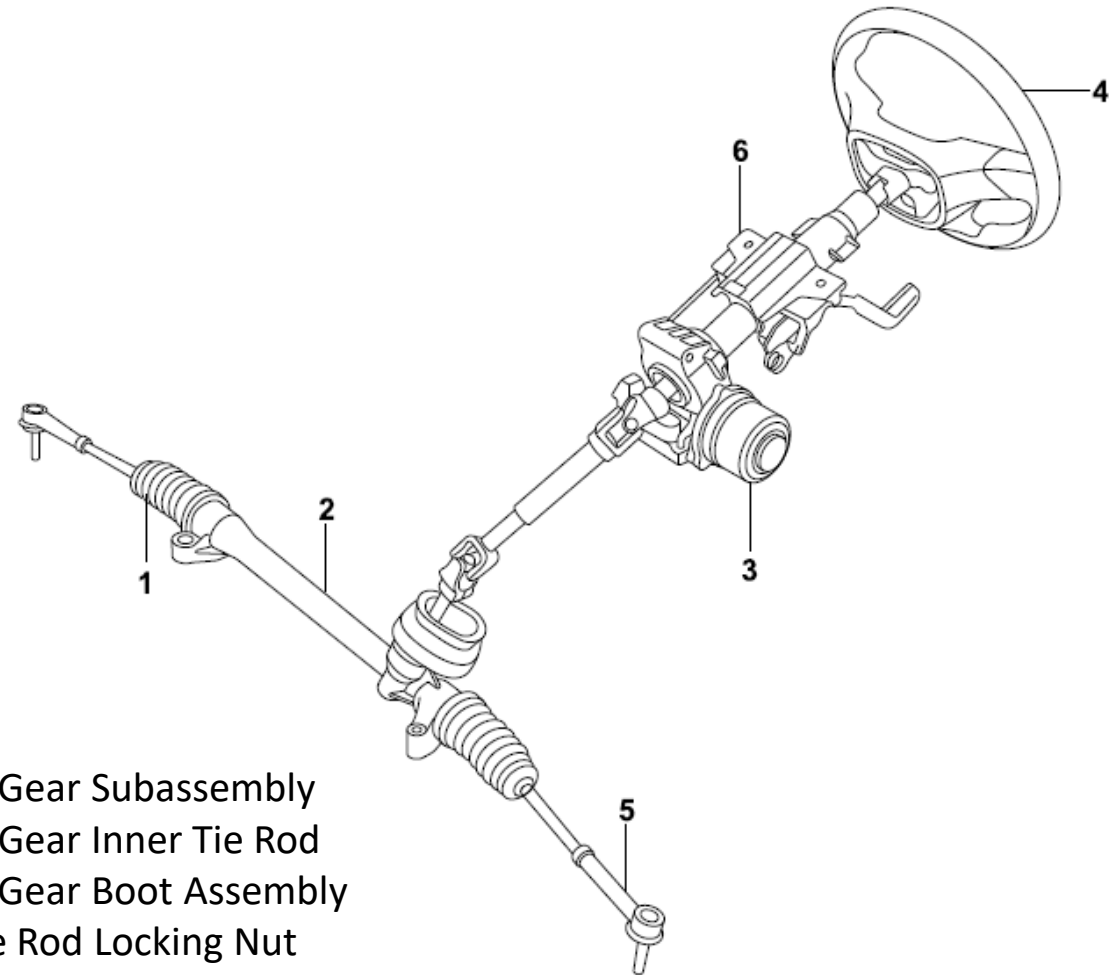
ระบบบังคับเลี้ยว(Steering System)

Steering System Component Layout



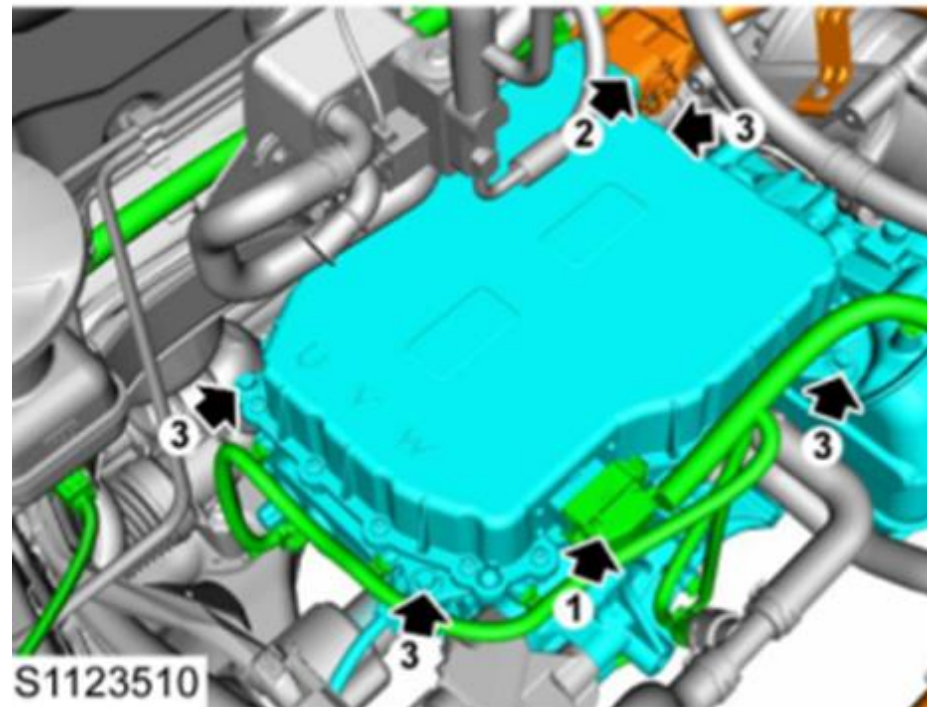
1. Steering Gear Subassembly
2. Steering Gear Inner Tie Rod
3. Steering Gear Boot Assembly
4. Outer Tie Rod Locking Nut
5. Steering Gear Outer Tie Rod
6. Nut - Outer Tie Rod Ball Head to Steering Knuckle
7. Steering Gear Bolt
8. Steering Gear Nut
9. Seal

เสตราจารย์สุวิทย์ วงษ์เย็น



1. Steering Gear Boot
2. Steering Gear Assembly
3. Steering Motor
4. Steering Wheel
5. Steering Gear Outer Tie Rod
6. Steering Column Assembly

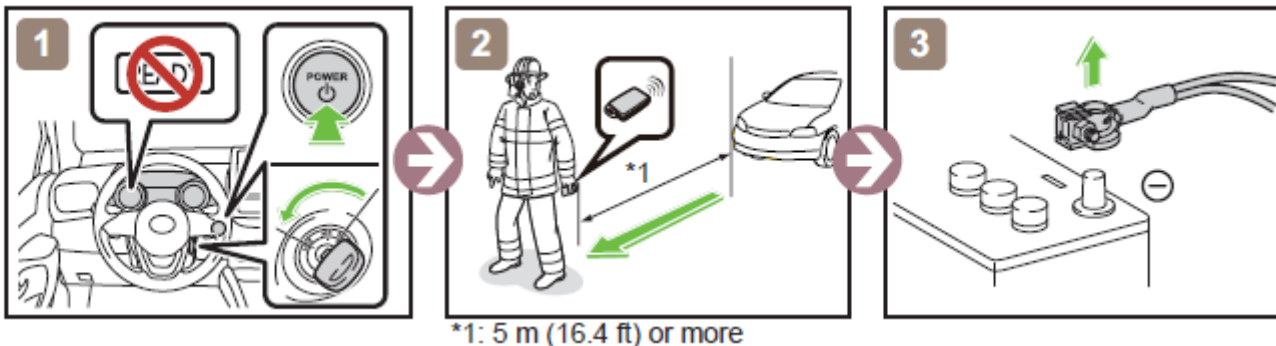
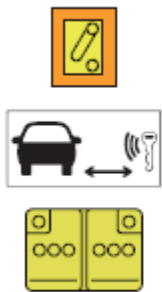
การปฏิบัติงานการตัดไฟฟ้าแรงสูงและถอดเปลี่ยนโมดูลแปลงผันพลังงานของรถยนต์ไฟฟ้า



งานตัดระบบไฟฟ้าแรงสูงของรถยนต์ไฟฟ้า(HV Battery disconnect): MG EP

ขั้นตอนการตัดระบบไฟฟ้าแรงสูงรถยนต์ไฟฟ้า(Service Procedures)

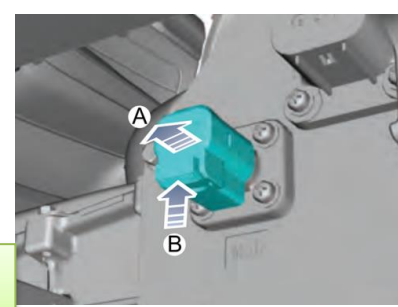
- 1.Ensure the START/STOP switch is OFF, (**READY OFF**)
- 2.Remove keys from the vehicle and place in a secure location at least 2metres from the vehicle and **wait 5 mins** before carrying out next step
- 3.Disconnect the **negative battery cable and protection reconnect pole**.
4. Raise the vehicle on a lift.
5. Remove 1 screw fixing the manual service cover plate to the bottom deflector, and remove the manual service cover plate
6. Loosen the clips and remove the **manual service disconnect (MSD)** from the high-voltage battery pack.
8. Fit special tool TEL00052to the **manual service disconnect(MSD)** base



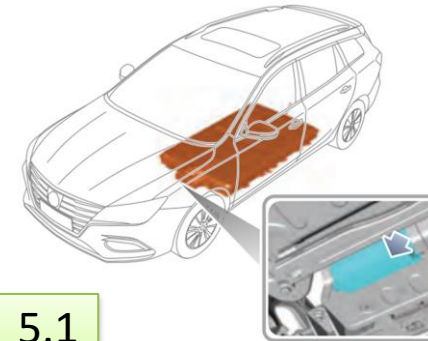
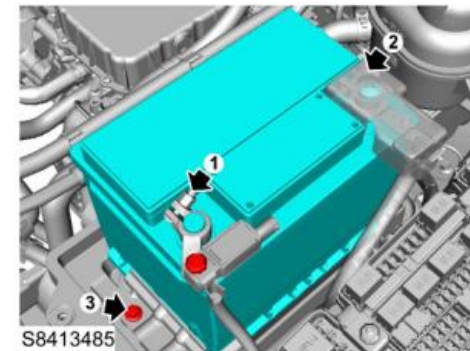
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ผู้ช่วยศาสตราจารย์สุวิทย์ วงษ์เย็น

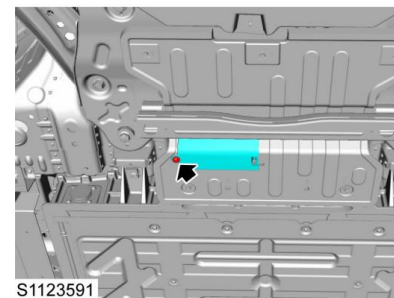
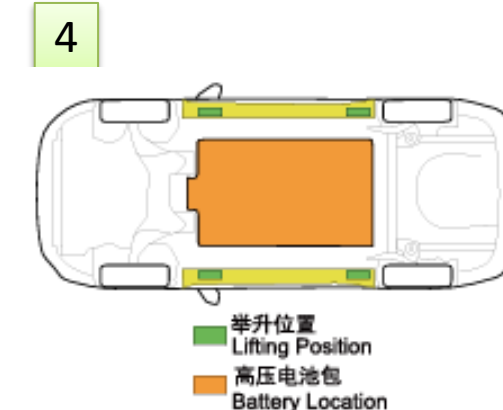
5.3



5.2



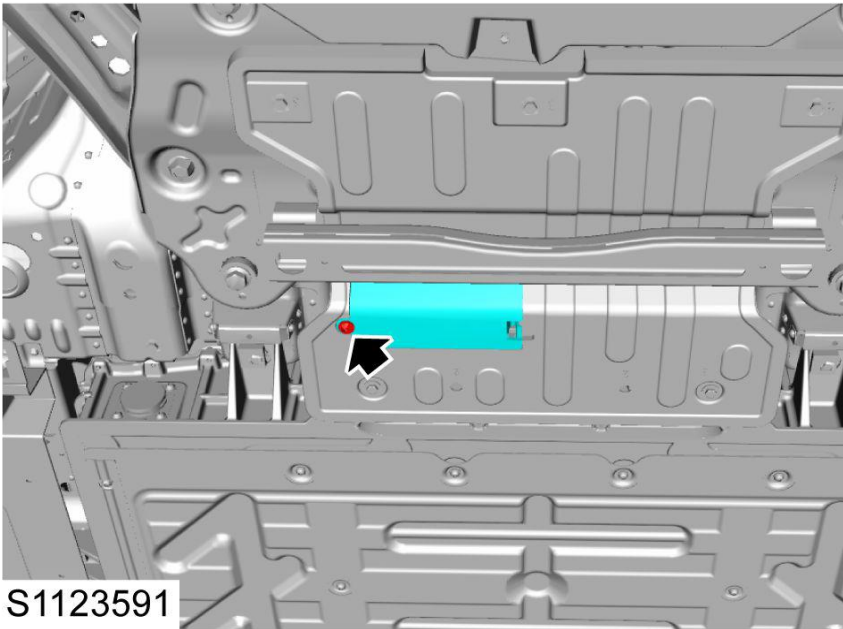
5.1



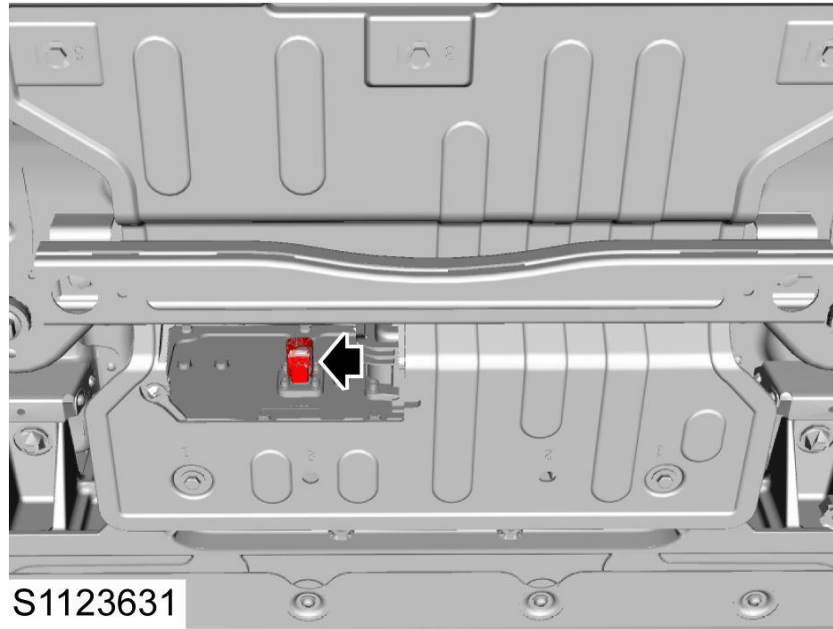
5

2

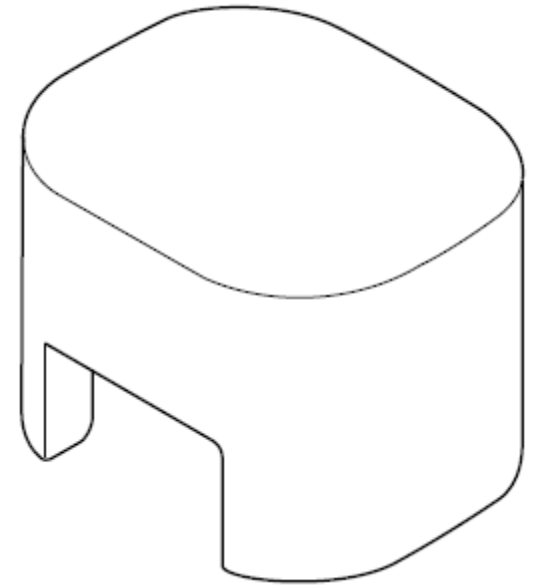
การถอด MSD (Manual Service Disconnect) : Remove



Remove 1 screw fixing the manual service cover plate to the bottom deflector



Loosen the clips and remove the manual service disconnect from the high-voltage battery pack.

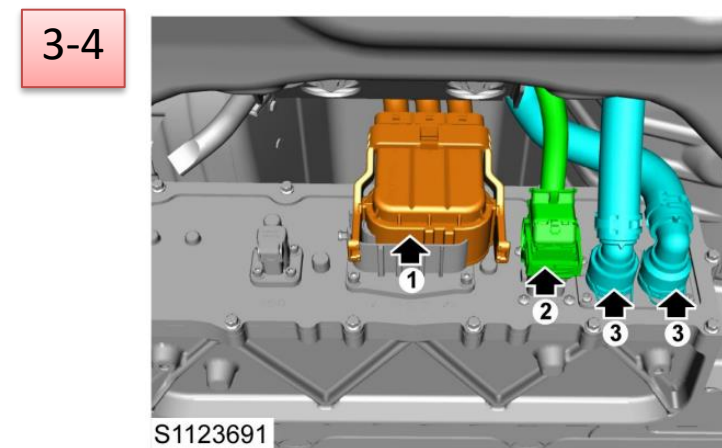
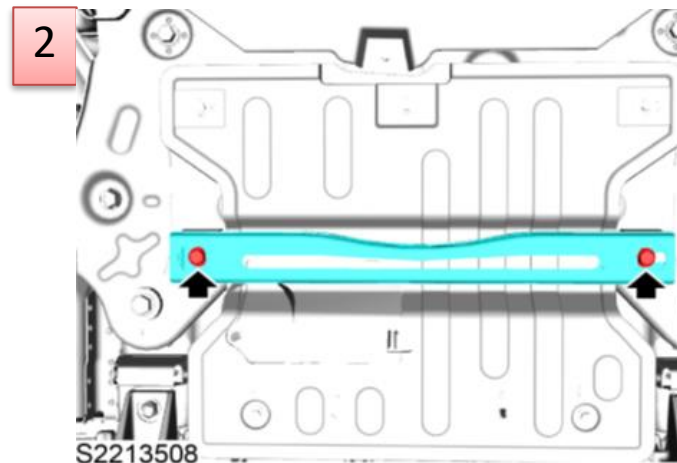
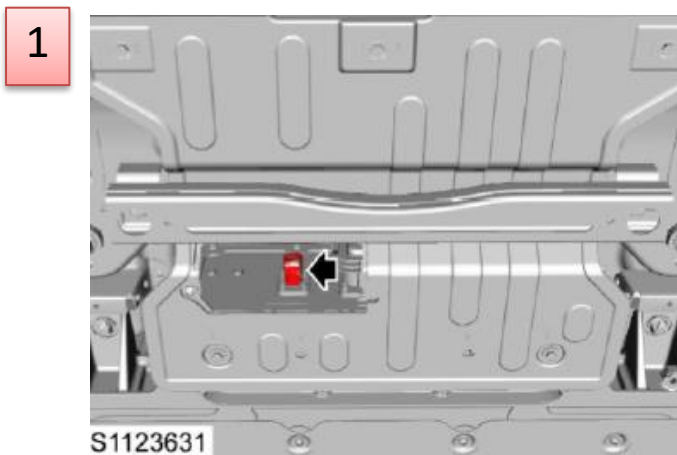


Fit special tool TEL00052 to the manual service disconnect base.

การถอดสายไฟแบตเตอรี่ไฟฟ้าแรงสูงจากรถยนต์ไฟฟ้า(High voltage battery connector remove)

การถอดแบตเตอรี่ไฟฟ้าแรงสูงจากออกจาก High voltage battery pack)รถยนต์ไฟฟ้า

1. Carry out high voltage battery isolation procedure - **see *Manual Service Disconnect (MSD)***.
2. Remove the bottom battery protection plate.
3. *Disconnect the high voltage harness connector (1) from the high-voltage battery pack.*
4. *Disconnect the low voltage harness connector (2) from the high-voltage battery pack.*
5. *Use a multi-meter (DCV, the range is greater than 400V) to measure if there is high voltage between terminals of high-voltage battery pack high-voltage connectors, between the terminals and the ground and between terminals in high-voltage connectors at the high-voltage harness end. If the voltage is zero, continue removing.*

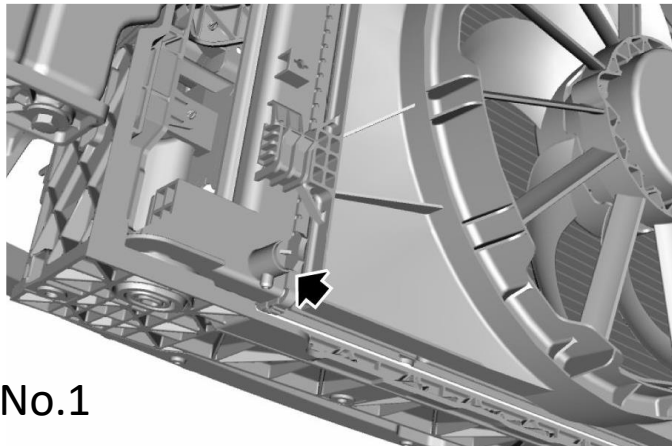


การถอดเปลี่ยนโมดูลแปลงพลังงานของรถยนต์ไฟฟ้า

Service Procedures

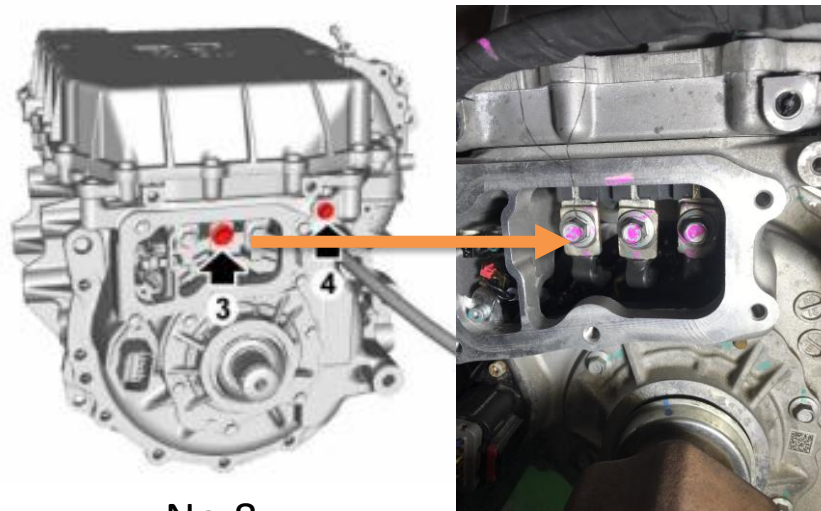
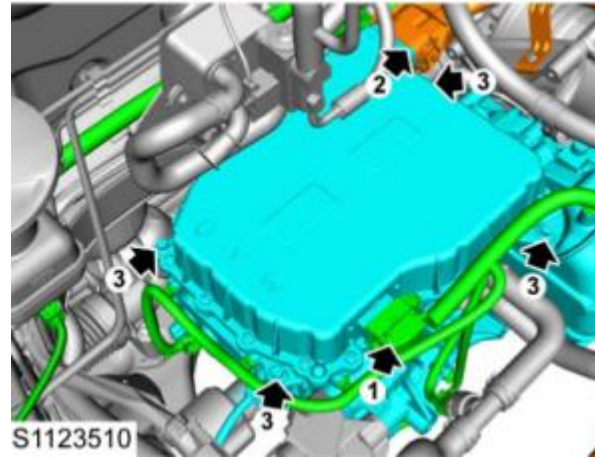
Power electronic box(PEB).

- 1.Drain the PEB coolant system and remove inlet/outlet cooling pipe.
- 2.Clean the coolant inside PEB.
- 3.Disconnected LV connector from PEB.
- 4.Disconnected high voltage harness PEB connector.
- 5.Remove electric drive transmission and suspension and release the buckle and Velcro straps. remove the upper section motor acoustic package cover.
- 6.Remove 4 bolts fixing the PEB assembly to the drive motor.(Tighten torque 26-34 Nm.)
- 7.Remove the high voltage interlock cover plate.
8. Remove 3 bolts fixing the PEB assembly to inside drive motor.(U,V,W busbar)
9. Remove the PEB assembly from drive motor.



No.1

No.3

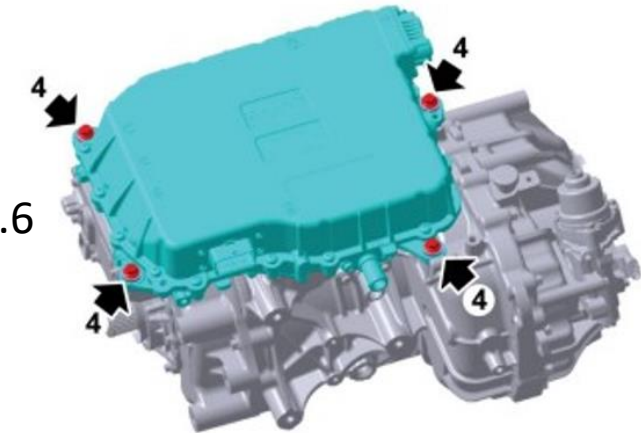


No.8

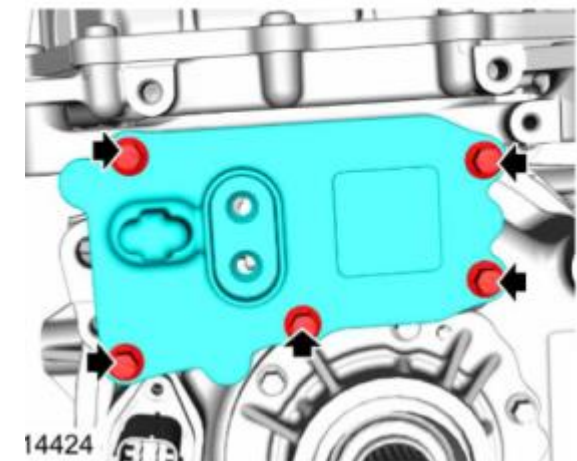
No.4



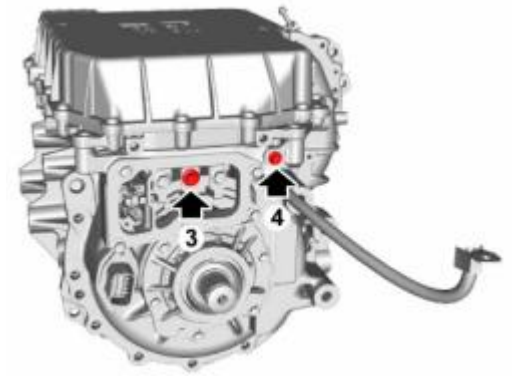
No.6



No.7



การติดตั้งโมดูลแปลงผันพลังงาน หรือ Power electronic box(PEB).



1. Refit the new seal rings.
2. Fix the PEB to the drive motor, fit 4 bolts ,tighten to 26-34 Nm , and check the torque.
3. Carry out a sealing test for water seal integrity.
4. Fix the PEB assembly to the inside of motor , fit 3 screws , and tighten to 14–16 Nm , and check the torque.
- 5. In a pollution-free environment with the ambient temperature of 15°C to 45°C and the ambient humidity of $RH \leq 75\%$, use the high-voltage insulation testing equipment to perform the following operations:**
 - a. Visually check if the PEB and high voltage harnesses are intact and free of defects; verify that the PEB grounding wire is properly fitted.**
 - b. Connect the positive terminal (red lead) of the high-voltage insulation testing equipment to the terminal of PEB harness connector position 1, 2, and 3 (U, V, W ; anyone of the three connectors); connect the negative terminal (black lead) of the testing equipment to the grounding wire (position 4) of the PEB housing.**
 - c. Start the testing equipment , maintain a testing voltage of 1,000 Vdc and wait for stable test values.**
 - d. Record the stable test values R_1, R_2 and R_3 .**
 - e. If the test value $R \geq 1 \text{ M}\Omega$, the test is passed . Otherwise , the parts need to be retested or repaired.**

High voltage battery torque

HV Battery Pack Subsystem Specifications Torque

Description	Value
Bolt-EDM Module to End Plate of Tray	5–7Nm
Bolt-Upper Cover to Tray	6–7Nm
Nut-Upper Cover to Tray	55–65Nm
Bolt-BMS Module to BMS Module Bracket	5–7Nm
Bolt-BMS Module Bracket to Tray	7.5–9.5Nm
Bolt-BMS Module Bracket to Tray	11–13Nm
Bolt-Bus-bar Bracket to Tray	5.5–7.5Nm
Bolt-Bus-bar to Battery Module	7.5–9.5Nm
Nut-Bus-bar to EDM Module	7.5–9.5Nm
Bolt-CMU Module to Battery Module	5.5–7.5Nm
Bolt-Battery Module to Tray	11–13Nm
Bolt-EDM Module to Tray	5–7Nm
Bolt-Water Pipe Bracket to Tray	5.5–7.5Nm
Bolt-GND Harness to Tray	7.5–9.5Nm
Nut-Energy storage system grounding to body	7–10Nm
Bolt-Deflector to Tray	9–12Nm
Bolt-Explosion Proof Valve to Tray	5.5–7.5Nm
Bolt-Plate to Tray	5–6Nm

HV Battery Management System

Specifications

Torque

Description	Value
Bolt-Energy storage system to body	55–65Nm
Bolt-Energy storage system grounding to body	7–10Nm
Bolt-CCU to CCU bracket	19–25Nm
Bolt-CCU Ground Harness to CCU	11-15Nm
Bolt-CCU Positive Harness to CCU	11-15Nm
Bolt-Fast charge to PDU	13–17Nm
Bolt-The PDU bracket to PDU	5–7Nm
Nut-Quick charge grounding to bracket	7–10Nm
Bolt-Charging Port to bracket	7–10Nm
Bolt-Battery Heater to body	7–10Nm
Bolt-Battery Heater grounding wire to body	7–10Nm
Bolt-Electric Vehicle Communication Controller to battery tray	7–10Nm

Inverter assembly torque

Power Drive System

Specifications

Torque

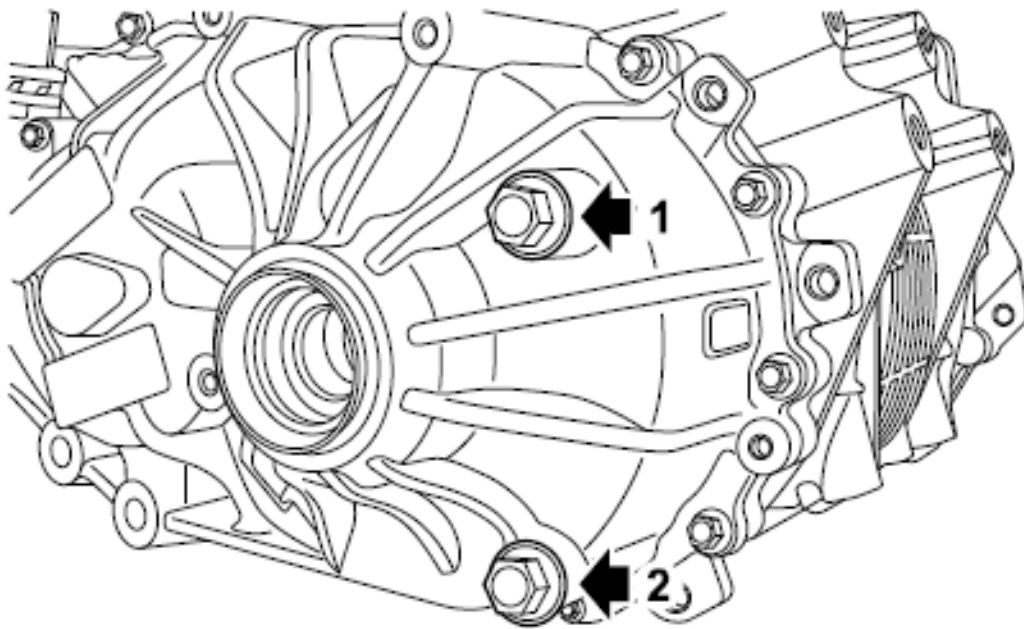
Description	Value
Bolt-PEB to TM	26-34Nm
Bolt-BUS—BAR to TM	14-16Nm
Bolt-Drive motor to EDU	19-25Nm
Bolt-High Voltage Interlock Cover Plate to Casing	7-10Nm
Bolt-Motor Rear End Cover Plate to Casing	7-10Nm
Bolt-Water Pipe Joint to Casing	7-10Nm

Lubrication & Maintenance : MG EP

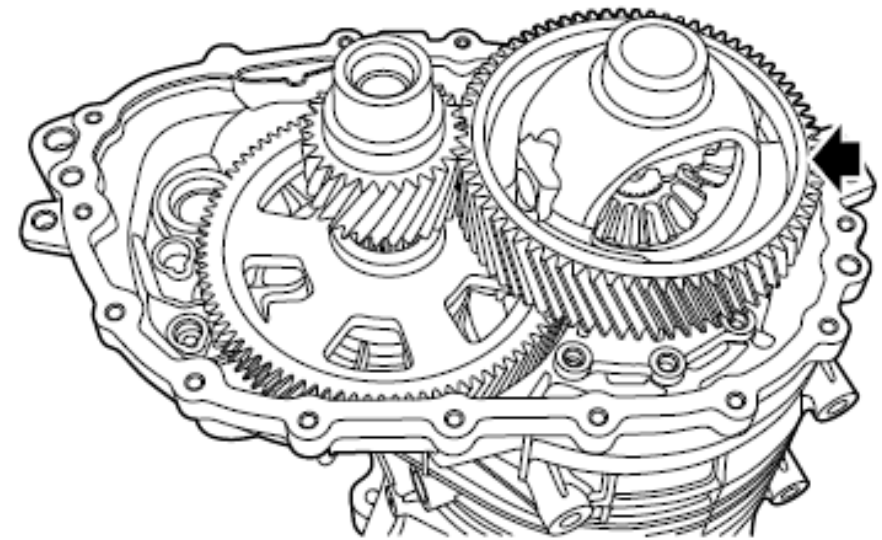
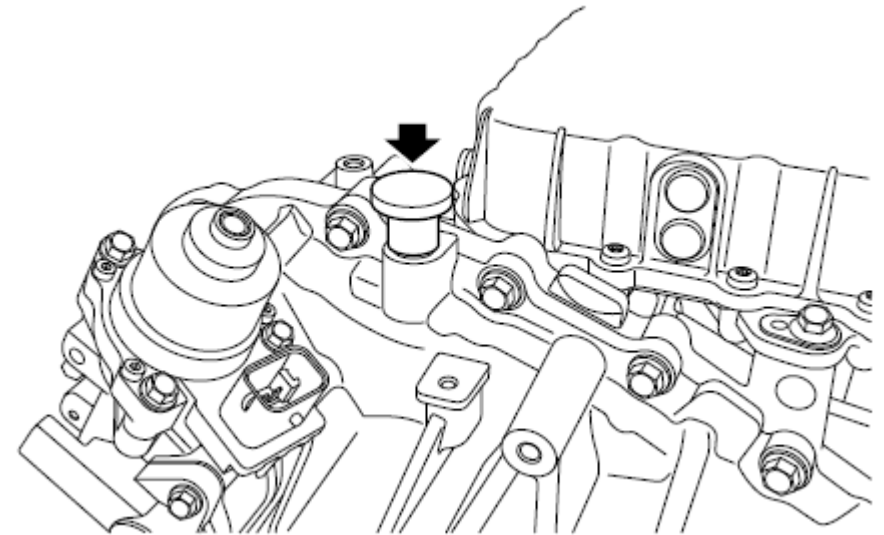
Component/System	Capacity
Electric Drive Transmission Oil:	
Refill from Wet State	0.8 L $\pm$ 0.05 L
Refill from Dry State	0.9 L $\pm$ 0.05 L
Cooling System - Motor Cooling	
Refill from Dry State	4.0 L
Cooling System - Battery Cooling	
Refill from Dry State	4.8 L
Brake System Brake Fluid	0.8 L
Windshield Washer Reservoir	2.5 L

Electric Drive Transmission

Oil - Drain and Refill



1 = Refill plug, tighten it to 30-40Nm
2 = Drain plug , tighten it to 25-35Nm,



•Drain and Refill $0.8 \pm 0.05L$
•Fill from Dry State $0.9 \pm 0.05L$

Thank You

“หวังว่าความรู้นี้จะประโยชน์
สำหรับผู้สนใจ”

ผู้ช่วยศาสตราจารย์สุวิทย์ วงษ์เย็น

